Statistical Modelling and Machine Learning in Longitudinal Data Analysis

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Signature: QUT Verified Signature

Date: 17/05/2021

To my family

Abstract

This thesis mainly focuses on statistical modelling and machine learning in longitudinal data analysis. Longitudinal data analysis often assumes that observations are correlated on the same subjects or clusters and independent among different subjects or clusters. Recently, there has been interest in the applied and methodological analysis of longitudinal data. To approach the correlations in longitudinal data, the current thesis focuses on two popular statistical modelling methods: the linear mixed model (LMM) and generalised estimating equations (GEE). Tree-based methods, the support-vector machine (SVM) and neural network (NN) methods are the machine learning methods discussed in this thesis.

First, GEE is a widely used method for analysing longitudinal data, and the sandwich method is often used to estimate the variance-covariance matrix of the regression coefficient estimators because it is robust against the misspecification of the correlation structure of the responses. However, the sandwich method relies on the inner product of residuals as an estimator for the true covariance of the responses, and this estimator becomes singular and sparse as the cluster size becomes larger. Our focus is to investigate whether the sandwich estimator and its modified versions are still valid when a sparse or singular matrix is used as an estimate of the true covariance of the responses.

The second main interest of this study is to investigate the performance of predictive responses in longitudinal data with LMM and six different machine learning methods: trees, bagging, the random forest, boosting, the SVM and the neural network. The tree method of combined random effects (RE-EM trees) is also included in the comparison. Indeed, statistical models have better performance than these machine learning methods when the mean function of models can be correctly specified.

The last part of the thesis explores the features derived from a mixture of time-window sizes, which can improve the classification accuracy of machine learning algorithms for sheep

behaviours. Preliminary analysis indicates that the time-window size of segmented signal data can significantly affect the classification accuracy of animal behaviours. Three machine learning algorithms, the random forest (RF), the SVM and the linear discriminant analysis (LDA) were applied to investigate the ability of these different approaches to classify sheep behaviour accurately. The results clearly show that the simultaneous inclusion of features derived from mixed time-window sizes significantly improved the behaviour classification accuracy, in contrast to those determined from a single unique time window size. Importantly, using features derived from time windows of different length may provide the context needed to accurately identify certain behaviours.

Preface

Declaration by author

I declare that the thesis has been composed by myself and that the work has not be submitted for any other degree or professional qualification. I confirm that the work submitted is my own, except where work which has formed part of jointly-authored publications has been included. My contribution and those of the other authors to this work have been explicitly indicated below. I confirm that appropriate credit has been given within this thesis where reference has been made to the work of others.

Publications during candidature

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Then the contributions of the PhD candidate and the co-authors to each of publications included in this thesis are presented.

Chapter 3: Performance of variance estimators in the analysis of longitudinal data with a large cluster size

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Contributor	Statement of contribution
Shuwen Hu (Candidate)	Wrote the manuscript, experimental design, conduct experiments, data analysis, revised the manuscript as suggested by co-authors and referees.
You-Gan Wang	Initiated the research concept, supervised research.
Liya Fu	Aided experimental design, supervised research, comments on the manuscript.

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Contributor	Statement of contribution
Shuwen Hu (Candidate)	Wrote the manuscript, experimental design, conduct experiments, data analysis, revised the manuscript as suggested by co-authors.
You-Gan Wang	Initiated the research concept, supervised research.
Christoper Drovandi	Aided experimental design, supervised research, comments on the manuscript.
Taoyun Cao	Revised manuscript, advised on interpretation of results.

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Contributor	Statement of contribution
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Sabine Schmoelzl	Data curation and editing.
Jody McNally	Sensor deployment and annotation.
Bryce Little	Annotation approach.
Daniel Smith	Conceptualisation.
Greg H.-Bishop	Conceptualisation.
You-Gan Wang	Methodology development, supervision.
Yutao Li	Conceptualisation, supervision, writing, reviewing and editing.

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▲ *The Master said: 'I am not one who was born in the possession of knowledge; I am one who is fond of antiquity, and earnest in seeking it there.' (Analects of Confucius: 7.20)*

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Nomenclature

Abbreviations

ACEMS	Australian Centre of Excellence for Mathematical and Statistical
AdaBoost	Adaptive Boosting
ANN	Artificial Neural Network
AR(1)	AugoRegressive model of order 1
BRT	Boosting Regression trees
CART	Classification And Regression Tree
CSIRO	Commonwealth Scientific and Industrial Research Organisation
EM	Expectation Maximisation
GEE	Generalised Estimating Equations
GLM	Generalised Linear Models
GMERT	Generalised Mixed-Effects Regression Trees
GUIDE	Generalised, Unbiased Interaction Detection and Estimation
KKT	Karush-Kuhn-Tucker
KNN	K-Nearest Neighbour
LDA	Linear Discriminant Analysis
LMM	Linear Mixed Model
LS-SVM	Least Squares Support Vector Machine
MEMS	MicroElectroMechanical System
MERT	Mixed-Effects Regression Trees
ML	Maximum Likelihood

ML	Machine Learning
MLP	MultiLayer Perceptron
NLSY	National Longitudinal Survey of Youth
NN	Neural Network
NR	Newton-Raphson
QDA	Quadratic discriminant analysis
QUT	Queensland University of Technology
RE-EM trees	Random Effects-Expectation Maximisation trees
REML	Restricted Estimation Maximum Likelihood
RF	Random Forest
RMSE	Rooted Mean Square Error
SVC	Support Vector Classifier
SVM	Support Vector Machine
SVR	Support Vector Regression
TRMSE	True Rooted Mean Square Error
VIM	Variable Importance Measure

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Chapter 1

Introduction

1.1 Overview

1.1.1 Longitudinal data

Repeated measurements data have become popular in recent years because they frequently appear in an increasing number of research areas, such as clinical trial studies, medical science and economics [Diggle et al., 2002, Heckman and Singer, 1986]. Using repeated measurements data means that there are multiple observations for each different subject. Longitudinal data or panel data are a special type of repeated measurements data in which multiple observations are collected over time. There are some advantages of studies with repeated measurements. First, the design of collecting longitudinal data is the only design that can obtain the information that concerns the individual pattern of change [Li, 2006]. Second, it requires a smaller number of subjects and the requisite time and costs can be much lower. Third, the power of analysis increases because the repeated measure designs control for factors that cause variability between subjects. However, there are also some challenges in the analysis of repeated measurements data. The main characteristic of longitudinal data is the correlation within the observations for each subject, which violates the common assumptions of classical statistical models. The estimates of the regression parameters are inefficient and the power to detect the differences in interests is lower when the correlation is ignored. Usually, the number of observations is not the same for each subject, which is known as unbalanced data. The other common phenomena in the longitudinal data are the data is incomplete as some patients may fail to be revisited because of certain factors relevant or irrelevant to the studies. Therefore, interest in longitudinal data

analysis has increased and appropriate models and corresponding analysis are needed.

We provide an intuitive description of the structure of longitudinal data in Table 1.1.

Table 1.1: The structure of longitudinal data

Subject/Cluster	Time	Covariates			Response
1	1	x_{111}	$\cdots$	x_{11p}	y_{11}
	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	n_1	x_{1n_11}	$\cdots$	x_{1n_1p}	y_{1n_1}
2	1	x_{211}	$\cdots$	x_{21p}	y_{21}
	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	n_2	x_{2n_21}	$\cdots$	x_{2n_2p}	y_{2n_2}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
K	1	x_{K11}	$\cdots$	x_{K1p}	y_{K1}
	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	n_K	x_{Kn_K1}	$\cdots$	x_{Kn_Kp}	y_{Kn_K}

In Table 1.1, y_{ij} signifies the observations of subject i at time j, where $i = 1, \cdots, K$ and $j = 1, \cdots, n_i$. Sometimes, longitudinal data are thought of as a special type of clustered data, which treats a subject as a cluster. We took this viewpoint for this thesis. The number of clusters is represented by K and the cluster size for the ith cluster is n_i. The number of covariates is p and $\boldsymbol{x}_{ij} = (x_{ij1}, \cdots, x_{ijp})$ is the vector of covariates. It is worth noting that the levels of statistical inference will be different when the scales K and n_i are different. For example, when $K = 1$ and $n_i = n$ is large, they are known as time series data. We will now review the development of the statistical analysis for the longitudinal data.

1.1.2 Statistical modelling

The idea of generalised linear models (GLM) was an important breakthrough in statistical modelling [McCullagh and Nelder, 1989, Nelder and Wedderburn, 1972]. The GLM can analyse both continuous and categorical data, which come from different distributions, such as normal, Poisson, binomial and gamma, etc. However, it still requires the assumption for the distribution of the response variables. There are many extensions of GLM, such as the quasi-likelihood method, which only needs the first two moments of the distribution for its responses

[Wedderburn, 1974]. Zeger and Liang [1992] gave an overview of methods for the analysis of longitudinal data: marginal models, transition models and random-effects models. Marginal models focus on the population average, which has the advantage of separately modelling the regression and within-subject correlation. The marginal model regression coefficients receive the same interpretation regardless of the number of repeated observations. The magnitude of the within-subject correlation does not alter the interpretations of the regression coefficients. However, the transition models and random-effects models aim to provide a prototype of the covariate effects and the within-subject correlation simultaneously. Transition models combine assumptions about the dependence of responses on the covariates and correlation into a single equation. It is usually assumed that the responses follow a stochastic process, which depends only on the values of the covariates measured by the subject in the transition models. Random-effects models operate under the assumption that the correlation among responses for an individual arise from heterogeneity among the subjects. The regression coefficients can explain the effect of the covariates both on the average of the population and on the subject's responses.

Liang and Zeger [1986] proposed a marginal model method—generalised estimating equations (GEE)—as an extension of the quasi-likelihood method to the multivariate correlated response. The advantage of this method is that consistent and asymptotically normal estimates can be obtained even when the correlation structure is misspecified. The corresponding variance-covariance estimator is called a 'sandwich estimator' or a 'robust estimator'. The GEE estimates are obtained using an iterative quasi-scoring algorithm. The variance-covariance matrix of responses is treated as signifying nuisance parameters, which means that the model fitting turns out to be easier. GEE is a flexible model that can accommodate different structures. The commonly used correlation structures include independent, autoregressive (for example, AR(1)), exchangeable and toeplitz. Wang [2014] provided a review of the recent developments in GEE.

In addition, Gardiner et al. [2009] examined the assumptions of three statistical models–the fixed-effects model, the random-effects model and GEE–in the analyses of longitudinal repeated-measures data. They also illustrated the similarities and differences of these methods in application. As we realised, whether the response variables are assumed to be normally distributed or not will cause different statistical analyses. Mixed-effects models, including both fixed effects and random effects, can handle dependence in longitudinal data [Laird and Ware, 1982]. There are various types of mixed-effects models, including linear mixed effects,

nonlinear mixed effects, and generalised linear mixed effects. We focused on the linear mixed-effects model (LMM) in this thesis. Linear mixed-effects models (LMM) are an extension of simple linear models by the inclusion of random effects, which used to account for correlation among measurements within the same subject. A LMM is a special form of random-effects model, wherein the normality assumption is required. The linear random intercept model is a special case of LMM wherein the random-effects covariates matrix is an identity matrix. It is worth mentioning that the linear random intercept model is equivalent to a linear marginal model with compound symmetry (also known as exchangeable correlation structure). An expectation-maximisation (EM) algorithm can be used to determine the maximum likelihood (ML) and restricted maximum likelihood estimation (REML) in the longitudinal data setting [Laird et al., 1987]. Lindstrom and Bates [1988] developed an efficient and computationally stable implementation of the Newton-Raphson algorithm for obtaining the parameters in mixed-effects models for longitudinal data.

1.1.3 Machine learning

Machine learning encompasses many widely used methods, such as classification and regression trees, the support-vector machine (SVM) and the neural network (NN). The goals of learning include understanding the structure of the data and obtaining a prediction. Supervised machine learning involves building a model for prediction, which is the focus of this thesis. There are inputs but no outputs in unsupervised machine learning. The most 'simple' and popular method is linear regression, which is used for predicting quantitative values. Linear discriminant analysis (LDA) [Fisher, 1936] and logistic regression [Cramer, 2002] were proposed to predict qualitative values. As mentioned in section 1.1.2, the GLM proposed by McCullagh and Nelder [1989] include both linear and logistic regressions as special cases. Because of the computational infeasibility of fitting nonlinear relationships, these methods were almost exclusively linear models during the early 1970s. LDA is a discriminant analysis that can separate a dataset into two or more classes. It assumes that the data within each class are drawn from a multivariate Gaussian distribution with a class-specific mean vector and a covariance matrix that is common to all classes.

The decision tree, bagging, the random forest (RF) and the boosting methods can be grouped together as they are all tree-based methods. The classification and regression tree (CART) is a

popular algorithm which was proposed by Breiman et al. [1984]. There are many advantages to tree-based methods. They can handle large data sets with many covariates without the necessity of considering collinearity problems. In addition, they can detect the potential interactions between covariates automatically and explore nonlinear relationships well without any distributional assumptions. A characteristic of trees is that they can be displayed graphically and are easy to explain. Bagging is the application of the bootstrap procedure to decision trees to lower the variance. The RF is a popular tree-based ensemble method that builds a large collection of de-correlated trees and then averages them based on the bagging [Breiman, 2001]. The different trees based on the bootstrapped data are independent in bagging. Boosting works in a similar way to bagging, but the difference is that the trees are constructed sequentially, which means that the growth of each tree depends on the trees that have already been constructed. Older trees can be seen to be built on the 'residuals' of formal trees.

The initial idea of the SVM is to construct a linear partition of the high-dimensional space into two sub-spaces for classification or regression [Schölkopf et al., 2002, Vapnik, 1998]. Intuitively, as the larger margins tend to provide fewer classification errors, a good separation is obtained by a hyperplane that has the largest distance to the nearest training data. It often happens that the data are not linearly separable. Then, a kernel function is used to transform the non-linearity system in the input space into the linearity system in the feature space. The artificial neural network (ANN) or, simply, the neural network (NN) is based on a collection of connected units or nodes, inspired by modelling the neurons in a biological brain. A multilayer perceptron (MLP) is an aspect of the feedforward ANN for classification and regression [Hastie et al., 2009].

These machine learning methods can be directly used in the prediction of longitudinal data analysis. These machine learning methods are not intended to deal with longitudinal data, the assumption of which is that observations are correlated on the same subjects and independent among different subjects. GEE and the LMM are popular methods for addressing correlations in the longitudinal data. However, GEE is a marginal method, which only focuses on the population average. Therefore, mixed-effects machine learning is proposed [Berger and Tutz, 2018, Cho, 2010, Hajjem et al., 2014, Kundu and Harezlak, 2019, Luts et al., 2012, Segal, 1992]. The key idea of mixed-effects machine learning is to dissociate fixed effects from random effects.

Segal [1992] was the first to apply regression trees to longitudinal data by modifying the split function to accommodate multiple responses. Hajjem et al. [2011] and Sela and Simonoff [2012] independently proposed the mixed-effects regression tree method or the random-effects expectation-maximisation (RE-EM) trees, through which the correlated observations are regarded as nested within clusters. They used a tree structure to estimate the mean function, but also incorporated object-specific random effects. Similarly, Hajjem et al. [2014] generalised the RF to longitudinal data, replacing the regression trees within each iteration of the mixed-effects regression trees algorithm with a forest of regression trees. As for the SVM algorithm, Cho [2010] proposed a mixed-effects least-squares support-vector regression for longitudinal data. In addition, a mixed-effects least SVM classifier was introduced by Luts et al. [2012].

1.2 Objectives and research questions

The main objective of this thesis is to investigate the performance of statistical modelling and machine learning methods in the analysis of longitudinal data. Specifically, GEE and the LMM are the statistical modelling methods discussed in this thesis to deal with correlations in the longitudinal data. The related machine learning methods discussed in this thesis are the tree-based method, the SVM, the NN and LDA.

The research questions are presented in this section. Each research questions motivates a chapter of this thesis. This section briefly introduces the literature-based status of each research question and the background of the research's contribute to the knowledge system. A detailed literature review of the factors surrounding each research question can be found in the introductory section of each of the corresponding chapters.

1. **How does the variance estimators perform in the analysis of longitudinal data with a large cluster size?**

GEE is a widely used method for analysing longitudinal data, and the sandwich method is often used to estimate the variance-covariance matrix of the regression coefficient estimators because it is robust to the misspecification of the correlation structure of the responses. Many studies have focused on improving the efficiency of the variance estimators when the sample size is not large enough [Kauermann and Carroll, 2001, Mancl and DeRouen, 2001, Pan, 2001, Wang and Long, 2011]. However, the sandwich method relies on the residual products as an

estimator for the true covariance of the responses, and the estimator becomes singular and sparse as the cluster size becomes larger. Our focus is to investigate whether the sandwich estimator and its modified versions are still valid when a sparse or singular matrix is used as an estimate of the true covariance of the responses.

2. **How do statistical modelling methods and machine learning methods perform in the analysis of longitudinal data?**

There have been very few comparison studies of statistical models and machine learning methods in the analysis of longitudinal data. Ahmed et al. [2010] presented a comparison study for the machine learning models for time series forecasting. Li and Wu [2015] claimed that the traditional LMM was inferior to the machine learning methods for both long-term and short-term prediction in milk protein data, which was apparently because the LMM is not sufficient to fit this data. It is of interest whether the machine learning methods always obtain a better performance than those of traditional statistical modelling in the longitudinal data analysis.

3. **How can sheep behaviours be accurately classified? How can their features be extracted according to the sensor data?**

This research question arises from a real-world case. Data were collected from accelerometers attached to the necks of 17 Merino sheep over a period of two days. A ground truth data set of behaviour recordings (grazing, ruminating, walking and standing) over the same period were also recorded. Our focus is to classify the sheep behaviours accurately. Primarily, the analysis indicates that the time-window size of the segmented signal data can significantly affect the classification accuracy of animal behaviours [Smith et al., 2016]. To date, there have been no studies evaluating the impact of a mixture of time-window size features on the accuracy of animal behaviour classification. Three machine learning approaches, random forest (RF), SVM and LDA are used to classify sheep behaviours. Apart from the common features derived from six basic statistics (minimum, maximum, mean, standard deviation, skewness and kurtosis) and the squared acceleration magnitude (acc) [Rahman et al., 2018], the features of the cumulative effects of measurements for a given time-window size are also worth investigation.

These are the two main research aims:

1. To validate or develop methods to obtain robust and efficient variance estimators when the cluster size is large.

2. To develop and compare the machine learning methods to analyse longitudinal data.

   The research objectives for aim 2 are as follows:

- 2.1 To compare the performance of statistical modelling methods and machine learning methods in the analysis of longitudinal data.
- 2.2 To apply the machine learning methods to real-world sensor data.

Figure 1.1 presents a diagrammatic representation of research motivation.

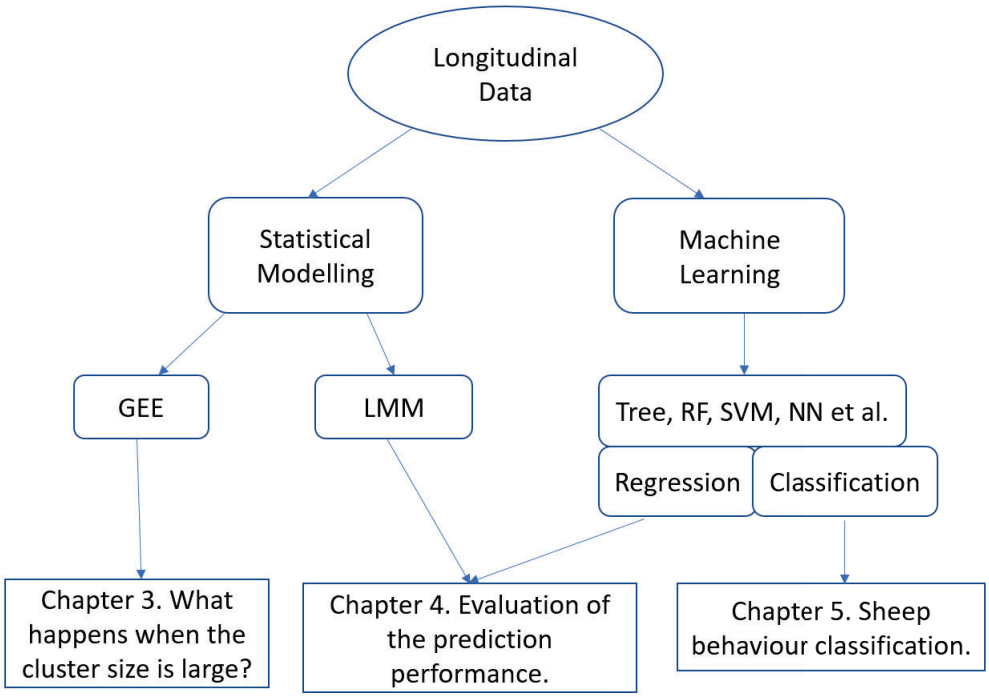

Figure 1.1: Diagrammatic representation of research motivation

1.3 Thesis structure

The main content of this thesis is divided into six chapters. Each of the main chapters within this thesis correspond to a journal publication (chapter 5) or a manuscript under review (chapters 3 and 4), and all are first-authored by the candidate. Hence, each chapter can be read independently of the others. Each chapter includes an introductory section that describes the problem of interest, the background content and the literature review. There may be some overlap in background information across different chapters because each chapter can be read

independently. However, the chapters are organised in such a way that they form a consistent whole that can be read in order.

Chapter 1 contains a brief introduction to the background and scientific problems. Chapter 2 provides a comprehensive literature review of the topics covered in this thesis, including longitudinal data analysis and relevant machine learning methods. An overview of the GEE and LMM methods for longitudinal data analysis is provided. The sandwich estimates, the various modified sandwich estimates and the smooth bootstrap methods are also described. The tree-based method, the SVM, the NN and the LDA are introduced at the end of this chapter. In chapter 3, the performance of variance estimators is investigated in the analysis of longitudinal data with a large cluster size. Our focus is to investigate whether the sandwich estimator and its modified versions are still valid when a sparse or singular matrix is used as an estimate of the true covariance of the responses. Chapter 4 investigates the performance of predicting responses in longitudinal data with LMM and six different machine learning methods: trees, bagging, the RF, boosting, the SVM and the NN. The RE-EM trees method is also included in the comparison. The motivation of this chapter is that there are few comparison studies of the performance of statistical modelling and machine learning in the longitudinal data analysis. Chapter 5 comprises a project on using machine learning method to accurately classify sheep behaviour. Three machine learning algorithms—RF, SVM and LDA are applied. The chapter also contains pre-processing sensor data and feature engineering. The results clearly show that the simultaneous inclusion of features derived from mixed-time window sizes significantly improves the behaviour classification accuracy, in contrast to those determined from a single, unique time-window size. In chapter 6, a summary of the contributions and limitations of this research and suggestions for future work is given.

Chapter 2

Literature Review

This chapter covers the theoretical background and reviews of longitudinal data analysis and the relevant machine learning methods. In subsection 2.1, the theoretical expression and the estimates of the generalized estimating equations (GEE) and the linear mixed-effects model (LMM) are introduced. In the following subsection 2.2, the relevant machine learning algorithm—the tree-based method, the support-vector machine (SVM), the neural network (NN) and linear discriminant analysis—are explained. Moreover, mixed-effects regression trees, the mixed-effects random forest (RF) and mixed-effects support-vector regression are also described.

2.1 Longitudinal data analysis

Longitudinal data appear frequently in economics, finance, medical science and other fields, in which it is often assumed that observations are correlated within the same subjects and independent among different subjects. Suppose that there are K subjects and that each subject is repeatedly observed n_i times. Let us denote the observed data as $(\boldsymbol{x}_{ij}, y_{ij})$, $i = 1, \cdots, K, j = 1, \cdots, n_i$. The marginal longitudinal generalised linear model only requires the first two moments of y_{ij}. Assume that the mean of y_{ij} is $\mu_{ij} = E(y_{ij}) = g(\boldsymbol{x}_{ij}^T \boldsymbol{\beta})$ and the variance of the response y_{ij} is $\text{Var}(y_{ij}) = \phi\sigma_{ij}^2$, where $i = 1, \cdots, K, j = 1, \cdots, n_i$, β is the $p \times 1$ vector of regression parameters, $g(\cdot)$ is a known link function, ϕ is the unknown dispersion parameter and σ_{ij}^2 is often expressed as a known function of μ_{ij}. Let $\boldsymbol{Y}_i = (y_{i1}, \cdots, y_{in_i})^T$ be the vector of responses for the ith subject and $\boldsymbol{X}_i = (\boldsymbol{x}_{i1}, \cdots, \boldsymbol{x}_{in_i})^T$ denote the associated $n_i \times p$ matrix of covariates. The marginal mean vector of $\boldsymbol{Y}_i$ is $\boldsymbol{\mu}_i(\boldsymbol{\beta}) = (\mu_{i1}(\boldsymbol{\beta}), \cdots, \mu_{in_i}(\boldsymbol{\beta}))^T$ and the

covariance matrix is $\boldsymbol{V}_i = \boldsymbol{A}_i^{1/2}\boldsymbol{R}_0(\boldsymbol{\alpha})\boldsymbol{A}_i^{1/2}$, where $\boldsymbol{A}_i(\boldsymbol{\beta}) = \text{diag}(\sigma_{ij})$, and the true structure of $\boldsymbol{R}_0$ is unknown and parameterised by $\boldsymbol{\alpha}$. Therefore, a working correlation matrix $\boldsymbol{R}_i(\boldsymbol{\alpha})$ can be used, which is $\boldsymbol{V}_i = \phi\boldsymbol{A}_i^{1/2}\boldsymbol{R}_i(\boldsymbol{\alpha})\boldsymbol{A}_i^{1/2}$. If the number of measurements from each cluster is the same—that is, $n_1 = n_2 = \cdots = n_K = n$—the data is called balanced data. Otherwise, the data are known as unbalanced longitudinal data.

2.1.1 Generalised estimating equations

Since the GEE was first proposed by Liang and Zeger [1986], significant literature on the improvement and expansion of the GEE method has revealed its popularity. An appealing characteristic of this approach is that consistent regression estimators can still be obtained, even if the working correlation structure is misspecified. Moreover, the GEE approach only requires first and second moment assumptions. The GEE method can be expressed as follows:

$$U(\boldsymbol{\beta}) = \sum_{i=1}^{K} \boldsymbol{D}_i(\boldsymbol{\beta})^T \boldsymbol{V}_i^{-1}(\boldsymbol{\theta})\boldsymbol{S}_i(\boldsymbol{\beta}) = \boldsymbol{0}_{p\times 1}. \tag{2.1}$$

Here, $\boldsymbol{D}_i(\boldsymbol{\beta}) = \partial\boldsymbol{\mu}_i(\boldsymbol{\beta})/\partial\boldsymbol{\beta}^T$ and $\boldsymbol{S}_i(\boldsymbol{\beta}) = \boldsymbol{Y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta})$. The true values are denoted as $\widetilde{\boldsymbol{\theta}} = (\widetilde{\boldsymbol{\beta}}, \widetilde{\boldsymbol{\alpha}}, \widetilde{\phi})$. An iterative algorithm is used to obtain the regression estimates:

$$\hat{\boldsymbol{\beta}}^{(m+1)} = \hat{\boldsymbol{\beta}}^{(m)} + \left(\sum_{i=1}^{K} \hat{\boldsymbol{D}}_i^{\,T}\hat{\boldsymbol{V}}_i^{-1}\hat{\boldsymbol{D}}_i\right)^{-1}\left(\sum_{i=1}^{K} \hat{\boldsymbol{D}}_i^{\,T}\hat{\boldsymbol{V}}_i^{-1}\hat{\boldsymbol{S}}_i\right). \tag{2.2}$$

Here, $\hat{\boldsymbol{D}}_i$ and $\hat{\boldsymbol{S}}_i$ are evaluated at $\hat{\boldsymbol{\beta}}^{(m)}$ and $\hat{\boldsymbol{V}}_i$ is evaluated at $\hat{\boldsymbol{\theta}}^{(m)} = (\hat{\boldsymbol{\beta}}^{(m)}, \hat{\boldsymbol{\alpha}}, \hat{\phi})$. Liang and Zeger [1986] provided the consistency and asymptotic normality of the regression estimators $\hat{\boldsymbol{\beta}}$. The scale parameter ϕ can usually be estimated using the moments of original or person residuals. The estimates for the unknown correlation parameter $\boldsymbol{\alpha}$ should be given consideration. Wang and Carey [2003] investigated the asymptotic relative efficiency of the GEE method for the mean parameters when the correlation parameters were estimated by various methods. They proposed that the independence working model maintains high efficiency only for very restricted cases and, in general, exhibits a substantial loss in efficiency even when the design is balanced. The popular choices for this working correlation matrix are the independence, AR(1), exchangeable (also called compound symmetry) and unstructured models. For certain working correlation structures, the parameters $\boldsymbol{\alpha}$ can be estimated by the residual-based moment method.

Therefore, it is necessary to choose the correlation structure that is the most plausible for the data collected.

Under mild and regular conditions, when $K \to \infty$, the asymptotic covariance matrix of $\hat{\boldsymbol{\beta}}$ is

$$\text{cov}(\hat{\boldsymbol{\beta}}) = \left(\sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{D}_i\right)^{-1} \left(\sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \text{cov}(\boldsymbol{Y}_i) \boldsymbol{V}_i^{-1} \boldsymbol{D}_i\right) \left(\sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{D}_i\right)^{-1}.$$

The 'sandwich method' is a well-known and robust approach for obtaining the variance estimators in longitudinal data, which uses the product of residuals $\boldsymbol{S}_i \boldsymbol{S}_i^T$ to estimate the true covariance matrix of $\boldsymbol{Y}_i$. The so-called sandwich estimator appears as follows:

$$\hat{\boldsymbol{V}}_{LZ} = \widehat{\text{cov}}(\hat{\boldsymbol{\beta}}) = \hat{\boldsymbol{Q}}^{-1} \hat{\boldsymbol{\Omega}}_{LZ} \hat{\boldsymbol{Q}}^{-1}. \tag{2.3}$$

Here:

$$\hat{\boldsymbol{Q}} = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{D}_i|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}},$$

and

$$\hat{\boldsymbol{\Omega}}_{LZ} = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{S}_i \boldsymbol{S}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{D}_i|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}.$$

Notice that in (2.3) and in the following covariance estimators, the values of $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and ϕ are replaced by their consistent estimates $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\alpha}}$, and $\hat{\phi}$, respectively. It was shown that under conditions of mild regularity, the estimator $\hat{\boldsymbol{\beta}}$ is consistent and asymptotically and normally distributed with a mean $\widetilde{\boldsymbol{\beta}}$ and a variance of $\text{cov}(\hat{\boldsymbol{\beta}})$ [Liang and Zeger, 1986]. In case the working correlation matrix $\boldsymbol{R}_i$ is specified correctly, we have $\boldsymbol{V}_i = \text{cov}(\boldsymbol{Y}_i)$, which result in:

$$\hat{\boldsymbol{V}}_N = \hat{\boldsymbol{Q}}^{-1} = (\sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{D}_i)^{-1}|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}}. \tag{2.4}$$

This is known as the model-based or naive estimate of $\text{cov}(\hat{\boldsymbol{\beta}})$. Generally, this model-based estimator would be the most optimal and the smallest covariance estimator.

The sandwich estimator is consistent, even if the working correlation is misspecified. However, the sandwich estimator underestimates the true variance of the parameters when the sample size is small. Many improved sandwich estimators have been proposed [Kauermann and Carroll, 2001, Mancl and DeRouen, 2001, Pan, 2001, Wang and Long, 2011].

For the sandwich method, $\boldsymbol{S}_i\boldsymbol{S}_i^T$ was considered the estimator of $\text{cov}(\boldsymbol{Y}_i)$. However, Pan [2001] noticed that $\boldsymbol{S}_i\boldsymbol{S}_i^T$ is not an efficient estimator of $\text{cov}(\boldsymbol{Y}_i)$ because it is based on the data from only one cluster. This led to the modified estimator, under the assumption that there is a common correlation structure across all clusters; hence, the marginal variance of y_{ij} should be correctly specified. Pan [2001] proposed that the working correlation matrix could borrow information from other different subjects:

$$\hat{\boldsymbol{R}}_P = \frac{1}{K}\sum_{i=1}^{K} \boldsymbol{A}_i^{-1/2}\boldsymbol{S}_i\boldsymbol{S}_i^T\boldsymbol{A}_i^{-1/2}.$$

In other words, the $\text{cov}(\boldsymbol{Y}_i)$ can be estimated by $\boldsymbol{A}_i^{1/2}\hat{\boldsymbol{R}}_P/\boldsymbol{A}_i^{1/2}$. Then, the robust variance estimator is as follows:

$$\hat{\boldsymbol{V}}_P = \hat{\boldsymbol{Q}}^{-1}\hat{\boldsymbol{\Omega}}_P\hat{\boldsymbol{Q}}^{-1}, \tag{2.5}$$

where

$$\hat{\boldsymbol{\Omega}}_P = \sum_{i=1}^{K} \boldsymbol{D}_i^T\boldsymbol{V}_i^{-1}\boldsymbol{A}_i^{1/2}\hat{\boldsymbol{R}}_P\boldsymbol{A}_i^{1/2}\boldsymbol{V}_i^{-1}\boldsymbol{D}_i.$$

It has been shown that these estimators are more efficient than sandwich estimators because they can utilise more information across different subjects.

Other modified sandwich estimators were proposed by Mancl and DeRouen [2001] and by Kauermann and Carroll [2001] to improve performance when the number of clusters is small. The idea was that $\boldsymbol{S}_i\boldsymbol{S}_i^T$ underestimates $\text{cov}(\boldsymbol{Y}_i)$; hence Mancl and DeRouen [2001] proposed that the covariance of $\boldsymbol{Y}_i$ can be estimated by:

$$\widehat{\text{cov}}(\boldsymbol{Y}_i)_{MD} = (\boldsymbol{I}_i - \boldsymbol{H}_{ii})^{-1}\boldsymbol{S}_i\boldsymbol{S}_i^T(\boldsymbol{I}_i - \boldsymbol{H}_{ii}^T)^{-1},$$

where $\boldsymbol{I}_i$ is an identity matrix and $\boldsymbol{H}_{ii} = \boldsymbol{D}_i(\sum_{i=1}^{K}\boldsymbol{D}_i^T\boldsymbol{V}_i^{-1}\boldsymbol{D}_i)^{-1}\boldsymbol{D}_i^T\boldsymbol{V}_i^{-1}$. Then, the bias

corrected variance estimator is as follows:

$$\hat{\boldsymbol{V}}_{MD} = \hat{\boldsymbol{Q}}^{-1}\hat{\boldsymbol{\Omega}}_{MD}\hat{\boldsymbol{Q}}^{-1}, \tag{2.6}$$

where

$$\hat{\boldsymbol{\Omega}}_{MD} = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1}(\boldsymbol{I}_i - \boldsymbol{H}_{ii})^{-1}\boldsymbol{S}_i\boldsymbol{S}_i^T(\boldsymbol{I}_i - \boldsymbol{H}_{ii}^T)^{-1}\boldsymbol{V}_i^{-1}\boldsymbol{D}_i.$$

A better understanding of the bias-correction estimates from Mancl and DeRouen [2001] was provided by Li [2006], based on the common results of ordinary linear regression.

Otherwise, Kauermann and Carroll [2001] suggested using the leverage-adjusted residual $(\boldsymbol{I} - \boldsymbol{H}_{ii})^{-1/2}\boldsymbol{S}_i$ to replace the residual $\boldsymbol{S}_i$. Then, the corresponding variance estimator is as follows:

$$\hat{\boldsymbol{V}}_{KC} = \hat{\boldsymbol{Q}}^{-1}\hat{\boldsymbol{\Omega}}_{KC}\hat{\boldsymbol{Q}}^{-1}, \tag{2.7}$$

where

$$\hat{\boldsymbol{\Omega}}_{KC} = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1}(\boldsymbol{I}_i - \boldsymbol{H}_{ii})^{-1/2}\boldsymbol{S}_i\boldsymbol{S}_i^T(\boldsymbol{I}_i - \boldsymbol{H}_{ii}^T)^{-1/2}\boldsymbol{V}_i^{-1}\boldsymbol{D}_i.$$

Combining the strength from Pan [2001] and Mancl and DeRouen [2001], Wang and Long [2011] proposed that the covariance of $\boldsymbol{Y}_i$ can be estimated as:

$$\widehat{\mathrm{cov}}(\boldsymbol{Y}_i)_{WL} = \boldsymbol{A}_i^{1/2}\{\sum_{i=1}^{K} \boldsymbol{A}_i^{-1/2}(\boldsymbol{I}_i - \boldsymbol{H}_{ii})^{-1}\boldsymbol{S}_i\boldsymbol{S}_i^T(\boldsymbol{I}_i - \boldsymbol{H}_{ii}^T)^{-1}\boldsymbol{A}_i^{-1/2}/K\}\boldsymbol{A}_i^{1/2}.$$

Then, the variance estimator is

$$\hat{\boldsymbol{V}}_{WL} = \hat{\boldsymbol{Q}}^{-1}\hat{\boldsymbol{\Omega}}_{WL}\hat{\boldsymbol{Q}}^{-1}, \tag{2.8}$$

where

$$\hat{\boldsymbol{\Omega}}_{WL} = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1}\widehat{\mathrm{cov}}(\boldsymbol{Y}_i)_{WL}\boldsymbol{V}_i^{-1}\boldsymbol{D}_i.$$

It is worth mentioning that these modified sandwich estimators are proposed to reduce bias when the cluster size is small. However, we are interested in whether the sandwich estimator and its modified versions are still valid when a sparse or singular matrix is used as an estimate of the true covariance of the responses in longitudinal data with a large cluster size.

Warton [2011] investigated a high-dimensional setting in which the number of clusters (K) was not large relative to the cluster size (n), based on Pan [2001]. If n is large and K is small, $\hat{\boldsymbol{R}}_P$ may be affected because $\boldsymbol{S}_i\boldsymbol{S}_i^T$ is a large ($n \times n$) sparse matrix whose rank is one. This leads to instability in $\widehat{\text{cov}}(\hat{\boldsymbol{\beta}})$. Hence, Warton [2011] proposed a regularisation approach:

$$\hat{\boldsymbol{R}}_W = \lambda\hat{\boldsymbol{R}}_P + (1-\lambda)\boldsymbol{R}_i(\boldsymbol{\alpha}),$$

where $\lambda \in [0, 1]$. The corresponding variance $\hat{\boldsymbol{V}}_W$ can be expressed using shrinkage estimators:

$$\hat{\boldsymbol{V}}_W = \lambda\hat{\boldsymbol{V}}_P + (1-\lambda)\hat{\boldsymbol{V}}_N. \tag{2.9}$$

A special case was considered when $\boldsymbol{R}_i(\boldsymbol{\alpha}) = \boldsymbol{I}$ because Warton [2011] claimed that the computation time would be affected in a high-dimensional setting. A cross-validation method was used to find the value of λ:

$$\hat{\lambda} = \arg\max_\lambda \sum_{g=1}^{G} \frac{-K_g}{2}\log|\hat{\boldsymbol{R}}_W^{\backslash g}| - \frac{1}{2}\text{tr}[\boldsymbol{S}_g(\hat{\boldsymbol{R}}_W^{\backslash g})^{-1}\boldsymbol{S}_g^T],$$

where $\hat{\boldsymbol{R}}_W^{\backslash g}$ means that the correlation matrix is estimated from all groups except the gth. The matrix of Pearson residuals has been split into roughly G equal parts $[\boldsymbol{S}_1^T, \cdots, \boldsymbol{S}_G^T]$, and $\boldsymbol{S}_g$ is a $K_g \times n$ matrix. It is noteworthy that there were n different variables in the setting of Warton [2011], which is different from the n measurements of a variable in typical longitudinal data. Xie and Yang [2003] established the asymptotic normality of the parameter estimators when neither the number of clusters (K) nor the cluster size (n) or both extended to infinity in the longitudinal data.

Unlike the previous modified estimators based on sandwich estimator, a smooth bootstrap method perturbs the GEE function to obtain the variance of parameters in longitudinal data [Li and Wang, 2008]. The following perturbed GEE version of $U(\boldsymbol{\beta})$ was proposed to obtain the

smooth bootstrap estimates:

$$U^B(\boldsymbol{\beta}) \quad = \quad \sum_{i=1}^{K} w_i \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1}(\boldsymbol{Y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta})), \tag{2.10}$$

where $w_1, ..., w_K$ are independent realisations from a distribution with an arbitrary mean, but with unit variance. Following Li and Wang [2008], we take $\mathrm{E}(w_k) = 1$. The w_i from $\mathrm{N}(1, 1)$, or $\mathrm{Pois}(1)$, can be used to obtain the smooth bootstrap estimators. The iterative, weighted least-squares method and the quasi-Fisher scoring algorithm were used to solve these equations and obtain estimators.

2.1.2 Linear mixed-effects models (LMMs)

Linear mixed models are an extension of simple linear models that allow both fixed and random effects, and that are particularly useful when there is dependence in the data:

$$y_{ij} = \boldsymbol{x}_{ij}^T \boldsymbol{\beta} + \boldsymbol{z}_{ij}^T \boldsymbol{b}_i + \epsilon_{ij}, i = 1, ..., K, j = 1, ..., n_i. \tag{2.11}$$

Here, y_{ij} is the response variable; $\boldsymbol{x}_{ij}$ is the vector of p fixed effects; $\boldsymbol{\beta}$ is the p parameter vector for the fixed effects; $\boldsymbol{z}_{ij}$ is the vectors of q random effects; $\boldsymbol{b}_i$ is the q parameter vector for random effects, which are assumed to be Gaussian distributions with a mean of zero and variance $\boldsymbol{\Psi}$; and ϵ_{ij} is the random error with $N(0, \sigma^2)$. Let response vector $\boldsymbol{Y_i}$ be the $n_i \times 1$ vector $(y_{i1}, \ldots, y_{in_i})^T$, in which y_{ij} is the jth measurement for the ith subject ($i = 1, \ldots, K$; $j = 1, \ldots, n_i$). $\boldsymbol{X_i}$ (of dimension $n_i \times p$) and $\boldsymbol{Z_i}$ (of dimension $n_i \times q$) are the separate fixed-effect and random-effect covariates. The formulation of the LMM is as follows:

$$\boldsymbol{Y_i} = \boldsymbol{X_i}\boldsymbol{\beta} + \boldsymbol{Z_i}\boldsymbol{b_i} + \boldsymbol{\epsilon_i}, \tag{2.12}$$
$$\boldsymbol{b_i} \sim N(0, \boldsymbol{\Psi}), \boldsymbol{\epsilon_i} \sim N(0, \sigma^2 \boldsymbol{\Lambda_i}).$$

The within-group errors $\boldsymbol{\epsilon_i}$ and the random effects $\boldsymbol{b_i}$ are assumed to be independent. It is a special case if $\boldsymbol{\Lambda_i} = \boldsymbol{I}$; then, it follows that $\boldsymbol{Y_i} \sim N(\boldsymbol{X_i}\boldsymbol{\beta}, \boldsymbol{\Sigma_i})$, where $\boldsymbol{\Sigma_i} = \sigma^2(\boldsymbol{\Lambda_i} + \boldsymbol{Z_i}\boldsymbol{\Psi}\boldsymbol{Z_i^T})$. The matrix form for the model is as follows:

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{Z}\boldsymbol{b} + \boldsymbol{\epsilon},$$

where $\boldsymbol{Y} = \begin{bmatrix} \boldsymbol{Y_1} \\ \boldsymbol{Y_2} \\ \vdots \\ \boldsymbol{Y_K} \end{bmatrix}$, $\boldsymbol{X} = \begin{bmatrix} \boldsymbol{X_1} \\ \boldsymbol{X_2} \\ \vdots \\ \boldsymbol{X_K} \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} \boldsymbol{b_1} \\ \boldsymbol{b_2} \\ \vdots \\ \boldsymbol{b_K} \end{bmatrix}$, $\boldsymbol{Z} = \text{diag}(\boldsymbol{Z_1}, \boldsymbol{Z_2}, ..., \boldsymbol{Z_K})$, $\Lambda = \text{diag}(\boldsymbol{\Lambda_1}, \boldsymbol{\Lambda_2}, ..., \boldsymbol{\Lambda_K})$, $\Sigma = \text{diag}(\boldsymbol{\Sigma_1}, \boldsymbol{\Sigma_2}, ..., \boldsymbol{\Sigma_K})$ and $\tilde{\boldsymbol{\Psi}} = \text{diag}(\boldsymbol{\Psi}, \boldsymbol{\Psi}, ..., \boldsymbol{\Psi})$. It follows that $\boldsymbol{Y}$ are independent, multivariate normal vectors with a mean of $\boldsymbol{X\beta}$ and the covariance matrix is $\boldsymbol{\Sigma} = \sigma^2(\boldsymbol{\Lambda} + \boldsymbol{Z\tilde{\Psi}Z^T})$. Then, the likelihood function is as follows:

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2|\boldsymbol{Y}) = p(\boldsymbol{Y}|\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2) = \prod_{i=1}^{K} p(\boldsymbol{Y_i}|\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2),$$

where

$$p(\boldsymbol{Y_i}|\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2) = (2\pi\sigma^2)^{-\frac{n_i}{2}} \exp\left(\frac{(\boldsymbol{Y_i} - \boldsymbol{X_i\beta})^{\boldsymbol{T}}\boldsymbol{\Sigma_i^{-1}}(\boldsymbol{Y_i} - \boldsymbol{X_i\beta})}{-2\sigma^2}\right)|\boldsymbol{\Sigma_i}|^{-\frac{1}{2}}.$$

In addition, $\boldsymbol{\theta}$ represents the parameters in $\tilde{\boldsymbol{\Psi}}$ and $\boldsymbol{\Lambda}$. An expectation-maximisation (EM) algorithm can be used to obtain both the maximum likelihood (ML) and restricted maximum likelihood (REML) estimation according to Laird et al. [1987]. The major process for the EM algorithm is outlined below in Table 2.1.

2.2 Machine learning methods

Machine learning methods focus on the development of computer programs that can access data and use those data to learn for themselves. Machine learning algorithms are often categorised as entailing supervised, semi-supervised, unsupervised and reinforcement learning. Supervised machine learning algorithms can apply what has been learned in the past to new data using labelled examples to predict future events. For example, random forests (RF), the support-vector machine (SVM) and the neural network (NN) are all supervised learning methods. Supervised learning problems can be further grouped into regression and classification problems. The regression algorithms attempt to estimate the mapping function from the input variables to the numerical or continuous output variables. Conversely, classification algorithms attempt

Table 2.1: EM algorithms for the maximum-likelihood (ML) estimates in LME

Step 1. Set the initial values for $\hat{\sigma}^2_{(r)} = 1$ and $\hat{\boldsymbol{\Psi}}_{(r)} = \boldsymbol{I}_q$, $r = 0$. Here $\boldsymbol{\Lambda}_i = \boldsymbol{I}$ is assumed.

Step 2. Obtain the estimates $\hat{\boldsymbol{\beta}}_{(r+1)}$ and $\hat{\boldsymbol{b}}_{i(r+1)}$

$$\hat{\boldsymbol{\beta}}_{(r+1)} = \left(\sum_{i=1}^{K} \boldsymbol{X}_i^T \hat{\boldsymbol{\Sigma}}_{i(r)}^{-1} \boldsymbol{X}_i\right)^{-1} \left(\sum_{i=1}^{K} \boldsymbol{X}_i^T \hat{\boldsymbol{\Sigma}}_{i(r)}^{-1} \boldsymbol{Y}_i\right),$$

$$\hat{\boldsymbol{b}}_{i(r+1)} = \hat{\boldsymbol{\Psi}}_{(r)} \boldsymbol{Z}_i^T \hat{\boldsymbol{\Sigma}}_{i(r)}^{-1} \left(\boldsymbol{Y}_i - \boldsymbol{X}_i \hat{\boldsymbol{\beta}}_{(r+1)}\right),$$

where $\hat{\boldsymbol{\Sigma}}_{i(r)} = \boldsymbol{Z}_i \hat{\boldsymbol{\Psi}}_{(r)} \boldsymbol{Z}_i^T + \hat{\sigma}^2_{(r)} \boldsymbol{I}_{n_i}$.

Step 3. Update $\hat{\sigma}^2_{(r+1)}$ and $\hat{\boldsymbol{\Psi}}_{(r+1)}$ using

$$\hat{\sigma}^2_{(r+1)} = \left(\sum_{i=1}^{K} n_i\right)^{-1} \sum_{i=1}^{K} \left[\hat{\boldsymbol{\epsilon}}_{i(r+1)}^T \hat{\boldsymbol{\epsilon}}_{i(r+1)} + \hat{\sigma}^2_{(r)} \left(n_i - \hat{\sigma}^2_{(r)} \text{trace}(\hat{\boldsymbol{\Sigma}}_{i(r)})\right)\right],$$

$$\hat{\boldsymbol{\Psi}}_{(r+1)} = K^{-1} \sum_{i=1}^{K} \left[\hat{\boldsymbol{b}}_{i(r+1)} \hat{\boldsymbol{b}}_{i(r+1)}^T + \left(\hat{\boldsymbol{\Psi}}_{(r)} - \hat{\boldsymbol{\Psi}}_{(r)} \boldsymbol{Z}_i^T \hat{\boldsymbol{\Sigma}}_{i(r)}^{-1} \boldsymbol{Z}_i \hat{\boldsymbol{\Psi}}_{(r)}\right)\right],$$

where $\hat{\boldsymbol{\epsilon}}_{i(r+1)} = \boldsymbol{Y}_i - \boldsymbol{X}_i \hat{\boldsymbol{\beta}}_{(r+1)} - \boldsymbol{Z}_i \hat{\boldsymbol{b}}_{i(r+1)}$.

Step 4. Repeat Steps 2 and 3 until the algorithm converges.

to estimate the mapping function from the input variables to the discrete or categorical output variables.

2.2.1 Linear discriminant analysis

Linear discriminant analysis (LDA) is a widely used statistical method for the analysis of data with categorical response variables [Fisher, 1936, Rao et al., 1973]. LDA entails the assumption that the explanatory variables follow Gaussian distribution with equal covariance matrices. Suppose the class-conditional density of Z in class $G = g$, as follows:

$$f_g(z) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp(-\frac{1}{2}(z - \mu_g)^T \boldsymbol{\Sigma}^{-1} (z - \mu_g)). \tag{2.13}$$

The Bayes classifier assigns an observation to the class for which the following equation is the largest:

$$\delta_g(z) = z^T \boldsymbol{\Sigma}^{-1} \mu_g - \frac{1}{2} \mu_g^T \boldsymbol{\Sigma}^{-1} \mu_g + \log \pi_g. \tag{2.14}$$

Here, π_g signifies the prior probability of class g. If the covariance matrix is not equal, then a quadratic discriminant analysis is needed.

2.2.2 Tree-based methods

The decision tree, bagging, the RF and boosting methods can be grouped together, as they are all tree-based approaches. The classification and regression tree (CART) is a popular algorithm, which was proposed by Breiman et al. [1984]. In the tree method, the training data are used to construct a tree, starting at the root node. The predicted space is divided into non-overlapping M regions ($R_1, R_2, ..., R_M$), determined by recursive splitting, which is a top-down and greedy approach [James et al., 2013]. Regression trees and classification trees are similar but have different ways of interpreting. The splitting we selected will cause the largest reduction in the mean square error (RSS) for the regression trees:

$$RSS = \sum_{m=1}^{M} \sum_{i \in R_m} (y_i - \bar{y}_{R_m})^2.$$

Here, $\bar{y}_{R_m}$ is the mean response value of all training observations in the R_m region. In each region, a constant $c_m, (m = 1, ..., M)$ would be the response. We can split recursively, until the mean square error reaches a threshold. The predicted response for each test data point is the mean of the training observations in the region to which that test point belongs. For each test data point that falls in the same region on a path starting from the root node and reaching a terminal (leaf) node, the prediction would be the same. The usual strategy for a single tree is to grow a large tree and then trim it by weakest-link pruning.

Otherwise, in classification trees, the classification error rate is a natural criterion for creating binary splits. The classification error rate is defined by:

$$E = 1 - \max_k(\hat{p}_{mk}),$$

where $\hat{p}_{mk}$ is the frequency of training observations in the mth region and from the kth class. The Gini index and entropy are the commonly used criteria because the classification error rate is not sufficiently sensitive for tree-growing. The Gini index measures the variance across the total classes:

$$G = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk}).$$

The entropy is calculated by:

$$D = -\sum_{k=1}^{K} \hat{p}_{mk} \log \hat{p}_{mk}.$$

Both the Gini index and the entropy can be used to evaluate the quality of a particular split. The predicted response for a test data point is the most commonly occurring class of the training observations in the region to which that test point belongs.

Bagging is the application of the bootstrap procedure to decision trees to lower the variance. There are three main steps: first, generate random sub-samples of the training data set with replacement; second, train the decision tree (regression tree or classification tree) method for each sample; and third, calculate the average prediction from each model using the test data. The average prediction should be the final prediction for each test data point. For the classification trees, the average prediction from each model indicates the most commonly occurring class among all the trees. Bagging, in contrast to the tree method, will improve the prediction accuracy at the cost of interpretability.

RF is a popular tree-based ensemble method that builds a large collection of de-correlated trees and then averages them based on the bagging [Breiman, 2001]. When building this algorithm, a random sample of features is chosen as split candidates from the full set of predictors rather than using all the features in the bagging. This forces each split to consider only a subset of the predictors, which is reasonable, especially when there is an extremely strong predictor in the training data set. After a certain number of trees have been grown, the predictor is obtained by the average (for regression) or the majority vote (for classification) [James et al., 2013]. This algorithm contains four main parameters: the total number of observations, the total number of predictor variables, randomly chosen features for determining the decision tree and the total number of decision trees.

Based on the bootstrapped data, the different trees are independent in the bagging. Boosting works in a similar way to bagging, but the difference is that the trees are constructed sequentially, which means that the growth of each tree depends on the trees that have already been constructed. Gradient tree boosting is a generalisation of boosting to arbitrarily differentiable loss functions [Friedman, 2002]. At each stage, the decision tree is chosen to minimise a loss function given the current model. Boosting trees have three parameters: the number of trees, the shrinkage parameter that controls the learning rate and the number of splits in each tree that determines the complexity of the boosted ensemble. The predictions are combined through

a weighted majority vote (classification) or a weighted sum (regression) to produce the final prediction.

2.2.3 Support-vector machine

The initial idea of a SVM is to construct a linear partition of the high-dimensional space into two sub-spaces for classification or regression [Vapnik, 1998]. The support-vector classifier (SVC) and support-vector regression (SVR) are described separately. Let $\boldsymbol{x_k} \in R^p$, for $k = 1, 2, ..., N$ and group indicators $y_k \in \{-1, 1\}$. If the data are linearly separable, a separating hyperplane $f(\boldsymbol{x_k}) = \boldsymbol{\omega}^T \boldsymbol{x_k} + b_0$ can be constructed then $\hat{y}_k = \text{sign}(f(\boldsymbol{x_k}))$, where $\boldsymbol{\omega}$ is weight vector and b_0 is the bias. The objection function is

$$\begin{aligned} &\min_{\boldsymbol{\omega}, b_0} \tfrac{1}{2}||\boldsymbol{\omega}||^2 \\ &s.t.\ y_k(\boldsymbol{\omega}^T x_k + b_0) \geq 1. \end{aligned} \tag{2.15}$$

Sometimes the data cannot be separated linearly; thus, a 'soft margin' must be introduced. The objection function is as follows:

$$\min_{\boldsymbol{\omega}, b_0} \frac{1}{2}||\boldsymbol{\omega}||^2 + C \sum_{k=1}^{N} \ell_{0/1}(y_k(\boldsymbol{\omega}^T x_k + b_0) - 1). \tag{2.16}$$

Here, $C > 0$ is a constant regularisation term in the Lagrange formulation and

$$\ell_{0/1} = \begin{cases} 1, & \text{if } z < 0 \\ 0, & \text{otherwise.} \end{cases}$$

While the $\ell_{0/1}$ is a non-convex and non-continuous function, there are some surrogate loss functions:

- hinge loss: $\ell_{hinge}(z) = \max(0, 1 - z)$
- exponential loss: $\ell_{exp}(z) = \exp(-z)$
- logistic loss: $\ell_{log}(z) = \log(1 + \exp(-z))$.

The hinge loss function is a popular alternative; then, the objection function becomes:

$$\begin{aligned} &\min_{\boldsymbol{\omega},b_0,\xi_k} \tfrac{1}{2}||\boldsymbol{\omega}||^2 + C\textstyle\sum_{k=1}^{N} \xi_k \\ &s.t.\ y_k(\boldsymbol{\omega}^T \boldsymbol{x_k} + b_0) \geq 1 - \xi_k \\ &\xi_k \geq 0,\ k = 1, ..., N. \end{aligned} \tag{2.17}$$

If we use the Lagrange algorithm, the Lagrange function is:

$$L(\boldsymbol{\omega}, b_0, \iota, \xi, \eta) = \frac{1}{2}||\boldsymbol{\omega}||^2 + C\sum_{k=1}^{N} \xi_k + \sum_{k=1}^{N} \iota_k(1 - \xi_k - y_k(\boldsymbol{\omega}_k^T \boldsymbol{x_k} + b_0)) - \sum_{k=1}^{N} \eta_k \xi_k. \tag{2.18}$$

Setting the respective derivatives to zero, we obtain:

$$\begin{aligned} &\boldsymbol{\omega} = \textstyle\sum_{k=1}^{N} \iota_k y_k \boldsymbol{x_k}, \\ &0 = \textstyle\sum_{k=1}^{N} \iota_k y_k, \\ &\iota_k = C - \eta_k. \end{aligned} \tag{2.19}$$

Then the dual objection function is as follows:

$$\begin{aligned} &\min_{\iota_k} \tfrac{1}{2} \textstyle\sum_{k=1}^{N} \sum_{k'=1}^{N} \iota_k \iota_{k'} y_k y_{k'} \boldsymbol{x_k}^T \boldsymbol{x_{k'}} - \sum_{k=1}^{N} \iota_k \\ &s.t.\ \textstyle\sum_{k=1}^{N} \iota_k y_k = 0 \\ &0 \leq \iota_k \leq C,\ k = 1, ..., N. \end{aligned} \tag{2.20}$$

In addition to the conditions in (2.19), the Karush-Kuhn-Tucker conditions include the following constraints:

$$\begin{aligned} &\iota_k[y_k(\boldsymbol{x_k}^T \boldsymbol{\omega} + b_0) - (1 - \xi_k)] = 0, \\ &\eta_k \xi_k = 0, \\ &y_k(\boldsymbol{x_k}^T \boldsymbol{\omega} + b_0) - (1 - \xi_k) \geq 0. \end{aligned} \tag{2.21}$$

The above equations uniquely characterise the solution to the primal and dual problem. From (2.19) we see that the solution for $\boldsymbol{\omega}$ has the form:

$$\hat{\boldsymbol{\omega}} = \sum_{k=1}^{N} \hat{\iota}_k y_k \boldsymbol{x_k}. \tag{2.22}$$

This includes nonzero coefficients $\hat{\iota}_k$ which adhere to (2.19). These nonzero $\hat{\iota}_k$ are called the support vectors, as $\hat{\boldsymbol{\omega}}$ is represented in terms of them alone. It often happens that the data are not linearly separable. A kernel function is then used to transform the nonlinear system in the input space to a linear system in the feature space. The dual function (2.20) has the form:

$$\begin{aligned} &\min_{\iota_k} \tfrac{1}{2} \textstyle\sum_{k=1}^{N} \sum_{k'=1}^{N} \iota_k \iota_{k'} y_k y_{k'} \langle h(\boldsymbol{x_k}), h(\boldsymbol{x_{k'}}) \rangle - \sum_{k=1}^{N} \iota_k \\ &s.t.\ \textstyle\sum_{k=1}^{N} \iota_k y_k = 0 \\ &0 \leq \iota_k \leq C,\ k = 1, ..., N. \end{aligned} \tag{2.23}$$

Here, $\kappa(x, x^{'}) = \langle h(x), h(x^{'}) \rangle$ is the kernel function. The popular kernel functions are as follows:

- dth-degree polynomial: $\kappa(x, x^{'}) = (1 + \langle x, x^{'} \rangle)^d$
- Radial basis: $\kappa(x, x^{'}) = \exp(-\gamma ||x - x^{'}||^2)$
- Sigmoid: $\kappa(x, x^{'}) = \tanh(\gamma_1 \langle x, x^{'} \rangle + \gamma_2$.

Then, the solution can be written thus:

$$\hat{f}(\boldsymbol{x}) = \sum_{k=1}^{N} \hat{\iota}_k y_k \kappa(\boldsymbol{x}, \boldsymbol{x}_k) + \hat{b}_0.$$

The SVMs can also be adapted for regression and we can discuss the linear regression model:

$$f(\boldsymbol{x}) = \boldsymbol{\omega}^T \boldsymbol{x} + b_0. \tag{2.24}$$

To estimate $\boldsymbol{\omega}$, we consider minimisation of:

$$\frac{1}{2} ||\boldsymbol{\omega}||^2 + C \sum_{k=1}^{N} \ell_\epsilon (y_k - f(\boldsymbol{x_k})). \tag{2.25}$$

Here:

$$\ell_\epsilon(z) = \begin{cases} 0, & \text{if } |z| < \epsilon \\ |z| - \epsilon, & \text{otherwise} \end{cases}$$

is an 'ϵ-insensitive' error measure. Then, the objective function can be written as:

$$\begin{aligned} &\min_{\boldsymbol{\omega}, b_0, \xi_k} \tfrac{1}{2}||\boldsymbol{\omega}||^2 + C\textstyle\sum_{k=1}^N (\xi_k + \hat{\xi}_k) \\ &s.t.\ f(\boldsymbol{x}_k) - y_k \leq \epsilon + \xi_k \\ &y_k - f(\boldsymbol{x}_k) \geq \epsilon + \hat{\xi}_k \\ &\xi_k \geq 0, \hat{\xi}_k \geq 0\ k = 1, ..., N. \end{aligned} \tag{2.26}$$

Similarly, the correspondingly SVR quadratic programming problems are as follows:

$$\min_{\iota_k^*, \iota_k} \epsilon \sum_{k=1}^N (\iota_k^* + \iota_k) - \sum_{k=1}^N y_k(\iota_k^* - \iota_k) + \frac{1}{2} \sum_{k,k'=1}^N (\iota_k^* - \iota_k)(\iota_{k'}^* - \iota_{k'}) \boldsymbol{x}_{\boldsymbol{k}}^T \boldsymbol{x}_{\boldsymbol{k}'}, \tag{2.27}$$

which meet the constraints

$$\begin{aligned} &0 \leq \iota_k, \iota_k^* \leq C, \\ &\textstyle\sum_{k=1}^N (\iota_k^* - \iota_k) = 0, \\ &\iota_k \iota_k^* = 0. \end{aligned} \tag{2.28}$$

If $\hat{\boldsymbol{\omega}}$, $\hat{b}_0$ are the estimates, the solutions are:

$$\hat{\boldsymbol{\omega}} = \sum_{k=1}^N (\hat{\iota}_k^* - \hat{\iota}_k) \boldsymbol{x}_{\boldsymbol{k}}, \tag{2.29}$$

$$\hat{f}(\boldsymbol{x}) = \sum_{k=1}^N (\hat{\iota}_k^* - \hat{\iota}_k) \boldsymbol{x}^T \boldsymbol{x}_{\boldsymbol{k}} + \hat{b}_0, \tag{2.30}$$

where $\hat{\iota}_k^*$, $\hat{\iota}_k$ are the nonzero values.

A similar kernel function can be used in the SVR to handle nonlinear boundaries. Then, the

corresponding solution is:

$$\hat{\boldsymbol{\omega}} = \sum_{k=1}^{N} (\hat{\iota}_k^* - \hat{\iota}_k) h(\boldsymbol{x_k}). \tag{2.31}$$

In addition:

$$\hat{f}(\boldsymbol{x}) = \sum_{k=1}^{N} (\hat{\iota}_k^* - \hat{\iota}_k) \kappa(\boldsymbol{x}, \boldsymbol{x}_k) + \hat{b}_0. \tag{2.32}$$

Here, the definitions of functions h and κ are the same as in (2.23).

Suykens and Vandewalle [1999] proposed the least-squares version of the support-vector machine, which called the least-squares support-vector machine (LS-SVM). The proposed optimisation problem is defined below:

$$\begin{aligned} &\min_{\boldsymbol{\omega}, b_0, e_k} \tfrac{1}{2}||\boldsymbol{\omega}||^2 + \tfrac{C}{2} \textstyle\sum_{k=1}^{N} e_k^2 \\ &s.t.\ y_k(\boldsymbol{\omega}^T h(\boldsymbol{x_k}) + b_0) = 1 - e_k, \\ &k = 1, ..., N. \end{aligned} \tag{2.33}$$

Here, $e_k = y_k - (\boldsymbol{\omega}^T h(\boldsymbol{x_k}) + b_0)$. The Lagrangian function is:

$$L(\boldsymbol{\omega}, b_0, \iota, e, \eta) = \frac{1}{2}||\boldsymbol{\omega}||^2 + \frac{C}{2} \sum_{k=1}^{N} e_k^2 - \sum_{k=1}^{N} \iota_k((\boldsymbol{\omega}^T h(\boldsymbol{x_k}) + b_0) + e_k - y_k). \tag{2.34}$$

Setting the respective derivatives to zero, we obtain:

$$\begin{aligned} &\boldsymbol{\omega} = \textstyle\sum_{k=1}^{N} \iota_k h(\boldsymbol{x_k}), \\ &0 = \textstyle\sum_{k=1}^{N} \iota_k, \\ &\iota_k = C e_k, \\ &y_k = \boldsymbol{\omega}^T h(\boldsymbol{x_k}) + b_0 + e_k, k = 1, ..., N. \end{aligned} \tag{2.35}$$

Then, the problem involves the solving of a set of linear equations instead of remaining a quadratic programming problem. Various kernel functions can also be used in LS-SVM to ensure that it is well-suited to deal with high-dimensional covariates.

2.2.4 Neural networks

A multilayer perceptron (MLP) is a class of feedforward artificial neural networks for classification and regression. There are at least three layers of nodes for an MLP: an input layer, a hidden layer and an output layer. Each node is a neuron that uses a nonlinear activation function, except for the input nodes. The differences between the MLP and linear perceptron is in the multiple layers of the MLP and its nonlinear activation function.

The formula is as follows:

$$\hat{y} = \sum_{k=0}^{NH} v_k g_k(\boldsymbol{x}).$$

Here, v_k are the weights, $g_k(\boldsymbol{x})$ are the hidden functions (or hidden units), NH is the number of hidden nodes, $\boldsymbol{x}$ is the input vector and $\hat{y}$ is the output. The hidden function g_0 takes a fixed value of one to allow a constant term in the equation. The sigmoid function—$g_k(u) = 1/(1 + \exp(-u))$—is commonly used. The general approach of this method is minimise the sum of squared errors by gradient descent, which is known as backpropagation (BP). BP is a popular algorithm for training feedforward neural networks, which computes the gradient of the loss function with respect to the weights of the network for a single input-output example [Rumelhart et al., 1986].

2.2.5 Mixed-effects machine learning

Segal [1992] was the first to apply regression trees to longitudinal data. Hajjem et al. [2011] and Sela and Simonoff [2012] independently proposed the mixed-effects regression trees (MERT) for clustered data or longitudinal data. The notation in an MERT model follows the LMM:

$$\boldsymbol{Y_i} = f(\boldsymbol{X_i}) + \boldsymbol{Z_i b_i} + \boldsymbol{\epsilon_i}.$$

Here, the $\boldsymbol{Y_i}, \boldsymbol{X_i}, \boldsymbol{Z_i}, \boldsymbol{b_i}$ and $\boldsymbol{\epsilon_i}$ are the same as in the LME. The difference is that a standard tree structure is used to estimate the fixed part of the model.

As we can see from Table 2.2, a standard tree algorithm is needed to estimate the fixed component. Similarly, Hajjem et al. [2014] proposed the mixed-effects random forest (MERF) method, which uses the RF method to estimate fixed effects. If the response variable follows a

Table 2.2: EM algorithms for the maximum likelihood estimates in MERT

Step 1. Set the initial values for $\hat{\boldsymbol{b}}_{i(r)} = 0$, $\hat{\sigma}^2_{(r)} = 1$ and $\hat{\boldsymbol{\Psi}}_{(r)} = \boldsymbol{I}_q$, $r = 0$. Here $\boldsymbol{\Lambda}_i = \boldsymbol{I}$ is assumed.
Step 2. Update $\boldsymbol{Y}^*_{i(r+1)}$, $\hat{f}(\boldsymbol{X}_i)_{(r+1)}$ and $\hat{b}_{i(r+1)}$
(1) $\boldsymbol{Y}^*_{i(r+1)} = \boldsymbol{Y}_i - \boldsymbol{Z}_i\hat{\boldsymbol{b}}_{i(r)}$,
(2) $\hat{f}(\boldsymbol{X}_i)_{(r+1)}$ is an estimate obtained from a standard tree algorithm with $\boldsymbol{Y}^*_{i(r+1)}$ as response and $\boldsymbol{X}_i$ as covariates,
(3) $\hat{\boldsymbol{b}}_{i(r+1)} = \hat{\boldsymbol{\Psi}}_{(r)}\boldsymbol{Z}_i^T\hat{\boldsymbol{\Sigma}}^{-1}_{i(r)}\left(\boldsymbol{Y}_i - \hat{f}(\boldsymbol{X}_i)_{(r+1)}\right)$,
where $\hat{\boldsymbol{\Sigma}}_{i(r)} = \boldsymbol{Z}_i\hat{\boldsymbol{\Psi}}_{(r)}\boldsymbol{Z}_i^T + \hat{\sigma}^2_{(r)}\boldsymbol{I}_{n_i}$.
Step 3. Update $\hat{\sigma}^2_{(r+1)}$ and $\hat{\boldsymbol{\Psi}}_{(r+1)}$ using
$\hat{\sigma}^2_{(r+1)} = \left(\sum_{i=1}^K n_i\right)^{-1}\sum_{i=1}^K\left[\hat{\boldsymbol{\epsilon}}^T_{i(r+1)}\hat{\boldsymbol{\epsilon}}_{i(r+1)} + \hat{\sigma}^2_{(r)}\left(n_i - \hat{\sigma}^2_{(r)}\text{trace}(\hat{\boldsymbol{\Sigma}}_{i(r)})\right)\right]$,
$\hat{\boldsymbol{\Psi}}_{(r+1)} = K^{-1}\sum_{i=1}^K\left[\hat{\boldsymbol{b}}_{i(r+1)}\hat{\boldsymbol{b}}^T_{i(r+1)} + \left(\hat{\boldsymbol{\Psi}}_{(r)} - \hat{\boldsymbol{\Psi}}_{(r)}\boldsymbol{Z}_i^T\hat{\boldsymbol{\Sigma}}^{-1}_{i(r)}\boldsymbol{Z}_i\hat{\boldsymbol{\Psi}}_{(r)}\right)\right]$,
where $\hat{\boldsymbol{\epsilon}}_{i(r+1)} = \boldsymbol{Y}_i - \hat{f}(\boldsymbol{X}_i)_{(r+1)} - \boldsymbol{Z}_i\hat{\boldsymbol{b}}_{i(r+1)}$.
Step 4. Repeat Steps 2 and 3 until the algorithm converges.

non-Gaussian distribution, generalised mixed-effects regression tree is proposed [Hajjem et al., 2014].

For SVM, a mixed-effects least-squares SVR was proposed by Cho [2010] and a mixed-effects least-squares SVC was introduced by Luts et al. [2012]. The mixed effects components were added into the LS-SVR and the optimisation problem was defined as follows:

$$\begin{aligned}
&\min_{\boldsymbol{\omega},b_0,\boldsymbol{b}_i} \tfrac{1}{2}||\boldsymbol{\omega}||^2 + \tfrac{\lambda_1}{2}\textstyle\sum_{i=1}^K \boldsymbol{b}_i^T\boldsymbol{\Psi}^{-1}\boldsymbol{b}_i + \tfrac{\lambda_2}{2}\textstyle\sum_{i=1}^K\sum_{j,k=1}^{n_i}\epsilon_{ij}\boldsymbol{\Lambda}^{-1}_{ijk}\epsilon_{ik} \\
&s.t.\ y_{ij} = b_0 + \boldsymbol{\omega}^T h(\boldsymbol{x}_{ij}) + \boldsymbol{b}_i^T\boldsymbol{z}_{ij} + \epsilon_{ij} \\
&i = 1, ..., K, j = 1, ..., n_i.
\end{aligned} \tag{2.36}$$

Here, y_{ij} is the jth response variable of the ith subject, $\boldsymbol{x}_{ij}$ connotes the corresponding fixed-effects covariates, $\boldsymbol{b}_i$ is a random-effects parameter vector from $N(\boldsymbol{0}, \boldsymbol{\Psi})$, $\boldsymbol{z}_{ij}$ refers to the random-effects covariates and ϵ_{ij} follows normal distribution $N(\boldsymbol{0}, \boldsymbol{\Lambda}_i)$, which has the same definition as LMM. In addition, the $h(\cdot)$ function is a transformation function and b_0 is the bias. The Lagrangian function is:

$$L(\boldsymbol{\omega}, b_0, \iota_{ij}, \boldsymbol{b}_i, \epsilon_{ij}) = \frac{1}{2}||\boldsymbol{\omega}||^2 + \frac{\lambda_1}{2}\sum_{i=1}^{K} \boldsymbol{b}_i^T \boldsymbol{\Psi}^{-1} \boldsymbol{b}_i + \frac{\lambda_2}{2}\sum_{i=1}^{K}\sum_{j,k=1}^{n_i} \epsilon_{ij} \boldsymbol{\Lambda}_{ijk}^{-1} \epsilon_{ik} +$$
$$+ \sum_{i=1}^{K}\sum_{j=1}^{n_i} \iota_{ij}(y_{ij} - b_0 - \boldsymbol{\omega}^T h(\boldsymbol{x}_{ij}) - \boldsymbol{b}_i^T \boldsymbol{z}_{ij} - \epsilon_{ij}). \tag{2.37}$$

Here, the ι_{ij}'s are Lagrange multipliers. The following computation is the same as in LS-SVR and the estimates $\hat{b}_0$ and $\hat{\iota}_{ij}$ can be obtained. For new given data points $(\boldsymbol{x}_o, \boldsymbol{z}_o)$, the prediction is as follows:

$$\hat{y}(\boldsymbol{x}_o, \boldsymbol{z}_o) = \hat{b}_0 + \sum_{i=1}^{K}\sum_{j=1}^{n_i} \hat{\iota}_{ij} \kappa(\boldsymbol{x}_{ij}, \boldsymbol{x}_o) + \hat{\boldsymbol{b}}_i \boldsymbol{z}_o. \tag{2.38}$$

Here, $\kappa(\cdot)$ is the kernel function and $\hat{\boldsymbol{b}}_i = \frac{1}{\lambda_1}\boldsymbol{\Psi}\boldsymbol{Z}_i^T \hat{\boldsymbol{\iota}}_i$. More details about the mixed-effects least-squares SVC can be found in Luts et al. [2012].

Chapter 3

Performance of Variance Estimators in Analysis of Longitudinal Data with a Large Cluster Size

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Abstract

Generalised estimating equations (GEE) is a widely used method for analysing longitudinal data, and the sandwich method is often used to estimate the variance-covariance matrix of the regression coefficient estimators because of the advantage that it is robust to the misspecification of the correlation structure of the responses. However, the sandwich method relies on the residual products as an estimator for the true covariance of the responses, and the estimator becomes singular and sparse when the cluster size (n) becomes larger. We carry out a large set of simulation studies to address the question, when the cluster size becomes large, whether the sandwich estimator and their modified versions are still valid when a sparse or singular matrix is used as an estimate of the true covariance of the responses. The performance of smooth bootstrap which involves perturbing the estimating functions is also worth to be investigated when the cluster size is large. Two datasets are analysed to illustrate these findings.

Keywords: Covariance estimate; Large cluster size; Generalised estimating equations; Smooth bootstrap; Regularised sandwich estimate.

3.1 Introduction

Longitudinal data appear frequently in economics, finance, medical science, and other fields, where it is often assumed that observations are correlated on the same subjects and independent among different subjects. Various methods have been developed to analyze longitudinal data. A popular one is generalised estimating equations (GEE) [Liang and Zeger, 1986], which has the advantage of using consistent mean estimators that can be obtained even if the working correlation structure is misspecified. The classic sandwich estimator can be used to get the estimates of the covariance matrix. Since the fitted mean values tend to close to the observed values than the true mean values when the sample size (K) is small, the product of raw residuals is biased downward for estimating the true covariance matrix [Paik, 1988]. Some modified sandwich estimators that can improve the efficiency of the variance estimator have been proposed [Kauermann and Carroll, 2001, Mancl and DeRouen, 2001, Pan, 2001, Wang and Long, 2011]. Wang et al. [2016] reviewed the recent development of modified sandwich estimators in longitudinal data with small samples. However, these modifications are all focused on correcting bias in small sample sizes, and we are interested in their performance under large cluster sizes. It is worth noting that the rank of the residual matrix (of dimension $n \times n$) is one. The residual matrix in the estimating equations became more sparse with the cluster sizes increasing. Xie and Yang [2003] investigated the cases when either the number of clusters (K), the cluster sizes (n), or both go to infinity in longitudinal data. They also established the asymptotic normality of the parameter estimators under these three scenarios. The modified sandwich method proposed by Wang and Long [2011] which combined the advantages of methods from Pan [2001] and Mancl and DeRouen [2001] are being used as a comparison in this work.

Warton [2011] proposed a regularised sandwich estimator when the number of clusters (K) is not large relative to the cluster sizes (n). He mentioned three specific issues: (i) the larger n leads to issues in the computation time and in the convergence of the algorithm because there is a correlation matrix with a larger dimension in the estimating equations; (ii) there is problem about the inference of parameters because the asymptotic distribution of parameters is not clear; (iii) the moment-based estimates of the correlation matrix will be singular for $n > K$, and the sandwich estimators will become numerically problematic. For the first problem and second problem, a simple independent correlation matrix was applied, and the resampling method was

used to obtain the inference of parameters, respectively. The regularised sandwich estimator was proposed to solve the third problem, which is the main contribution in Warton [2011]. Specific real data about the effect of a nutrient treatment on nematode communities were used as the example. There are 12 ponds ($K = 12$) and the abundance of 26 species of the nematode in each pond was recorded ($n = 26$). This setting that there are n different variables differs from the typical longitudinal data that there are n measurements of a variable.

Perturbing the estimating functions to obtain bootstrapped parameter estimates are different from previous modified sandwich methods. Sherman and Cessie [1997] claimed that the correlation structure within a block can be unknown as the correlation structure could be maintained by boosting these blocks. Cheng et al. [2013] proposed a theoretical study of using a cluster bootstrap for inference in GEE and showed that a consistent approximation of the distribution of the regression estimator and the confidence sets can be obtained by the cluster bootstrap methods. El Karoui and Purdom [2018] considered the linear regression case, claiming that the two commonly used methods of bootstrap, residual bootstrap and pairs bootstrap, yield either highly conservative or highly anti-conservative confidence intervals. The suggestion is that rather than using the generic resampling procedures for high-dimensional problems, the problems should be treated differently. According to Li and Wang [2008], the smooth bootstrap estimators were shown to be more efficient in terms of variance estimation and confidence interval coverage in a small cluster size. The variance estimators from smooth bootstrap method relied on the parameter estimates from perturbing the estimating functions without using the $n \times n$ residual matrix. Hence, the performance of the smooth bootstrap method under large cluster sizes is also worth to be investigated.

Motivated by these papers, we are interested in the performance of these modified sandwich estimators when the cluster sizes (n) increases in the longitudinal data with a finite sample size. As noticed, the sandwich estimator involves a singular matrix as an estimator for the true covariance when the cluster size is large because there is a $n \times n$ sparse matrix in the estimating equations. The smooth bootstrap method is expected to suppress modified sandwich estimators and regularised sandwich estimators because it bypasses this dilemma.

In the current work, we investigate the performance of different methods, the sandwich method, the improved sandwich method, the regularised sandwich method, and the smooth bootstrap method, to obtain the covariance of parameter estimators in longitudinal data with

large cluster sizes. The rest of the present work is organized as follows: The sandwich estimators and their derivatives are discussed in section 3.2. In section 3.3, simulation studies are carried out to evaluate the performance of various methods in obtaining the covariance of parameter estimators in longitudinal data with the cluster size increasing. Two real datasets are used for illustration in section 3.4. Some conclusions and discussion are given in section 3.5.

3.2 Covariance estimators

Consider y_{ij} is the jth measurement for the ith subject and $\boldsymbol{x}_{ij}$ is the corresponding covariate vector, where $i = 1, \ldots, K$, $j = 1, \ldots, n_i$. The response variable y_{ij} has a mean μ_{ij} and variance $\phi\sigma_{ij}^2$. Here, $\mu_{ij} = g(x_{ij}, \boldsymbol{\beta})$, where $g(\cdot)$ is a known link function, $\boldsymbol{\beta}$ is a $p \times 1$ vector of the parameters, and its true value is $\widetilde{\boldsymbol{\beta}}$. The scale parameter ϕ is unknown, and σ_{ij}^2 is often expressed as a known function of μ_{ij}. Let $\boldsymbol{Y}_i$ be the $n_i \times 1$ vector $(y_{i1}, \ldots, y_{in_i})^T$ and $\boldsymbol{X}_i$ be the $n_i \times p$ matrix $(\boldsymbol{x}_{i1}, \ldots, \boldsymbol{x}_{in_i})^T$ for the ith subject. We denote $\boldsymbol{\mu}_i = (\mu_{i1}, ..., \mu_{in_i})$ as the marginal mean vector and the covariance matrix of $\boldsymbol{Y}_i$ is $\phi\boldsymbol{A}_i^{1/2}\boldsymbol{R}_0(\boldsymbol{\alpha})\boldsymbol{A}_i^{1/2}$, where $\boldsymbol{A}_i = \text{diag}(\sigma_{ij}^2)$ and the structure of $\boldsymbol{R}_0$ is unknown and parameterised by $\boldsymbol{\alpha}$. Instead, some working correlation matrix $\boldsymbol{R}_i(\boldsymbol{\alpha})$ can be used, so we let the working covariance matrix be $\boldsymbol{V}_i = \phi\boldsymbol{A}_i^{1/2}\boldsymbol{R}_i(\boldsymbol{\alpha})\boldsymbol{A}_i^{1/2}$. If the number of measurements from each cluster is the same, that is, $n_1 = n_2 = \cdots = n_K = n$, the data is called balanced data. Without a loss of generality, we will consider the balanced longitudinal dataset in this work.

The popular GEE method proposed by Liang and Zeger [1986] can be expressed as,

$$U(\boldsymbol{\beta}) = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{S}_i = \boldsymbol{0}_{p\times 1}, \tag{3.1}$$

where $\boldsymbol{D}_i = \partial\boldsymbol{\mu}_i(\boldsymbol{\beta})/\partial\boldsymbol{\beta}^T$ and $\boldsymbol{S}_i = \boldsymbol{Y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta})$. The asymptotically consistent estimator $\hat{\boldsymbol{\beta}}$ can be obtained from (3.1), even though the given $\boldsymbol{R}_i(\boldsymbol{\alpha})$ is misspecified. Wang and Carey [2003] investigated advisable methods to estimate $\boldsymbol{\alpha}$ for some well-known correlation structures. In general, the choice for how to estimate $\boldsymbol{\alpha}$ will affect the asymptotic efficiency of $\hat{\boldsymbol{\beta}}$. The various methods of how to obtain the variance of $\hat{\boldsymbol{\beta}}$ are described specifically in this section.

3.2.1 Sandwich estimator

The sandwich method is a well-known method in longitudinal data. It was shown that under mild regularity conditions, the estimator $\hat{\boldsymbol{\beta}}$ is consistent and asymptotically normally distributed with a mean $\widetilde{\boldsymbol{\beta}}$ and a variance $\text{cov}(\hat{\boldsymbol{\beta}})$ [Liang and Zeger, 1986]. The so called sandwich estimator is as follows:

$$\hat{\boldsymbol{V}}_{LZ} = \widehat{\text{cov}}(\hat{\boldsymbol{\beta}}) = \hat{\boldsymbol{Q}}^{-1}\hat{\boldsymbol{\Omega}}_{LZ}\hat{\boldsymbol{Q}}^{-1}, \tag{3.2}$$

where

$$\hat{\boldsymbol{Q}} = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{D}_i|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}},$$

$$\hat{\boldsymbol{\Omega}}_{LZ} = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \widehat{\text{cov}}(\boldsymbol{Y}_i) \boldsymbol{V}_i^{-1} \boldsymbol{D}_i|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}},$$

and $\widehat{\text{cov}}(\boldsymbol{Y}_i) = \boldsymbol{S}_i\boldsymbol{S}_i^T$. Here, $\boldsymbol{S}_i\boldsymbol{S}_i^T$ was applied to estimate $\text{cov}(\boldsymbol{Y}_i)$ to obtain a consistent estimator of $\text{cov}(\hat{\boldsymbol{\beta}})$. Notice that in (3.2) and the following covariance estimators, the values of $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and ϕ are replaced by their consistent estimates $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\alpha}}$, and $\hat{\phi}$, respectively. The advantage of the sandwich estimator is that even if the working correlation matrix $\boldsymbol{R}_i$ is not specified correctly, $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{V}}_{LZ}$ are still consistent estimators of $\boldsymbol{\beta}$ and $\text{cov}(\hat{\boldsymbol{\beta}})$. In case $\boldsymbol{R}_i$ is specified correctly, we have $\boldsymbol{V}_i = \text{cov}(\boldsymbol{Y}_i)$ and that is

$$\hat{\boldsymbol{V}}_N = \hat{\boldsymbol{Q}}^{-1} = \left(\sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \boldsymbol{D}_i \right)^{-1} \Bigg|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}},$$

which is called the model-based or naive estimate of $\text{cov}(\hat{\boldsymbol{\beta}})$. Generally speaking, this model-based estimator would be the optimal and smallest covariance estimator.

3.2.2 Modified sandwich estimator

In the sandwich method, $\boldsymbol{S}_i\boldsymbol{S}_i^T$ is considered the estimator of $\text{cov}(\boldsymbol{Y}_i)$. However, Pan [2001] noticed that $\boldsymbol{S}_i\boldsymbol{S}_i^T$ is not an efficient estimator of $\text{cov}(\boldsymbol{Y}_i)$ because it is based on the data from only one cluster. This leads to the modified estimator under the assumption that there is a

common correlation structure across all clusters, hence the marginal variance of y_{ij} should be correctly specified. Pan [2001] proposed the working correlation matrix can borrow information from other different subjects:

$$\hat{\boldsymbol{R}}_P = \frac{1}{K}\sum_{i=1}^{K} \boldsymbol{A}_i^{-1/2}\boldsymbol{S}_i\boldsymbol{S}_i^T\boldsymbol{A}_i^{-1/2},$$

that is, the $\text{cov}(\boldsymbol{Y}_i)$ can be estimated by $\boldsymbol{A}_i^{1/2}\hat{\boldsymbol{R}}_P\boldsymbol{A}_i^{1/2}$. Then the robust variance estimator is

$$\hat{\boldsymbol{V}}_P = \hat{\boldsymbol{Q}}^{-1}\hat{\boldsymbol{\Omega}}_P\hat{\boldsymbol{Q}}^{-1},$$

where

$$\hat{\boldsymbol{\Omega}}_P = \sum_{i=1}^{K} \boldsymbol{D}_i^T\boldsymbol{V}_i^{-1}\boldsymbol{A}_i^{1/2}\hat{\boldsymbol{R}}_P\boldsymbol{A}_i^{1/2}\boldsymbol{V}_i^{-1}\boldsymbol{D}_i.$$

Usually, these two assumptions needed in Pan [2001] were satisfied. It has been shown that these estimators are more efficient than the sandwich estimator because more information across different subjects can be used. The simulated data were generated from the linear model and a logistic model with $n = 3$, $K = 10$ or 40 and $K = 20$ or 40, respectively, which is a traditional longitudinal data setting with small cluster sizes.

Another modified sandwich estimator was proposed by Mancl and DeRouen [2001] to improve the performance when the number of clusters is small. The idea was that the $\boldsymbol{S}_i\boldsymbol{S}_i^T$ underestimates the $\text{cov}(\boldsymbol{Y}_i)$, so they proposed that the covariance of $\boldsymbol{Y}_i$ can be estimated by

$$\widehat{\text{cov}}(\boldsymbol{Y}_i)_{MD} = (\boldsymbol{I}_i - \boldsymbol{H}_{ii})^{-1}\boldsymbol{S}_i\boldsymbol{S}_i^T(\boldsymbol{I}_i - \boldsymbol{H}_{ii}^T)^{-1},$$

where $\boldsymbol{I}_i$ is an identity matrix and $\boldsymbol{H}_{ii} = \boldsymbol{D}_i(\sum_{i=1}^{K}\boldsymbol{D}_i^T\boldsymbol{V}_i^{-1}\boldsymbol{D}_i)^{-1}\boldsymbol{D}_i^T\boldsymbol{V}_i^{-1}$. Then, the bias corrected variance estimator is

$$\hat{\boldsymbol{V}}_{MD} = \hat{\boldsymbol{Q}}^{-1}\hat{\boldsymbol{\Omega}}_{MD}\hat{\boldsymbol{Q}}^{-1},$$

where

$$\hat{\boldsymbol{\Omega}}_{MD} = \sum_{i=1}^{K} \boldsymbol{D}_i^T\boldsymbol{V}_i^{-1}(\boldsymbol{I}_i - \boldsymbol{H}_{ii})^{-1}\boldsymbol{S}_i\boldsymbol{S}_i^T(\boldsymbol{I}_i - \boldsymbol{H}_{ii}^T)^{-1}\boldsymbol{V}_i^{-1}\boldsymbol{D}_i.$$

Combining the strength from Pan [2001] and Mancl and DeRouen [2001], Wang and Long [2011] proposed that the covariance of Y_i can be estimated as:

$$\widehat{\text{cov}}(\boldsymbol{Y}_i)_{WL} = \boldsymbol{A}_i^{1/2} \left\{ \sum_{i=1}^{K} \boldsymbol{A}_i^{-1/2} (\boldsymbol{I}_i - \boldsymbol{H}_{ii})^{-1} \boldsymbol{S}_i \boldsymbol{S}_i^T (\boldsymbol{I}_i - \boldsymbol{H}_{ii}^T)^{-1} \boldsymbol{A}_i^{-1/2} / K \right\} \boldsymbol{A}_i^{1/2}.$$

Then, the variance estimator is

$$\hat{\boldsymbol{V}}_{WL} = \hat{\boldsymbol{Q}}^{-1} \hat{\boldsymbol{\Omega}}_{WL} \hat{\boldsymbol{Q}}^{-1},$$

where

$$\hat{\boldsymbol{\Omega}}_{WL} = \sum_{i=1}^{K} \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1} \widehat{\text{cov}}(\boldsymbol{Y}_i)_{WL} \boldsymbol{V}_i^{-1} \boldsymbol{D}_i.$$

The performance of this combined estimator is considered when the cluster size is large.

3.2.3 Regularised sandwich estimator

Warton [2011] investigated a high-dimensional setting in which the number of clusters (K) is not large relative to the cluster size (n) based on Pan [2001]. If n is large and K is small, $\hat{\boldsymbol{R}}_P$ may be affected because $\boldsymbol{S}_i \boldsymbol{S}_i^T$ is a large ($n \times n$) sparse matrix that has rank one. This leads to the instability in $\widehat{\text{cov}}(\hat{\boldsymbol{\beta}})$. Hence, he proposed a regularisation approach:

$$\hat{\boldsymbol{R}}_W = \lambda \hat{\boldsymbol{R}}_P + (1 - \lambda) \boldsymbol{R}_i(\boldsymbol{\alpha}),$$

where $\lambda \in [0, 1]$. The corresponding variance $\hat{\boldsymbol{V}}_W$ can be expressed as shrinkage estimators:

$$\hat{\boldsymbol{\Omega}}_W = \lambda \hat{\boldsymbol{\Omega}}_P + (1 - \lambda) \hat{\boldsymbol{Q}},$$
$$\hat{\boldsymbol{V}}_W = \lambda \hat{\boldsymbol{V}}_P + (1 - \lambda) \hat{\boldsymbol{V}}_N.$$

In his paper, a special case was considered when $\boldsymbol{R}_i(\boldsymbol{\alpha}) = \boldsymbol{I}$ because he claimed the computation time would be affected in the high-dimensional setting. A cross-validation method was

used to find the value of λ:

$$\hat{\lambda} = \arg\max_{\lambda} \sum_{m=1}^{M} \frac{-K_m}{2} \log|\hat{\boldsymbol{R}}_W^{\backslash m}| - \frac{1}{2}\mathrm{tr}[\boldsymbol{S}_m(\hat{\boldsymbol{R}}_W^{\backslash m})^{-1}\boldsymbol{S}_m^T],$$

where $\hat{\boldsymbol{R}}_W^{\backslash m}$ is the correlation matrix estimated from all groups except the mth. The matrix of Pearson residuals has been split into M roughly equal parts $[\boldsymbol{S}_1^T, \cdots, \boldsymbol{S}_M^T]$, and $\boldsymbol{S}_m$ is a $K_m \times n$ matrix. As we have described in the introduction, a specific setting different from traditional longitudinal data is used in the current work. The performance of the regularised sandwich estimators is worth investigating in the longitudinal data with large cluster sizes.

3.2.4 Smooth bootstrap

Different from the previous estimators based on sandwich estimators, a smooth bootstrap method chooses another way by perturbing the GEE function to obtain the variance of parameters in longitudinal data [Li and Wang, 2008]. The perturbed GEE version of $U(\boldsymbol{\beta})$ was proposed to obtain the smooth bootstrap estimates:

$$U^B(\boldsymbol{\beta}) = \sum_{i=1}^{K} w_i \boldsymbol{D}_i^T \boldsymbol{V}_i^{-1}(\boldsymbol{Y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta})), \tag{3.3}$$

where $w_1, ..., w_K$ are independent realisations from a distribution with arbitrary mean, but unit variance. Following Li and Wang [2008], we take $\mathrm{E}(w_k) = 1$. The w_i from $\mathrm{N}(1, 1)$, or $\mathrm{Pois}(1)$, can be used to obtain the smooth bootstrap estimators. The iterative weighted least-squares method and quasi-Fisher scoring algorithm were used to solve these equations to get estimators. The iterative procedure for $\tilde{\boldsymbol{\beta}}_B^{(l)}$ can be described as follows:

Step1 $\tilde{\boldsymbol{\beta}}_B^{(0)} = \hat{\boldsymbol{\beta}}$;

Step2 $\tilde{\boldsymbol{\beta}}_B^{(1)} = \tilde{\boldsymbol{\beta}}_B^{(0)} + (\sum_{i=1}^K w_i \hat{\boldsymbol{Q}}_i)^{-1}\{\sum_{i=1}^K w_i \boldsymbol{D}_i^T(\tilde{\boldsymbol{\beta}}_B^{(0)})\boldsymbol{V}_i^{-1}(\tilde{\boldsymbol{\beta}}_B^{(0)}, \hat{\boldsymbol{\alpha}})(\boldsymbol{Y}_i - \boldsymbol{\mu}_i(\tilde{\boldsymbol{\beta}}_B^{(0)}))\}$;

Step3 $\tilde{\boldsymbol{\beta}}_B^{(2)} = \tilde{\boldsymbol{\beta}}_B^{(1)} + (\sum_{i=1}^K w_i \hat{\boldsymbol{Q}}_i)^{-1}\{\sum_{i=1}^K w_i \boldsymbol{D}_i^T(\tilde{\boldsymbol{\beta}}_B^{(1)})\boldsymbol{V}_i^{-1}(\tilde{\boldsymbol{\beta}}_B^{(1)}, \hat{\boldsymbol{\alpha}})(\boldsymbol{Y}_i - \boldsymbol{\mu}_i(\tilde{\boldsymbol{\beta}}_B^{(1)}))\}$;

Then for any l, the covariance of the estimators is given by

$$\hat{\boldsymbol{V}}_B(\tilde{\boldsymbol{\beta}}_B^{(l)}) = \hat{\boldsymbol{Q}}^{-1}\left(\sum_{i=1}^{K} \hat{\boldsymbol{D}}_i^T \hat{\boldsymbol{V}}_i^{-1}(\boldsymbol{Y}_i - \hat{\boldsymbol{\mu}}_i)(\boldsymbol{Y}_i - \hat{\boldsymbol{\mu}}_i)^T \hat{\boldsymbol{V}}_i^{-1} \hat{\boldsymbol{D}}_i\right)\hat{\boldsymbol{Q}}^{-1}. \tag{3.4}$$

The smooth bootstrap estimators are more efficient in terms of the variance estimation and confidence interval coverage based on their simulation results and real data analysis. However, we are interested in the performance of the smooth bootstrap method when the cluster size is large. Because the smooth bootstrap method bypasses the sandwich estimators, it is expected to be a more efficient method, and our extensive simulation can support this.

3.3 Simulation studies

3.3.1 Simulation setup

To investigate the performance of the different methods, we carry out extensive simulation studies. The sample size is chosen to be $K = 30$. As for the cluster size, $n = 5$, 10, 20, 30, 50, 100, 200, 500 and 1000 are considered separately to see the performances when n increases. The number of simulations is 1000, and the bootstrap times is 1000. Simulations are considered for three types of response: normal, overdispersed Poisson, and binary.

The normal data are generated from the following linear model:

$$y_{ij} = \mu_{ij} + \epsilon_{ij}, \;\; i = 1, \ldots, K, \;\; j = 1, \ldots, n,$$

where $\mu_{ij} = \beta_0 + \beta_1 x_{ij}$, the true value of $\boldsymbol{\beta} = (\beta_0, \beta_1)^T = (0, 1)^T$, and x_{ij} are sampled from the uniform distribution $(j, j+1)$ that has a time trend. For each i, $(\epsilon_{i1}, \ldots, \epsilon_{in})$ are correlated with first order autogressive (AR(1)) structure, with the correlation coefficient being 0.5. The *rmvnorm* function of the R-package *mvtnorm* was used to generate the correalted normal data [Genz et al., 2020].

For overdispersed Poisson data, y_{ij} is sampled from a Poisson distribution with a mean μ_{ij} and variance $a\mu_{ij} + b\mu_{ij}^2$. Here, $\mu_{ij} = \exp(\beta_0 + \beta_1 x_{ij})$, where $\boldsymbol{\beta} = (\beta_0, \beta_1)^T = (0, 1)^T$ and x_{ij} are sampled from the uniform distribution $(j, j+1)$ and then log transformed. A gamma frailty model [Henderson and Shimakura, 2003] is used to generate the AR(1) overdispersed Poisson data. In our simulations, $a = 1$ and $b = 1$. The first observation y_{i1} for subject i is generated from a Pois$(\mu_{i1} t_1)$, where t_1 is a value generated from a gamma$(1, 1)$. For the following observations ($2 \leq j \leq n$), y_{ij} are generated from Pois$(\eta_j t_j)$, where $\eta_j = \mu_{ij} + \rho\sqrt{\frac{\mu_{ij}+\mu_{ij}^2}{\mu_{i(j-1)}+\mu_{i(j-1)}^2}}(y_{i(j-1)} - \mu_{i(j-1)})$ and $t_j = \frac{\mu_{ij}^2+\rho^2(\mu_{ij}+\mu_{ij}^2)}{(\mu_{ij}+\mu_{ij}^2)(1-\rho^2)-\mu_{ij}}$. Within-subject correlations are

AR(1) with a correlation coefficient of 0.5.

For binary data, y_{ij} is sampled from a Bernoulli distribution with $\pi_{ij} = 1/(1 + \exp(-\mu_{ij}))$ with $\mu_{ij} = \beta_0 + \beta_1 x_{ij}$, where $\boldsymbol{\beta} = (\beta_0, \beta_1)^T = (1, 1)^T$ and x_{ij} are uniformly distributed on (0,1). According to Fu et al. [2018], the correlation value ρ between any two pairs y_{ij} and y_{il} must satisfy the following constraints,

$$\max_{i,j\neq l}\left\{-\sqrt{\frac{p_{ij}p_{il}}{q_{ij}q_{il}}}, -\sqrt{\frac{q_{ij}q_{il}}{p_{ij}p_{il}}}\right\} \leq \rho \leq \min_{i,j\neq l}\left\{\sqrt{\frac{p_{ij}p_{il}}{q_{ij}q_{il}}}, \sqrt{\frac{q_{ij}q_{il}}{p_{ij}p_{il}}}\right\},$$

where $p_{ij} = p(y_{ij} = 1)$ and $q_{ij} = 1 - p_{ij}$. To ensure the constraints are met, the true correlation structure AR(1) with coefficient 0.3 is chosen. The R-package *SimCorMultRes* was used to generate correlated binary data [Touloumis, 2016].

3.3.2 Simulation results

The objective of our simulation studies focuses on the bias of variance estimators when the cluster size exceeds the number of clusters. We compare various different variance estimators: the model-based estimator (denoted as M-B, also called the naive estimator), the sandwich estimator of Liang and Zeger [1986] (denoted as LZ), the regularised estimator from Warton [2011] (denoted as W), the modified sandwich estimator from Wang and Long [2011] (denoted as WL), and the estimator from Li and Wang [2008], which perturbs the estimating equations using random values from a Poisson distribution Pois(1) (denoted as B-p). The performance of the smooth bootstrap method using random values from N$(1, 1)$ was omitted in our simulation because Li and Wang [2008] claimed that the performance of using random values form Pois(1) was better. The 'true' standard error of $\hat{\boldsymbol{\beta}}$ can be obtained in the simulations. The ratios of these estimated standard errors to the 'true' standard error will be calculated as the criteria.

Figures 3.1, 3.2 and 3.3 present the average ratio based on the 1000 simulations for the normal case, Poisson case, and binary case respectively. Here, the cluster sizes increased from 5 to 1000 in the number of clusters: $K = 30$. these three different cases. The quantitative bias and variance metrics are presented in Tables 3.1, 3.2 and 3.3. According to Figure 3.1(a), the model-based estimators are defensible because the correlation structure is specified correctly. In the norm and Poisson case (Figure 3.1(b) and Figure 3.2(b)), the model-based estimators perform worse against misspecification when the number of cluster sizes increased. The evidence shows

that the bias of sandwich estimators may be reduced when the cluster size increased in a limited range under this condition (see Figures 3.1(b) and 3.2(b)). When the cluster sizes n exceed the number of clusters K, the sandwich estimators, the improved sandwich estimators and the regularised sandwich estimators are still valid. The mean of the smooth bootstrap estimators are closer to the true value than the other estimators. However, it has larger variation than modified sandwich estimators and regularised sandwich estimators in the binary case. The results for the normal case when $n > K$ are similar to the results for the Poisson cases (see Figure 3.1 and Figure 3.2). Generally, the modified sandwich method and regularised sandwich method perform well in the binary case.

3.4 Data analysis example

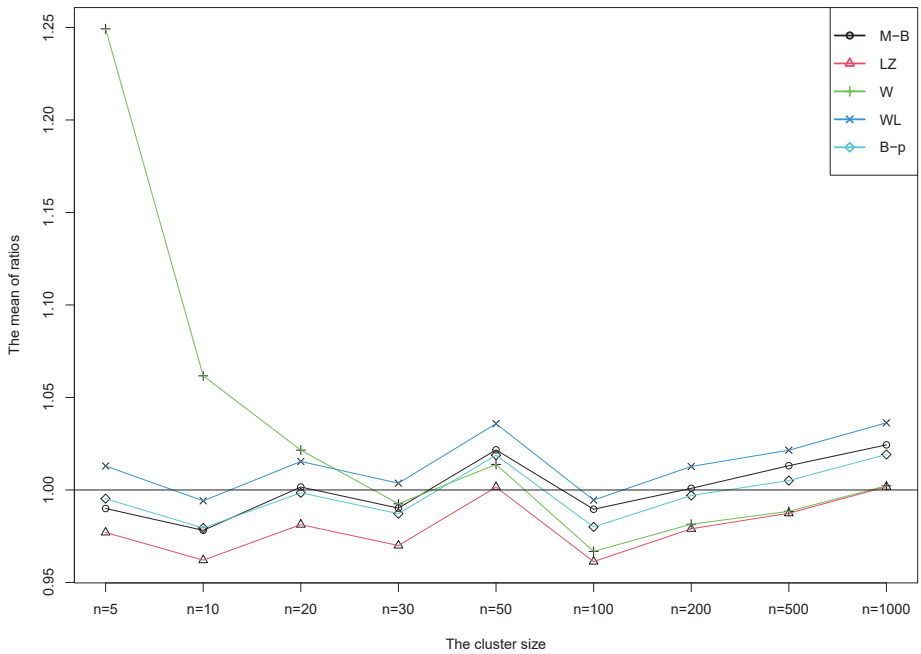
In this section, we illustrate the proposed methods using two datasets: children growth data from the Berkeley Growth Study [Tuddenham, 1954] and protein milk data from Diggle et al. [2002].

3.4.1 Berkeley growth study data

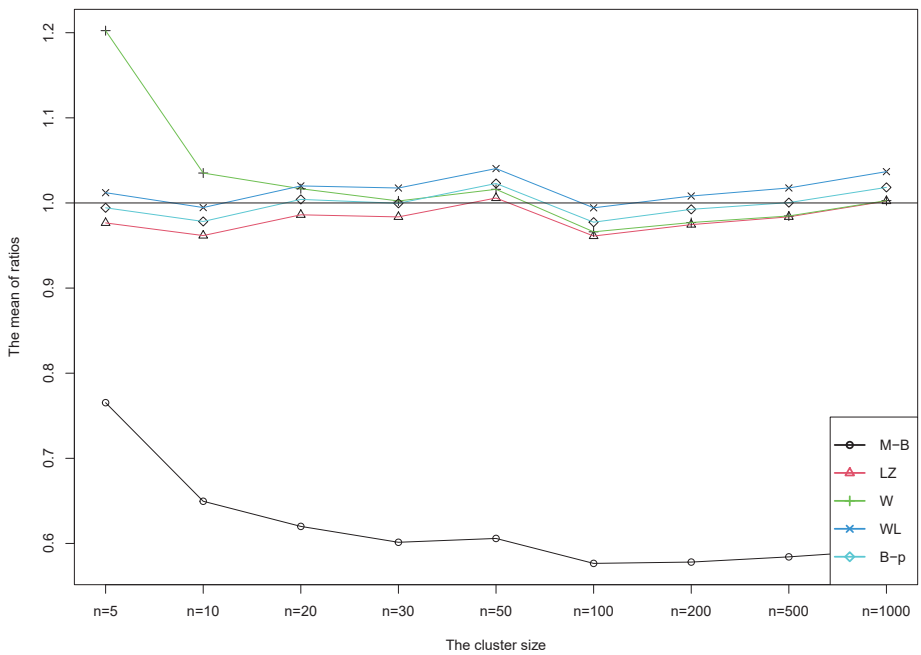
The Berkeley Growth Study, which includes the heights of 54 girls and 39 boys measured at a set of 31 ages, is analyzed [Tuddenham, 1954]. The heights were measured at unequal intervals. There are four measurements when the children was one year of age, annual measurements from two to eight years of age and two measurements per year until 18 years of age (Figure 3.4). We can see that the heights of the boys and girls are almost equal when the children are one year old. Then, the heights of the girls are larger than the boys, especially between eight and 13 years old. Finally, the height of the boys overtake the girls. We can also see from the figure that this is not just a linear trend between age and height. Therefore, a quadratic term and a cubic term are added in our model.

We use the following marginal model:

$$\begin{aligned} E(Y_{ij}) = {} & \beta_1 \text{Girl}_i + \beta_2 \text{Age}_{ij} * \text{Girl}_i + \beta_3 \text{Age}_{ij}^2 * \text{Girl}_i + \beta_4 \text{Age}_{ij}^3 * \text{Girl}_i \\ & + \beta_5 \text{Boy}_i + \beta_6 \text{Age}_{ij} * \text{Boy}_i + \beta_7 \text{Age}_{ij}^2 * \text{Boy}_i + \beta_8 \text{Age}_{ij}^3 * \text{Boy}_i, \end{aligned}$$

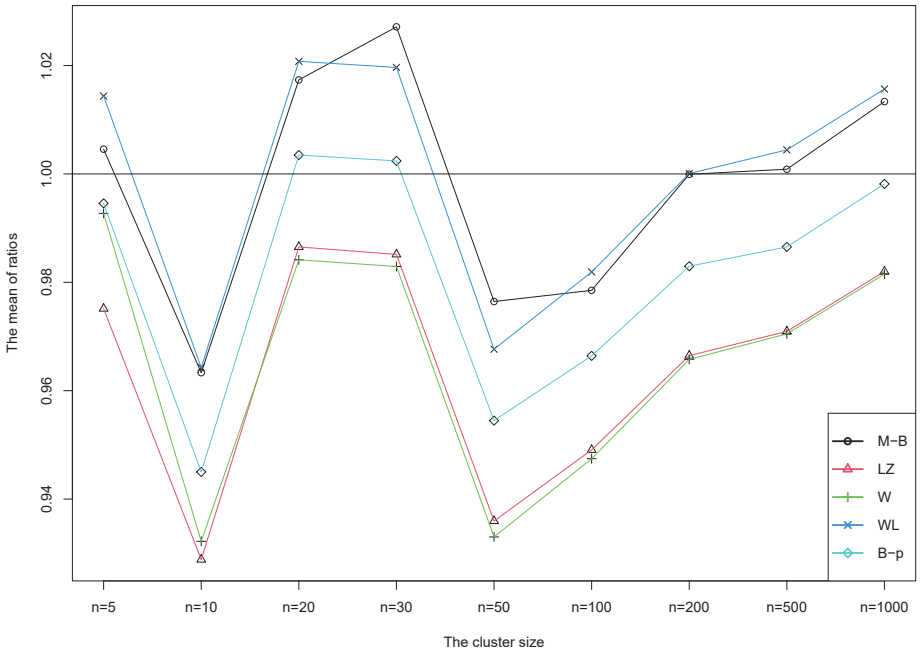


(a) AR(1) working correlation matrix

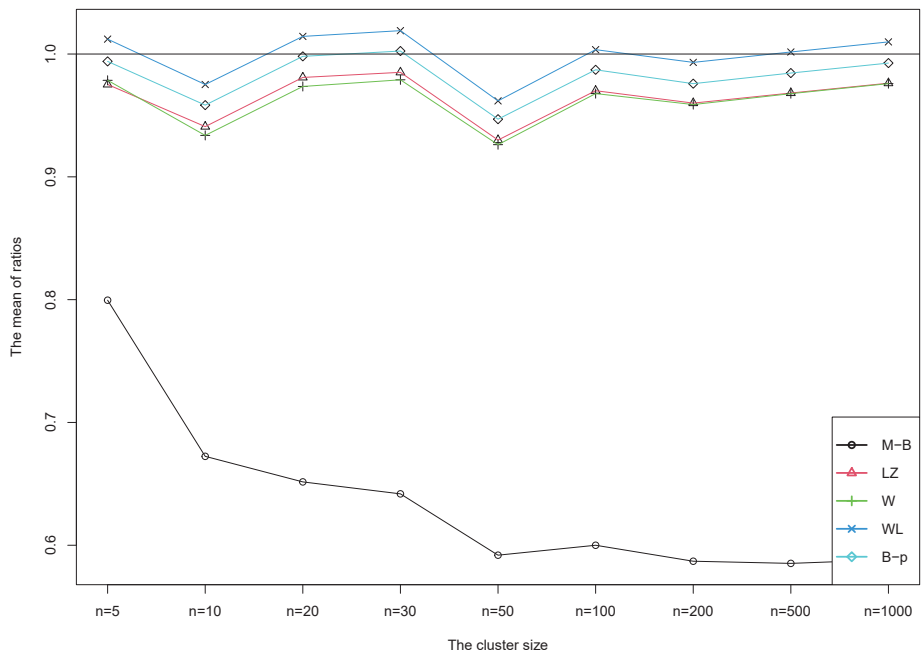


(b) Exchangeable working correlation matrix

Figure 3.1: The ratios of standard error by different methods versus the true standard error for norm data with sample size 30. The cluster sizes are 5, 10, 20, 30, 50, 100, 200, 500 and 1000 separately. M-B: model-based method, LZ: sandwich method, WL: method from Wang and Long [2011], W: regularised method from Warton [2011], B-p: bootstrap method with Poisson distribution

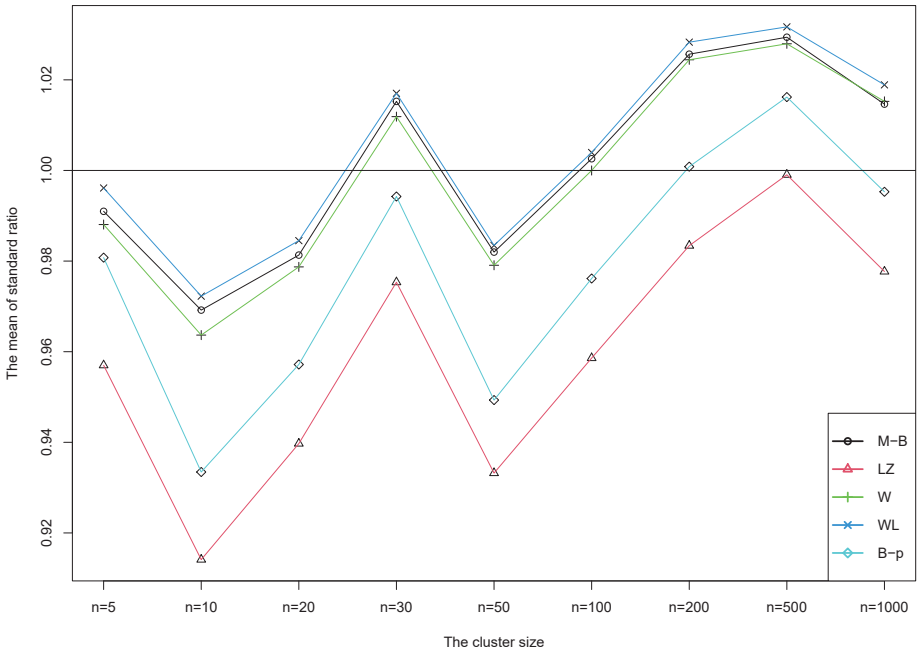


(a) AR(1) working correlation matrix

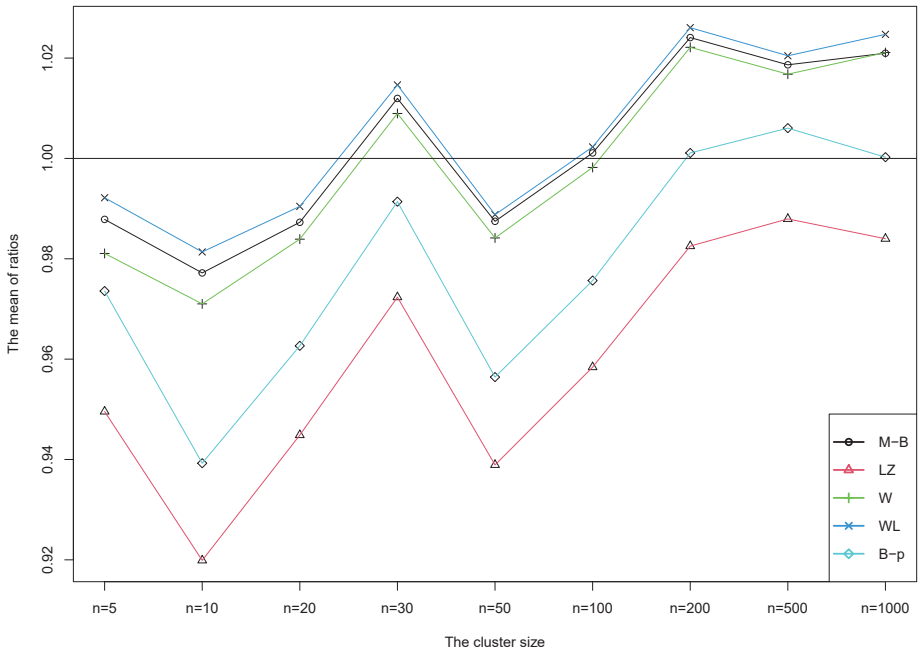


(b) Exchangeable working correlation matrix

Figure 3.2: The ratios of standard error by different methods versus the true standard error for Poisson data with sample size 30. The cluster sizes are 5, 10, 20, 30, 50, 100, 200, 500 and 1000 separately. M-B: model-based method, LZ: sandwich method, WL: method from Wang and Long [2011], W: regularised method from Warton [2011], B-p: bootstrap method with Poisson distribution



(a) AR(1) working correlation matrix



(b) Exchangeable working correlation matrix

Figure 3.3: The ratios of standard error by different methods versus the true standard error for binomial data with sample size 30. The cluster sizes are 5, 10, 20, 30, 50, 100, 200, 500 and 1000 separately. M-B: model-based method, LZ: sandwich method, WL: method from Wang and Long [2011], W: regularised method from Warton [2011], B-p: bootstrap method with Poisson distribution

Table 3.1: The mean and deviation of ratios of standard error by different methods versus the true standard error for norm data with sample size 30

(a) AR(1)

	Mean					SD				
	M-B	LZ	W	WL	B-P	M-B	LZ	W	WL	B-P
n=5	0.990	0.977	1.249	1.013	**0.995**	0.064	0.138	**0.188**	0.132	0.142
n=10	0.978	0.962	1.062	**0.994**	0.979	0.062	0.131	0.132	0.131	**0.134**
n=20	1.002	0.981	1.022	1.015	**0.998**	0.058	0.130	0.125	0.133	**0.134**
n=30	0.990	0.970	0.993	**1.004**	0.987	0.052	0.124	0.121	0.127	**0.128**
n=50	1.022	**1.002**	1.014	1.036	1.019	0.045	0.135	0.133	**0.139**	**0.139**
n=100	0.990	0.961	0.967	**0.995**	0.980	0.033	0.129	0.128	**0.133**	**0.133**
n=200	**1.001**	0.979	0.981	1.013	0.997	0.024	0.130	0.129	**0.134**	0.133
n=500	1.013	0.987	0.988	1.021	**1.005**	0.015	0.131	0.131	0.136	**0.137**
n=1000	1.024	**1.002**	**1.002**	1.036	1.019	0.011	0.134	0.134	**0.138**	**0.138**

(b) Exchangeable

	Mean					SD				
	M-B	LZ	W	WL	B-P	M-B	LZ	W	WL	B-P
n=5	0.765	0.977	1.202	1.012	**0.994**	0.057	0.138	**0.175**	0.140	0.141
n=10	0.650	0.962	1.035	**0.995**	0.978	0.034	0.131	0.127	**0.135**	0.134
n=20	0.620	0.986	1.017	1.020	**1.004**	0.023	0.132	0.129	**0.137**	0.136
n=30	0.601	0.984	1.002	1.018	**1.000**	0.019	0.125	0.122	**0.129**	0.128
n=50	0.606	**1.006**	1.016	1.040	1.023	0.014	0.136	0.134	**0.140**	0.139
n=100	0.577	0.961	0.966	**0.994**	0.977	0.010	0.129	0.128	**0.134**	0.133
n=200	0.578	0.975	0.977	1.008	**0.993**	0.007	0.130	0.129	**0.134**	**0.134**
n=500	0.584	0.984	0.985	1.018	**1.000**	0.004	0.132	0.132	0.136	**0.137**
n=1000	0.592	**1.002**	1.003	1.037	1.018	0.003	0.134	0.134	**0.138**	**0.138**

where Y_{ij} is the height for the ith child in the jth observation and the variable Girl$_i$ and Boy$_i$ are both 0/1 indicator variables for Gender being girl or boy. The Age variable shows the age in years from birth. The working correlation structure used in the estimation is AR(1). According to Table 3.4, all covariates are statistically significant except for the parameter β_4. For the analysis of girls data, i.e. for parameters β_1 to β_4, the regularised sandwich estimator has the smallest variance. The sandwich estimator and smooth bootstrap estimator have nearly the same performance for the analysis of boys data. The estimators from the model-based method

Table 3.2: The mean and deviation of ratios of standard error by different methods versus the true standard error for Poisson data with sample size 30

(a) AR(1)

	Mean					SD				
	M-B	LZ	W	WL	B-P	M-B	LZ	W	WL	B-P
n=5	**1.005**	0.975	0.993	1.014	0.995	0.108	0.178	0.151	0.169	**0.183**
n=10	0.963	0.929	0.932	**0.964**	0.945	0.113	0.161	0.146	0.161	**0.164**
n=20	1.017	0.987	0.984	1.021	**1.003**	0.103	0.175	0.167	0.177	**0.178**
n=30	1.027	0.985	0.983	1.020	**1.002**	0.110	0.176	0.170	0.180	**0.181**
n=50	**0.976**	0.936	0.933	0.968	0.954	0.091	0.157	0.153	0.160	**0.162**
n=100	0.979	0.949	0.947	**0.982**	0.966	0.073	0.162	0.161	**0.167**	**0.167**
n=200	**1.000**	0.966	0.966	1.000	0.983	0.059	0.159	0.159	**0.165**	0.163
n=500	**1.001**	0.971	0.970	1.004	0.987	0.039	0.153	0.153	**0.158**	0.155
n=1000	1.013	0.982	0.982	1.016	**0.998**	0.030	0.140	0.140	0.145	**0.148**

(b) Exchangeable

	Mean					SD				
	M-B	LZ	W	WL	B-P	M-B	LZ	W	WL	B-P
n=5	0.800	0.975	0.979	1.012	**0.994**	0.094	0.187	0.166	0.186	**0.191**
n=10	0.672	0.941	0.934	**0.975**	0.958	0.062	0.172	0.159	**0.176**	**0.176**
n=20	0.652	0.981	0.974	1.014	**0.998**	0.044	0.177	0.171	**0.183**	0.181
n=30	0.642	0.985	0.979	1.019	**1.002**	0.043	0.188	0.182	**0.194**	0.193
n=50	0.592	0.930	0.926	**0.962**	0.947	0.033	0.163	0.160	**0.168**	0.167
n=100	0.600	0.970	0.968	**1.004**	0.987	0.026	0.168	0.167	**0.174**	0.171
n=200	0.587	0.960	0.959	**0.993**	0.976	0.020	0.159	0.159	**0.165**	0.163
n=500	0.585	0.968	0.968	**1.002**	0.985	0.012	0.152	0.152	**0.157**	0.156
n=1000	0.588	0.976	0.976	1.010	**0.993**	0.010	0.140	0.140	**0.145**	0.144

are larger than the other methods for the variables Girl_i and Boy_i.

3.4.2 Protein of milk data

In this data set, milk was collected weekly from 79 Australian cows, and the protein content was investigated. There are three diets: 25 cows received a barley diet, 27 cows received a mixture of barley and lupins, and 27 cows received a diet of lupins only. The observation period of each

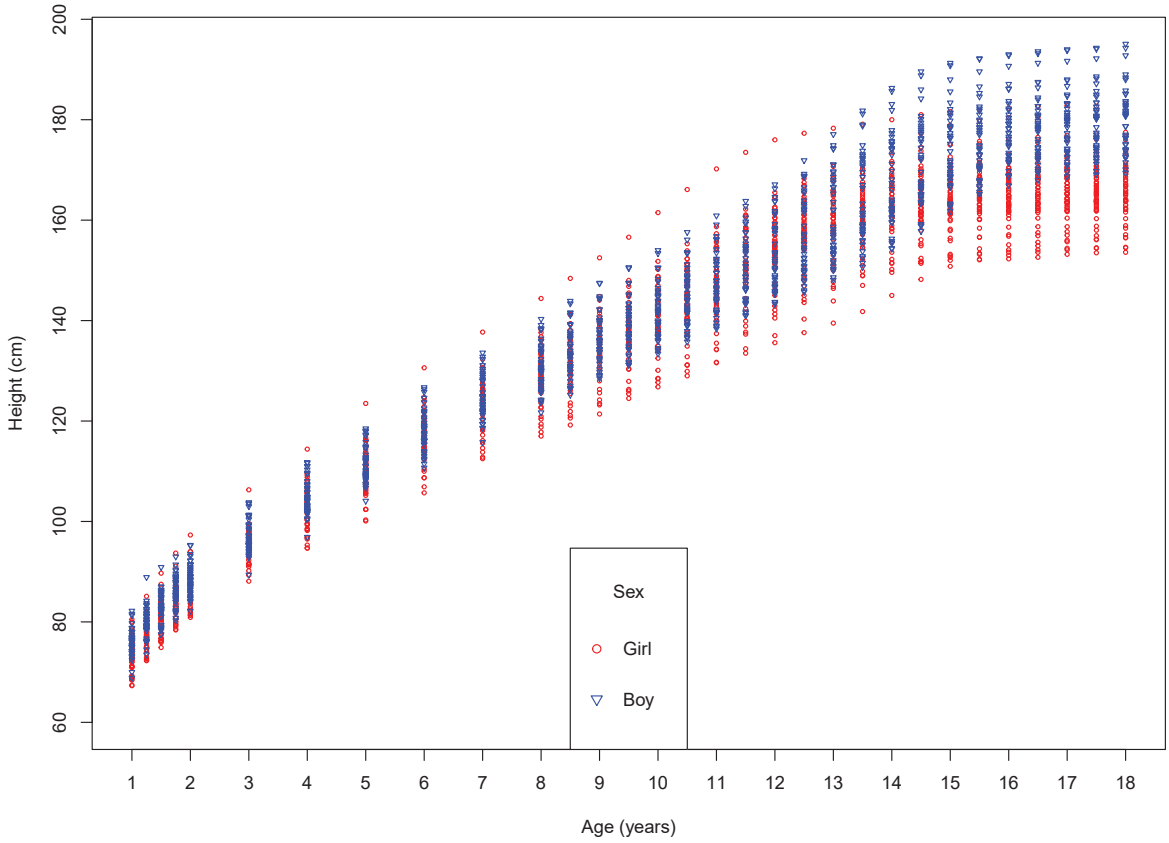


Figure 3.4: The heights of 54 girls and 39 boys measured at 31 different ages respectively

Table 3.3: The mean and deviation of ratios of standard error by different methods versus the true standard error for binary data with sample size 30

(a) AR(1)

	Mean					SD				
	M-B	LZ	W	WL	B-P	M-B	LZ	W	WL	B-P
n=5	0.991	0.957	0.988	**0.996**	0.981	0.129	0.204	0.132	0.135	**0.209**
n=10	0.969	0.914	0.964	**0.972**	0.933	0.162	0.220	0.155	0.165	**0.225**
n=20	0.981	0.940	0.979	**0.985**	0.957	0.101	0.176	0.105	0.107	**0.182**
n=30	1.015	0.975	1.012	1.017	**0.994**	0.090	0.165	0.092	0.094	**0.169**
n=50	0.982	0.933	0.979	**0.983**	0.949	0.064	0.145	0.069	0.070	**0.148**
n=100	1.003	0.959	**1.000**	1.004	0.976	0.043	0.139	0.049	0.049	**0.144**
n=200	1.026	0.983	1.024	1.028	**1.001**	0.032	0.135	0.040	0.041	**0.138**
n=500	1.029	**0.999**	1.028	1.032	1.016	0.020	0.129	0.033	0.034	**0.134**
n=1000	1.015	0.978	1.015	1.019	**0.995**	0.014	0.133	0.030	0.031	**0.137**

(b) Exchangeable

	Mean					SD				
	M-B	LZ	W	WL	B-P	M-B	LZ	W	WL	B-P
n=5	0.988	0.950	0.981	**0.992**	0.974	0.117	0.191	0.120	0.123	**0.195**
n=10	0.977	0.920	0.971	**0.981**	0.939	0.156	0.219	0.151	0.162	**0.224**
n=20	0.987	0.945	0.984	**0.990**	0.963	0.099	0.175	0.101	0.103	**0.180**
n=30	1.012	0.972	1.009	1.015	**0.991**	0.086	0.163	0.090	0.091	**0.168**
n=50	0.987	0.939	0.984	**0.989**	0.956	0.061	0.145	0.067	0.068	**0.150**
n=100	**1.001**	0.958	0.998	1.002	0.976	0.042	0.137	0.048	0.048	**0.142**
n=200	1.024	0.983	1.022	1.026	**1.001**	0.031	0.134	0.039	0.040	**0.138**
n=500	1.019	0.988	1.017	1.020	**1.006**	0.022	0.126	0.033	0.033	**0.131**
n=1000	1.021	0.984	1.021	1.025	**1.000**	0.017	0.136	0.030	0.031	**0.139**

cow is not necessarily the same, and each cow is observed for at least 12 weeks and at most 19 weeks, which means this is an unbalance data set.

It appears from Figure 3.5 that barely gives higher values than the mixture, which has higher values than lupins alone. The mean response profiles are approximately parallel, showing an initial sharp decline associated with a settling-in period, followed by an approximately constant mean response through the following period and a slow rise towards the end (Figure 3.6).

Table 3.4: Parameter estimates with standard errors and length(cm) of 95% confidence interval from a Gaussian model with AR(1) correlation structure for growth data

Variable	Methods	Estimate	SE	Z	p-value	Length of CI
Girl_i (β_1)	M-B	69.547	1.054	65.98	0.000	4.132
	LZ	69.547	0.579	120.08	0.000	2.270
	Warton	69.547	0.560	124.15	0.000	2.196
	WL	69.547	0.573	121.34	0.000	2.247
	B-p	69.550	0.570	122.02	0.000	2.234
$\text{Girl}_i * \text{Age}_{ij}$ (β_2)	M-B	8.703	0.285	30.57	0.000	1.116
	LZ	8.703	0.258	33.75	0.000	1.011
	Warton	8.703	0.237	36.67	0.000	0.930
	WL	8.703	0.243	35.83	0.000	0.952
	B-p	8.703	0.255	34.19	0.000	0.998
$\text{Girl}_i * \text{Age}_{ij}^2$ (β_3)	M-B	-0.105	0.036	-2.96	0.003	0.139
	LZ	-0.105	0.033	-3.16	0.002	0.130
	Warton	-0.105	0.029	-3.67	0.000	0.112
	WL	-0.105	0.029	-3.58	0.000	0.115
	B-p	-0.105	0.032	-3.27	0.001	0.126
$\text{Girl}_i * \text{Age}_{ij}^3/100$ (β_4)	M-B	-0.224	0.122	-1.84	0.066	0.477
	LZ	-0.224	0.121	-1.85	0.064	0.474
	Warton	-0.224	0.101	-2.21	0.027	0.396
	WL	-0.224	0.104	-2.16	0.031	0.406
	B-p	-0.224	0.117	-1.92	0.055	0.458
Boy_i (β_5)	M-B	67.364	0.896	75.20	0.000	3.511
	LZ	67.364	0.465	144.92	0.000	1.822
	Warton	67.364	0.476	141.51	0.000	1.866
	WL	67.364	0.487	138.30	0.000	1.909
	B-p	67.395	0.482	139.92	0.000	1.888
$\text{Boy}_i * \text{Age}_{ij}$ (β_6)	M-B	9.017	0.242	37.27	0.000	0.949
	LZ	9.017	0.188	47.86	0.000	0.739
	Warton	9.017	0.202	44.71	0.000	0.791
	WL	9.017	0.206	43.68	0.000	0.809
	B-p	9.011	0.181	49.90	0.000	0.708
$\text{Boy}_i * \text{Age}_{ij}^2$ (β_7)	M-B	-0.101	0.030	-3.36	0.001	0.118
	LZ	-0.101	0.021	-4.79	0.000	0.083
	Warton	-0.101	0.024	-4.16	0.000	0.096
	WL	-0.101	0.025	-4.06	0.000	0.098
	B-p	-0.100	0.021	-4.85	0.000	0.081
$\text{Boy}_i * \text{Age}_{ij}^3/100$ (β_8)	M-B	-0.547	0.103	-5.29	0.000	0.405
	LZ	-0.547	0.072	-7.65	0.000	0.280
	Warton	-0.547	0.086	-6.37	0.000	0.337
	WL	-0.547	0.088	-6.21	0.000	0.345
	B-p	-0.551	0.071	-7.79	0.000	0.277

Following Diggle et al. [2002], the mean response profiles model was used:

$$\mu_{it} = \begin{cases} \beta_{0i} + \beta_1 t & \text{if } t \leq 3, \\ \beta_{0i} + 3\beta_1 + \beta_2(t-3) & \text{if } t > 3, \end{cases}$$

where $i = 1, 2, 3$ denotes three different treatment group. The difference between Diggle et al. [2002] is that the quadratic term is removed because it is not statistically significant. The main interest is the coefficient β_{01}, β_{02}, β_{03}, β_1, and β_2. To illustrate the case when $n > K$, each five cows are randomly selected from three different diets ($K = 15$) as subset data and 1000 sampling is done to obtain the mean estimates. The regression parameters and their variance estimators under the AR(1) correlation structure are presented in Table 3.5. β_1 and β_2 represent how the protein of milk varied over time. These two covariates are significant in the complete data set. All the variance estimates provide comparable results. However, in the subset data set, the variance estimates for the regression coefficient for the time covariate are not significant.

In the complete data analysis, the sandwich estimator has the largest variance and the model-based estimator has the smallest variance except in the parameter β_2. The regularised sandwich estimator and the smooth bootstrap estimator are also defensible, which seems to have consistent findings with the simulation studies. However, in the subset data analysis, which means the cluster size is larger, the model-based estimator has the smallest variance except in the parameters β_{02} and β_2. In these two parameters, smooth bootstrap estimators have the smallest variance.

3.5 Discussion

In the current work, we compared the effects of large cluster sizes on the modified sandwich method and the smooth bootstrapping method in analyzing longitudinal data. In our extensive simulation studies, the sample size was fixed to 30, and the cluster size increased from 5 to 1000. The sandwich estimator exhibits considerable bias and underestimates the sampling variance when the cluster size is relatively small. Various modified sandwich estimators were developed to correct for the bias in K. However, in the longitudinal data with large cluster sizes, the sandwich estimators are still valid based on our simulations. The improved sandwich

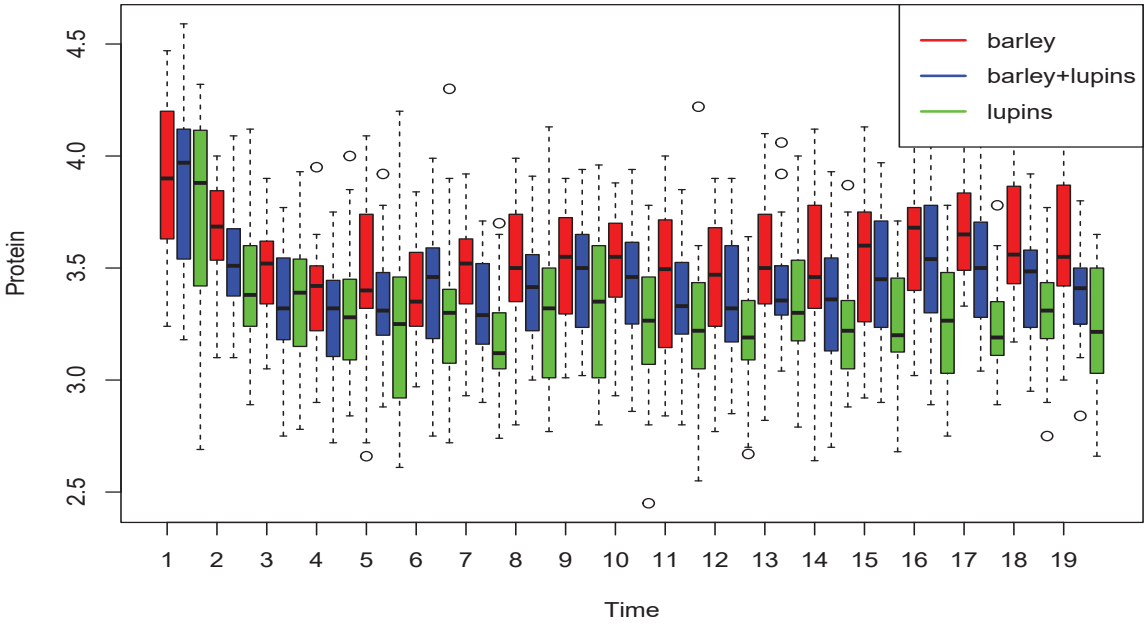


Figure 3.5: The protein of milk measured at 19 weeks for three different diets respectively

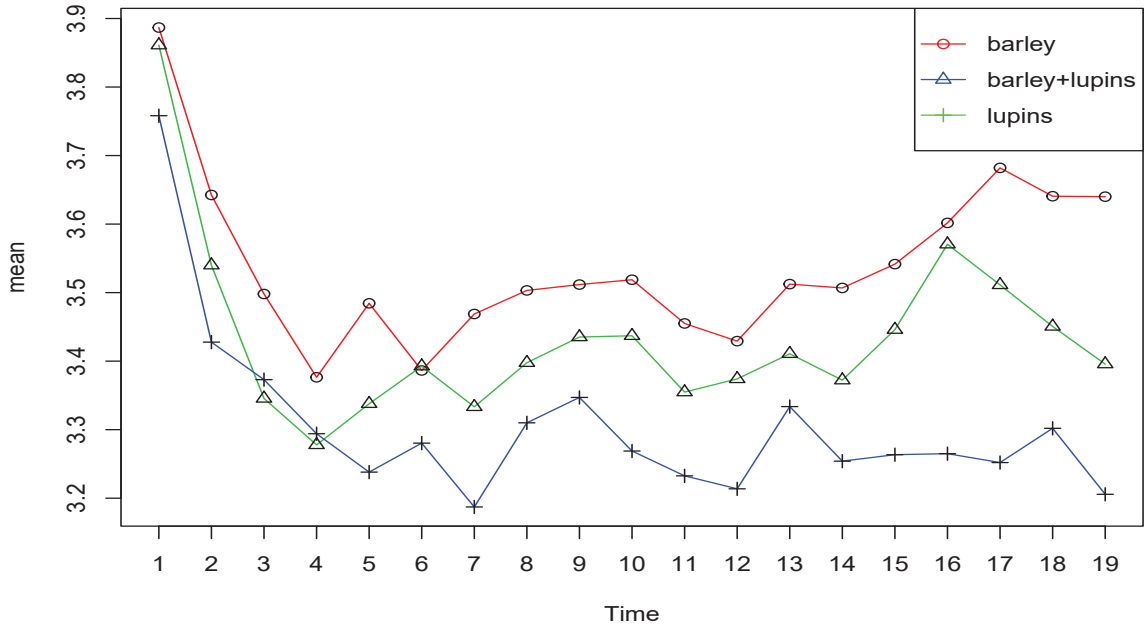


Figure 3.6: The mean protein of milk measured at 19 weeks for three different diets respectively

Table 3.5: Parameter estimates with standard errors and length of 95% confidence interval from a Gaussian model with AR(1) correlation structure for milk data

Variable	Methods	Estimate	SE	Z	p-value	Length of CI
Complete Data (K=79)						
Diet(=barley) (β_{01})	M-B	4.152	0.052	80.09	0.000	0.203
	LZ	4.152	0.069	60.13	0.000	0.271
	Warton	4.152	0.067	62.29	0.000	0.261
	WL	4.152	0.067	62.08	0.000	0.262
	B-p	4.151	0.066	63.27	0.000	0.257
Diet(=mixed) (β_{02})	M-B	-0.095	0.038	-2.54	0.011	0.147
	LZ	-0.095	0.048	-1.97	0.048	0.189
	Warton	-0.095	0.046	-2.08	0.038	0.180
	WL	-0.095	0.048	-2.00	0.045	0.187
	B-p	-0.095	0.048	-2.00	0.046	0.187
Diet(=lupins) (β_{03})	M-B	-0.202	0.038	-5.37	0.000	0.148
	LZ	-0.202	0.055	-3.69	0.000	0.215
	Warton	-0.202	0.046	-4.39	0.000	0.181
	WL	-0.202	0.048	-4.24	0.000	0.187
	B-p	-0.201	0.054	-3.71	0.000	0.212
Week ≤ 3 (β_1)	M-B	-0.237	0.018	-13.40	0.000	0.069
	LZ	-0.237	0.020	-12.08	0.000	0.077
	Warton	-0.237	0.020	-11.71	0.000	0.079
	WL	-0.237	0.020	-11.91	0.000	0.078
	B-p	-0.237	0.020	-11.81	0.000	0.079
Week > 3 (β_2)	M-B	0.007	0.003	2.31	0.021	0.012
	LZ	0.007	0.003	2.36	0.018	0.011
	Warton	0.007	0.002	2.74	0.006	0.010
	WL	0.007	0.002	2.75	0.006	0.010
	B-p	0.007	0.003	2.34	0.019	0.011
Subset Data (1000 resampling, K=15)						
Diet(=barley) (β_{01})	M-B	4.211	0.110	38.413	0.000	0.430
	LZ	4.211	0.147	28.706	0.000	0.575
	Warton	4.211	0.128	32.999	0.000	0.500
	WL	4.211	0.130	32.500	0.000	0.508
	B-p	4.148	0.126	33.021	0.000	0.492
Diet(=mixed) (β_{02})	M-B	-0.099	0.080	-1.249	0.212	0.312
	LZ	-0.099	0.088	-1.129	0.259	0.345
	Warton	-0.099	0.089	-1.114	0.265	0.350
	WL	-0.099	0.109	-0.911	0.362	0.428
	B-p	-0.099	0.078	-1.264	0.206	0.306
Diet(=lupins) (β_{03})	M-B	-0.215	0.080	-2.691	0.007	0.313
	LZ	-0.215	0.101	-2.131	0.033	0.395
	Warton	-0.215	0.090	-2.397	0.017	0.351
	WL	-0.215	0.110	-1.963	0.050	0.429
	B-p	-0.213	0.085	-2.514	0.012	0.332
Week ≤ 3 (β_1)	M-B	-0.246	0.039	-6.240	0.000	0.155
	LZ	-0.246	0.045	-5.517	0.000	0.175
	Warton	-0.246	0.043	-5.702	0.000	0.169
	WL	-0.246	0.041	-6.050	0.000	0.160
	B-p	-0.234	0.044	-5.327	0.000	0.172
Week > 3 (β_2)	M-B	0.001	0.006	0.134	0.894	0.023
	LZ	0.001	0.007	0.116	0.908	0.027
	Warton	0.001	0.006	0.130	0.896	0.024
	WL	0.001	0.006	0.131	0.896	0.023
	B-p	0.006	0.005	1.131	0.258	0.021

estimators and regularised sandwich estimators perform more robustly, which means they have smaller variations compared to the smooth bootstrap estimators. However, the smooth bootstrap method is more efficient in the longitudinal data analysis with large cluster sizes but has a larger variation. The core in the smooth bootstrap method is the choice of $w_i, i = 1, ..., K$, which is generated from the Pois(1) distribution. Li and Wang [2008] discussed that this resampling method only captures the first-order and second-order moments. How to choose the appropriate w_i for the specific data still needs further investigations.

There are some possible issues need to explore. First, there are many other modified sandwich estimators that are not covered here [Kauermann and Carroll, 2001, Wang et al., 2016]. Second, we compared the regularised sandwich estimator and smooth bootstrapping estimator through extensive simulation studies. However, it is still worth investigating how to select an appropriate method for various aspects of the real applications. A novel theory on the methods for the selection criteria is necessary for the longitudinal dataset with a large cluster size.

Chapter 4

Machine Learning with Mixed-effects in Analyzing Longitudinal Data

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Abstract

In this work, we consider response predictions in longitudinal studies, and investigate the well known statistical mixed-effects model and six different machine learning approaches: decision trees, bagging, random forest, boosting, support-vector machine and neural network. To incorporate random effects in machine learning, we pay particular attention to the combined approach of the random effects and the traditional tree methods. Our extensive simulation studies showed that: (i) when the linear mixed model is correctly specified the statistical model has better performance over the machine learning methods; (ii) the neural network and support-vector machine methods become superior when the random effect component is misspecified; and (iii) if the mean function is misspecified, the RE-EM trees and support-vector machine with a polynomial kernel perform better regardless of whether the random effect component is correctly specified. We also analyzed two real datasets for demonstration.

Keywords: Mixed-effects model; Machine learning; Longitudinal data; Comparison study; Regression tree; support-vector machine; Neural network.

4.1 Introduction

Longitudinal data, which occur frequently in economics, finance, medical science and other fields, are measured repeatedly for each subject. The circumstances under which the measurements are taken cannot be exactly the same. For example, students could be sampled in different classrooms or patients by different doctors. Therefore, the assumption of longitudinal data is that measurements are correlated for the same subjects but independent among different subjects. If the number of measurements from each subject is the same, the datasets are said to contain balanced data; otherwise, the datasets contain unbalanced data. Laird and Ware [1982] introduced the random effects models for longitudinal data because they claimed that a general multivariate model with unrestricted covariance structure was not suited for the analysis of unbalanced data. Mixed-effects models that include both fixed and random effects can handle the correlation in longitudinal data. The fixed effects are parameters related to the levels of the entire population or certain repeatable experimental factors, while the random effects are related to individual experimental units randomly chosen from a population [Pinheiro and Bates, 2000]. An expectation-maximisation (EM) algorithm can be used to determine the maximum likelihood (ML) and restricted maximum likelihood estimation (REML) in the longitudinal data setting [Laird et al., 1987]. Lindstrom and Bates [1988] developed an efficient and computationally stable implementation of the Newton-Raphson (NR) algorithm for obtaining the parameters in mixed-effects models for longitudinal data.

Among the six machine learning methods (trees, bagging, random forest, boosting, support-vector machine and neural network) addressed in this work, the trees method was the most broadly adopted for longitudinal data [Berger and Tutz, 2018, Hajjem et al., 2014, Kundu and Harezlak, 2019, Segal, 1992]. Sela and Simonoff [2012] presented the random effects expectation-maximisation (RE-EM) tree, which combined the structure of mixed-effects models with tree-based methods. They showed that the RE-EM tree had improved predictive power over traditional linear models with random effects and regression trees without random effects. However, Fu and Simonoff [2015] proposed what they claimed were unbiased RE-EM trees by using conditional inference trees instead of classification and regression trees (CARTs). In addition, Loh et al. [2013] had proposed an unbiased regression tree for longitudinal data based on a generalised, unbiased interaction detection and estimation (GUIDE) approach rather

than the traditional CART. Later, Eo and Cho [2014] combined the decision tree and mixed-effects methods for longitudinal data based on GUIDE. A framework for predicting longitudinal change in glycemic control measured by hemoglobin A1c (HbA1c) using mixed effect machine learning was presented by Ngufor et al. [2019].

Li and Wu [2015] claimed that the traditional linear mixed model was inferior to the machine learning methods for both long- and short-term prediction in milk protein data, which is apparently because the linear mixed model is not sufficient to fit this data. This milk protein data was also illustrated by Diggle et al. [2002] using a piecewise model at breakpoint three with an exponential correlation function. However, we noticed that the quadratic term was not necessary for this analysis, and a piecewise mixed-effects model would have better performance. Yang et al. [2016] illustrated the mathematical programming for a piecewise linear regression analysis. They showed that the piecewise regression method achieved better prediction performance than a number of state-of-the-art regression methods, such as random forest (RF), support-vector regression (SVR), K-nearest neighbour (KNN) and so on. Kohli et al. [2018] investigated the estimation of a piecewise mixed-effects model with unknown breakpoints using maximum likelihood. They found that the maximum likelihood estimates were reliable and accurate under the conditions that the observed variables had small residual variance. Based on these, we are interested in comparing the prediction performances between mixed-effects models and machine learning methods for longitudinal data.

In this work, we compared the performances of a piecewise mixed-effects model and six machine learning methods (tree, bagging, random forest, boosting, support-vector machine and neural network) in the prediction of longitudinal data. The remainder of this work is organized as follows. Section 4.2 describes the various methods that we compared in this work. In section 4.3, a description is made of the extensive simulations that were carried out to evaluate the performance of the different methods. Two different kinds of real data (milk protein and wages) were considered as case studies in section 4.4. Section 4.5 presents some conclusions and further discussion.

4.2 Models

In this section, the details of the linear mixed-effects model, tree-based method (including the RE-EM tree), support-vector machine and neural network are introduced. All these machine learning methods can be applied in regression and classification, and we have focused on regression problems in this work.

4.2.1 Linear mixed-effects models

Linear mixed-effects models are an extension of simple linear models by the inclusion of random effects that are used to account for the correlation among measurements within the same subject.

Let response vector $\boldsymbol{Y_i}$ be the $n_i \times 1$ vector $(y_{i1}, \ldots, y_{in_i})^T$, in which y_{ij} is the jth measurement for the ith subject ($i = 1, \ldots, K$, $j = 1, \ldots, n_i$). $\boldsymbol{X_i}$ (of dimension $n_i \times p$) and $\boldsymbol{Z_i}$ (of dimension $n_i \times q$) are the separate fixed-effect and random-effect covariates. $\boldsymbol{\beta}$ is a p-dimensional vector of the fixed effect, and $\boldsymbol{b_i}$ is a q-dimensional vector of the random effect, which are assumed to be Gaussian distributions with mean zero and variance $\boldsymbol{\Psi}$. The formulation of the linear mixed-effects model is as follows:

$$\boldsymbol{Y_i} = \boldsymbol{X_i\beta} + \boldsymbol{Z_i b_i} + \boldsymbol{\epsilon_i}, \tag{4.1}$$
$$\boldsymbol{b_i} \sim N(0, \boldsymbol{\Psi}), \boldsymbol{\epsilon_i} \sim N(0, \sigma^2 \boldsymbol{\Lambda_i}).$$

The within-groups errors $\boldsymbol{\epsilon_i}$ and the random effects $\boldsymbol{b_i}$ are assumed to be independent. It is a special case if $\boldsymbol{\Lambda_i} = \boldsymbol{I}$. Then, it follows that $\boldsymbol{Y_i} \sim N(\boldsymbol{X_i\beta}, \boldsymbol{\Sigma_i})$, where $\boldsymbol{\Sigma_i} = \sigma^2(\boldsymbol{\Lambda_i} + \boldsymbol{Z_i \Psi Z_i^T})$. The matrix form for the model is as follows:

$$\boldsymbol{Y} = \boldsymbol{X\beta} + \boldsymbol{Zb} + \boldsymbol{\epsilon},$$

where $\boldsymbol{Y} = \begin{bmatrix} \boldsymbol{Y_1} \\ \boldsymbol{Y_2} \\ \vdots \\ \boldsymbol{Y_K} \end{bmatrix}$, $\boldsymbol{X} = \begin{bmatrix} \boldsymbol{X_1} \\ \boldsymbol{X_2} \\ \vdots \\ \boldsymbol{X_K} \end{bmatrix}$, $\boldsymbol{b} = \begin{bmatrix} \boldsymbol{b_1} \\ \boldsymbol{b_2} \\ \vdots \\ \boldsymbol{b_K} \end{bmatrix}$, $\boldsymbol{Z} = \mathbf{diag}(\boldsymbol{Z_1}, \boldsymbol{Z_2}, ..., \boldsymbol{Z_K})$, $\boldsymbol{\Lambda} = \mathbf{diag}(\boldsymbol{\Lambda_1}, \boldsymbol{\Lambda_2}, ..., \boldsymbol{\Lambda_K})$, $\boldsymbol{\Sigma} = \mathbf{diag}(\boldsymbol{\Sigma_1}, \boldsymbol{\Sigma_2}, ..., \boldsymbol{\Sigma_K})$ and $\tilde{\boldsymbol{\Psi}} = \mathbf{diag}(\boldsymbol{\Psi}, \boldsymbol{\Psi}, ..., \boldsymbol{\Psi})$. It follows that $\boldsymbol{y}$ are independent multivariate normal vectors with mean $\boldsymbol{X\beta}$ and the covariance matrix is $\boldsymbol{\Sigma} = \sigma^2(\boldsymbol{\Lambda} + \boldsymbol{Z}\tilde{\boldsymbol{\Psi}}\boldsymbol{Z^T})$. Then, the likelihood function is

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2|\boldsymbol{Y}) = p(\boldsymbol{Y}|\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2) = \prod_{i=1}^{K} p(\boldsymbol{Y_i}|\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2),$$

where

$$p(\boldsymbol{Y_i}|\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2) = (2\pi\sigma^2)^{-\frac{n_i}{2}} \exp(\frac{(\boldsymbol{Y_i} - \boldsymbol{X_i\beta})^{\boldsymbol{T}} \boldsymbol{\Sigma_i^{-1}} (\boldsymbol{Y_i} - \boldsymbol{X_i\beta})}{-2\sigma^2})|\boldsymbol{\Sigma_i}|^{-\frac{1}{2}}$$

and $\boldsymbol{\theta}$ represents the parameters in $\tilde{\boldsymbol{\Psi}}$ and $\boldsymbol{\Lambda}$. An EM algorithm can be used to obtain both the ML and REML estimation according to Laird et al. [1987]. The *lme* function of the R-package *nlme* was used to fit the linear mixed model [Pinheiro et al., 2020].

4.2.2 Tree-based method

The decision tree, bagging, random forest, and boosting methods can be grouped together as they are all tree-based methods. CART (classification and regression tree) is a popular algorithm which was proposed by Breiman et al. [1984]. In the tree method, the training data is used to construct a data tree starting at the root node. The predicted space is divided into non-overlapping M regions ($R_1, R_2, ..., R_M$) determined by recursive splitting, which is a top-down and greedy approach [James et al., 2013]. In each region, a constant c_m would be the response. The splitting we choose will cause the largest reduction in the mean square error. We can split recursively until the mean square error reaches some threshold. The predicted response for a test data point is the mean of the training observations in the region to which that test point belongs. For each test data point that falls in the same region on a path starting from the root node until reaching a terminal (leaf) node, the prediction would be the same. A usual strategy to

fit a single tree is to grow a large tree and then trim it by weakest link pruning. The R-package *tree* was used to implement the above process in this work [Ripley, 2019].

Trees can be displayed graphically and are easy to explain but can be subject to overfitting. Also, trees are not robust, which means small changes in the training data can cause very different series of splits. Ensemble decision tree methods, including bagging, random forests and boosting, combine many decision trees to produce better predictive performances than a single decision tree.

Bagging is the application of the bootstrap procedure to decision trees in order to lower the variance. There are three main steps: firstly, generate random sub-samples of the training data set with replacement; secondly, train the decision tree method on each sample; and thirdly, calculate the average prediction from each model using the test data. The average prediction would be the final prediction for each test data point. Bagging will improve the prediction accuracy compared to the tree method at the cost of interpretability.

Random forest is a popular tree-based ensemble method that builds a large collection of decorrelated trees and then averages them based on the bagging [Breiman, 2001]. When building this algorithm, a random sample of features is chosen as split candidates from the full set of predictors rather than using all the features in bagging. This forces each split to consider only a subset of the predictors, which is reasonable, especially when there is a very strong predictor in the training data set. After a certain number of trees are grown, the predictor is obtained by the average (for regression) or the majority vote (for classification) [James et al., 2013]. This algorithm contains four main parameters: total number of observations, total number of predictor variables, randomly chosen features for determining the decision tree and the total number of decision trees. The R-package *randomForest* was used to implement the algorithm of bagging and random forest [Liaw and Wiener, 2002].

The different trees based on the bootstrapped data are independent in bagging. Boosting works in a similar way to bagging, but the difference is the trees are constructed sequentially, which means that the growth of each tree depends on the trees that have already been constructed. It is a forward stagewise approach. Boosting regression trees (BRT) have three parameters: the number of trees, the shrinkage parameter that controls the learning rate and the number of splits in each tree that determines the complexity of the boosted ensemble. The BRT algorithm has three main steps: firstly, a regression tree is fitted; secondly, another tree is

fitted to the residuals of the first tree; and thirdly, the model is updated to have two trees with a shrinkage parameter (this last step is repeated hundreds or thousands of times). The final model is a linear combination of these trees. The R-package *gbm* was used [Greenwell et al., 2019].

4.2.3 Mixed-effects regression tree

Segal [1992] was the first to apply regression trees to longitudinal data. The mixed-effects tree method we have used in this work, the RE-EM tree, was proposed by Sela and Simonoff [2012]. The notation in an RE-EM tree follows the linear mixed-effects model:

$$\boldsymbol{Y_i} = f(\boldsymbol{X_i}) + \boldsymbol{Z_i b_i} + \boldsymbol{\epsilon_i},$$

in which the $\boldsymbol{Y_i}, \boldsymbol{X_i}, \boldsymbol{Z_i}, \boldsymbol{b_i}$ and $\boldsymbol{\epsilon_i}$ are the same with equation (4.1). If f is a linear function, $f(\boldsymbol{X_i}) = \boldsymbol{X_i}\boldsymbol{\beta}$, then the model is a linear mixed model. Generally, this f function can be estimated by a tree method when the random effects $\boldsymbol{b_i}$ are known. However, when neither the fixed effects nor the random effects are known, an iterative two-step process is utilised. Firstly, the random effects $\hat{\boldsymbol{b}}_{\boldsymbol{i}}$ is given as zero initially, and a regression tree is used to estimate function f based on $\boldsymbol{Y_i} - \boldsymbol{Z_i}\hat{\boldsymbol{b}}_{\boldsymbol{i}}$. A linear mixed-effects model is then fitted to estimate the random effects based on the tree regression results: $y_{ij} = Z_{ij}\boldsymbol{b_i} + I(\boldsymbol{X_{ij}} \in g_p)\mu_p + \epsilon_{ij}$, in which $I(\boldsymbol{X_{ij}} \in g_p)\mu_p$ means the estimated value for y_{ij} at terminal node g_p. The algorithm will not stop until the estimates of random effects $\hat{\boldsymbol{b}}_{\boldsymbol{i}}$ converge. We used R package *REEMtree* [Sela and Simonoff, 2012] in this work.

4.2.4 Support-vector machine

The initial idea of a support-vector machine (SVM) is to construct a linear partition of the high-dimensional space into two sub-spaces for classification or regression [Schölkopf et al., 2002]. We will focus on the regression application in this work. Given the training data $(\boldsymbol{x_1}, y_1), (\boldsymbol{x_2}, y_2), ..., (\boldsymbol{x_N}, y_N)$, the prediction is shown as a linear function $f(\boldsymbol{x}) = \boldsymbol{\omega^T x} + b_0$, and the error function is

$$\frac{1}{2}||\boldsymbol{\omega}||^2 + C\sum_{k=1}^{N} \ell_\epsilon(f(\boldsymbol{x_k}) - y_k),$$

where $\ell_\epsilon(z) = \max\{0, |z| - \epsilon\}$ is the ϵ-insensitive loss. After minimising the error function, the solution is

$$f(\boldsymbol{x}) = \sum_{k=1}^{N} (\hat{\alpha}_k - \alpha_k) \boldsymbol{x_k^T} \boldsymbol{x} + b_0, \tag{4.2}$$

where $\hat{\alpha}_k$ and α_k are Lagrange multipliers. The nonzero Lagrange multipliers that indicate the training vector makes $(\hat{\alpha}_k - \alpha_k) \neq 0$ in (4.2) are called support-vectors. Obviously, the non-support-vectors do not contribute directly to the solution . It often happens that the data are not linearly separable. A kernel function is then used to transform the nonlinear system in the input space to a linear system in the feature space. Popular kernel functions are polynomial kernel, radial kernel and so on [James et al., 2013]. We will also include the support-vector regression with a polynomial kernel in our simulations. The *svm* function of the R-package *e1071* was used [Meyer et al., 2019].

4.2.5 Neural network

In this work, we consider the neural network as a multilayer perceptron (MLP), which is a class of feedforward artificial neural network. The multilayer perceptron is a popular network for classification and regression. The formula is as follows:

$$\hat{y} = \sum_{k=0}^{NH} v_k g_k(\boldsymbol{x}),$$

where v_k are the weights, $g_k(\boldsymbol{x})$ are the hidden functions (or hidden units), NH is the number of hidden nodes, $\boldsymbol{x}$ is the input vector and $\hat{y}$ is the output. Here, the hidden function g_0 takes a fixed value of one to allow a constant term in the equation. The sigmoid function is commonly used: $g_k(u) = 1/(1 + \exp(-u))$. Also, the tangent hyperbolicus function $\tanh x = (e^x - e^{-x})/(e^x + e^{-x})$ is used in the simulation. The general approach to fit this method is minimizing the sum-of-squared errors by gradient descent, which is called back-propagation. The R-package *neuralnet* was used to implement the neural network algorithm [Fritsch et al., 2019].

4.3 Simulation studies

In this section, we describe the investigation of the performances of linear mixed-effects models and machine learning methods through extensive simulations.

4.3.1 Design of simulations

We generated data from the mixed model:

$$y_{ij} = \mu_{ij} + \boldsymbol{b}_i + \epsilon_{ij},$$

where $\mu_{ij} = \beta_0 + \beta_1 x_{ij} + \beta_2 t_i + \beta_3 x_{ij}^2$ is the overall mean response, $\boldsymbol{b}_i$ is the random effects from the normal distribution $N(0, \sigma_b^2)$ and ϵ_{ij} come from an iid normal distribution $N(0, \sigma^2)$. In addition, x_{ij} is sampled from the uniform distribution $(-1, 1)$, and t_i is sampled from the uniform distribution $(0, 1)$. In our simulation, there are two different true mean function: $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)^T = (0.5, 1, 1, 0)^T$ and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)^T = (0.5, 1, 1, -5)^T$, which demonstrate the true mean function is linear and quadratic respectively. Also $\sigma = 1$ and $\sigma_b = 1$ in this simulation.

There is another way to generate the correlated data from the following model:

$$y_{ij} = \mu_{ij} + \epsilon_{ij}, \;\; i = 1, \ldots, K, \;\; j = 1, \ldots, n,$$

where $\mu_{ij} = \beta_0 + \beta_1 x_{ij} + \beta_2 t_i + \beta_3 x_{ij}^2$, x_{ij} is sampled from the uniform distribution $(0, 1)$, and t_i is sampled from the binary distribution. In order to compare the performances between different methods under a scenario in which the mean function was correctly specified and misspecified, we have set different values for $\boldsymbol{\beta}$. In the model in which the mean function was correctly specified, $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)^T = (0, 0.5, 1, 0)^T$. This indicates the true mean function of the simulated data was linear (without a quadratic term). In contrast, $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)^T = (0, 0.5, 1, -5)^T$ was used when the mean function was misspecified, which indicates the data were generated from a quadratic model. We used the linear mean function in various prediction models. We had two different scenarios for ϵ_{ij}: the first is ϵ_{ij} were correlated with an exchangeable structure, in other words, $\text{cor}(\epsilon_{ij}, \epsilon_{ij'}) = 0.5$ if $j \neq j'$; and the second is for each i, $(\epsilon_{i1}, \ldots, \epsilon_{in})$ were correlated with an autoregressive AR(1) structure that also had a

correlation coefficient of 0.5. It is worth noting that when the data were generated from the linear mean function with first scenario (i.e., the correlation structure was exchangeable), the linear mixed-effects model with a random intercept is the true model. Otherwise, when the data were correlated with the AR(1) structure, the linear mixed-effects model is not the true model even if the mean function of simulated data is linear.

There are few references about how to measure the predictive power of methods for longitudinal data. The stratified cross-validation method cannot be used directly because the observations from longitudinal data contain sequences. Based on Sela and Simonoff [2012], three different ways were utilised: (i) predicting the future 30% of observations based on the previous 70% of observations for K different subjects, denoted as *future observation*; (ii) predicting another new $K/2$ objects based on the previous K different subjects, denoted as *new object*; and (iii) predicting a future 30% of observations for new $K/2$ objects based on the previous K different subjects and the previous 70% of observations in the new $K/2$ subjects, denoted as *future new observation*. In this case, there were 100 subjects (K = 100), and each subject was observed 10 times (n = 10). The results from 1000 simulations for each method are presented in Tables 4.1 and 4.2.

We also proposed another one-step prediction and two-step prediction in order to see the performances of different methods in real-time prediction. In this case, $K = 100$ and $n = 5$. This was a bit similar to the future observation method described above but did not just consist of the overall prediction. We can obtain the prediction performance at the time of each observation. In the one-step prediction, we used the first observation to predict the second observation; the first two observations were then used to predict the third, and so on. In contrast, the first observation was used to predict the third observation in the two-step prediction. The results based on 1000 simulations are presented in Tables 4.3, 4.4, 4.5 and 4.6.

To measure the prediction performance of the different methods, the root mean square error (RMSE) was used:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{K}\sum_{j=1}^{n}(\hat{y}_{ij} - y_{ij})^2}{K * n}},$$

where y_i is the measured value, and $\hat{y}_i$ is the predicted value. Because the true values of μ_i are known in the simulations and the curious is about how close the predictions would be to the true values without noise, we defined another metric, the true root mean square error (TRMSE), to

measure the prediction performance as follows:

$$\text{TRMSE} = \sqrt{\frac{\sum_{i=1}^{K}\sum_{j=1}^{n}(\hat{y}_{ij} - \mu_{ij})^2}{K * n}}.$$

4.3.2 Simulation results

The objective of our extensive simulations was to compare the prediction performances among the different methods in the longitudinal data. In Table 4.1, we can see that the linear mixed model performed the best when the mean function was specified correctly in terms of predicting future observations and future new observations. However, the support vector machine and neural network methods have better performance when we need to predict the observations from new objects. It was expected that the support vector regression with polynomial kernel and neural network with hyperbolic tangent activation function would also have better performance if the mean function was misspecified. However, the RE-EM trees was the best when the mean function was misspecified, which means that it is necessary to combine the random effects with the machine learning methods. The TRMSE values that measured the differences between the predictions and true values were usually smaller than the RMSE values. According to the TRMSE values, the support-vector machine with a linear kernel had the best performance if the mean function was correctly specified, and the tree method was the best method when the mean function was misspecified (see Table 4.1).

The performance of different methods in simulated data generated from the exchangeable and AR(1) correlation structure is presented in Table 4.2. In Table 4.2(a), because the linear mixed model is the true model when the correlation structure is exchangeable, it was not a surprise to see that the linear mixed model performed the best when the mean model was specified correctly and in terms of predicting future observations and future new observations. The support-vector machine with a linear kernel and the neural network also had good performances when predicting the observations from new object. However, when the correlation structure was AR(1), which means that the random effect component was misspecified, the support vector machine and neural network method had better performance.

If the mean function was misspecified, the RE-EM trees and support-vector machine with a polynomial kernel had the advantages regardless of whether the random effect component is misspecified or not (see Table 4.2(b)). It was not a surprise to see that the support-vector

Table 4.1: The performances of different methods in simulated data generated from a mixed-effects model

	Correct specification of the mean						Misspecification of the mean					
	future		new		fu-new		future		new		fu-new	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	**1.059**	0.927	1.699	0.943	**1.059**	0.931	1.880	1.707	2.229	1.725	1.879	1.704
tree	1.418	0.461	1.480	0.434	1.426	0.371	1.494	0.565	**1.518**	0.546	1.495	0.526
re-em	1.100	0.970	1.742	1.018	1.089	0.963	**1.142**	1.016	1.780	1.082	**1.114**	0.990
bag	1.324	0.704	1.589	0.731	1.347	0.679	1.413	0.688	1.580	0.699	1.433	0.659
rf	1.303	0.643	1.558	0.660	1.333	0.615	1.314	0.674	1.576	0.689	1.343	0.633
boost	1.303	1.164	1.814	1.143	1.245	1.086	1.303	1.167	1.825	1.148	1.244	1.081
svm	1.406	0.156	**1.420**	0.149	1.406	0.122	2.058	1.512	2.068	1.508	2.056	1.505
svmk	1.429	0.308	1.445	0.303	1.428	0.281	2.083	1.548	2.094	1.542	2.078	1.534
nn	1.406	0.166	**1.420**	0.155	1.406	0.127	1.711	0.978	2.301	1.814	2.297	1.812
nntanh	1.406	0.164	**1.420**	0.154	1.406	0.127	1.765	1.059	2.291	1.802	2.287	1.799

future: future observations; new: new objects; fu-new: future new observations;
RMSE: root mean square error; TRMSE: true root mean square error;
lme: linear mixed-effects model; re-em: RE-EM trees with random intercept; tree: decision tree method;
bag: bagging method; rf: random forest method; boost: boosting method;
svm: support-vector regression with linear kernel; svmk: support-vector regression with polynomial kernel;
nn: neural network method with logistic activation function;
nntanh: neural network method with hyperbolic tangent activation function.

machine with a polynomial kernel had smaller RMSE values than when a linear kernel was used if the mean function was misspecified. The results according to RMSE values are a slightly different from the conclusions according to RMSE values. The support-vector machine with a linear kernel had the best performance if the mean function was correctly specified, and the support-vector machine with a polynomial kernel was the best method when the mean function was misspecified.

The results from the one-step and two-step predictions are presented in Tables 4.3, 4.4, 4.5 and 4.6, respectively. Again, the linear mixed model had the best performance both in the one-step and two-step predictions when the mean function was correctly no matter how the correlated data was generated. It is noted that in the simulated data generated from the mixed-effects model, random forest and mixed-effects trees had better performance when the mean function was misspecified. We can also conclude that the RE-EM trees and support-vector machine with a polynomial kernel performed well when the mean function was misspecified. The performances between the one-step and two-step predictions were different when the mean function was specified correctly while the correlation structure was different, see Table 4.5(a)(ii)

Table 4.2: The performances of different methods in simulated data which correlated with the exchangeable and AR(1) structure

(a) Correct specification of the mean

	EXC						AR(1)					
	future		new		fu-new		future		new		fu-new	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	**0.748**	0.661	1.205	0.671	**0.750**	0.660	1.026	0.406	1.062	0.348	1.016	0.369
tree	1.005	0.178	1.014	0.167	1.007	0.153	1.012	0.163	1.013	0.151	1.009	0.142
re-em	0.763	0.677	1.213	0.687	0.765	0.675	1.036	0.431	1.072	0.378	1.027	0.397
bag	1.148	0.590	1.162	0.589	1.156	0.590	1.161	0.587	1.161	0.588	1.159	0.586
rf	0.999	0.143	1.009	0.133	1.001	0.114	1.008	0.127	1.008	0.115	1.004	0.100
boost	1.154	0.596	1.134	0.532	1.105	0.479	1.164	0.592	1.133	0.529	1.109	0.479
svm	0.997	0.115	**1.006**	0.105	0.999	0.089	**1.004**	0.092	**1.005**	0.083	**1.002**	0.069
svmk	0.999	0.135	1.007	0.121	1.000	0.103	1.006	0.115	1.007	0.102	1.003	0.086
nn	0.996	0.115	**1.006**	0.107	0.999	0.090	**1.004**	0.094	**1.005**	0.084	**1.002**	0.071
nntanh	0.996	0.115	**1.006**	0.107	0.999	0.090	**1.004**	0.093	**1.005**	0.083	**1.002**	0.070

(b) Misspecification of the mean

	EXC						AR(1)					
	future		new		fu-new		future		new		fu-new	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	1.717	1.585	1.886	1.599	1.714	1.584	1.794	1.570	1.814	1.515	1.802	1.513
tree	1.122	0.535	1.127	0.527	1.113	0.514	1.128	0.535	1.126	0.523	1.122	0.512
re-em	**0.823**	0.744	1.341	0.889	**0.803**	0.719	**1.059**	0.710	1.145	0.561	1.075	0.506
bag	1.146	0.579	1.153	0.581	1.144	0.578	1.150	0.579	1.154	0.578	1.149	0.578
rf	1.225	0.724	1.231	0.723	1.221	0.720	1.227	0.724	1.229	0.719	1.229	0.717
boost	1.154	0.597	1.131	0.534	1.097	0.480	1.160	0.596	1.129	0.530	1.103	0.477
svm	1.810	1.518	1.815	1.517	1.806	1.514	1.811	1.518	1.811	1.512	1.812	1.510
svmk	1.070	0.384	**1.062**	0.352	1.056	0.357	1.070	0.376	**1.064**	0.359	**1.063**	0.355
nn	1.527	1.161	1.525	1.156	2.319	2.096	1.526	1.158	1.528	1.159	1.530	1.152
nntanh	1.595	1.247	1.600	1.248	2.312	2.088	1.596	1.249	1.596	1.244	1.591	1.231

EXC: exchangeable correlation structure; AR(1): first auto-regressive correlation structure;
The other notations are the same as Table 4.1.

and Table 4.6(a)(ii). In the one-step prediction, the linear mixed model was still comparable but not for the two-step prediction. The neural network method had the best performance when the random effect component was misspecified in the two-step prediction.

Table 4.3: The one-step prediction for different methods in simulated data generated from a mixed-effects model

(a) Correct specification of the mean								
	T2		T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	**1.345**	1.243	1.156	0.817	**1.117**	0.869	**1.092**	0.894
tree	1.596	0.844	1.518	0.738	1.466	0.624	1.441	0.552
re-em	1.493	1.380	**1.243**	0.912	1.178	0.933	1.146	0.949
bag	1.482	0.666	1.427	0.659	1.398	0.669	1.374	0.680
rf	1.439	0.592	1.388	0.596	1.362	0.610	1.342	0.622
boost	1.664	1.213	1.522	1.175	1.446	1.181	1.398	1.182
svm	1.428	0.254	1.415	0.206	1.408	0.185	1.406	0.173
svmk	1.472	0.447	1.446	0.369	1.436	0.344	1.431	0.329
nn	1.441	0.326	1.419	0.242	1.410	0.209	1.408	0.190
nntanh	1.442	0.322	1.419	0.237	1.409	0.202	1.408	0.187
(b) Misspecification of the mean								
	T2		T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	2.390	2.342	1.981	1.634	1.938	1.654	1.922	1.665
tree	1.623	0.849	1.542	0.706	1.516	0.641	1.502	0.603
re-em	1.687	1.535	**1.358**	1.003	**1.266**	1.001	**1.215**	1.005
bag	1.538	0.729	1.489	0.692	1.470	0.684	1.446	0.678
rf	**1.490**	0.720	1.414	0.666	1.382	0.658	1.354	0.655
boost	1.743	1.319	1.543	1.195	1.461	1.182	1.402	1.180
svm	2.086	1.557	2.079	1.535	2.061	1.522	2.064	1.516
svmk	2.148	1.639	2.120	1.591	2.091	1.563	2.091	1.553
nn	1.746	1.046	1.721	0.997	1.715	0.989	1.716	0.987
nntanh	1.809	1.136	1.789	1.101	1.769	1.075	1.771	1.071

4.4 Application to real data

4.4.1 Case study 1: milk protein data

In this data set, milk was collected weekly from 79 Australian cows and analyses for its protein content. There are three diets: 25 cows received a barley diet, 27 cows a mixture of barley and lupins, and 27 cows a diet of lupins only. The observation period of each cow is not necessarily the same and each cow is observed for between 12 weeks and 19 weeks.

It appears from the Figure 3.6 that barely gives higher values than the mixture, which in turn have higher values than lupins alone. The mean response profiles are approximately parallel, showing an initial sharp decline associated with a settling-in period, followed by an

Table 4.4: The two-step prediction for different methods in simulated data generated from a mixed-effects model

(a) Correct specification of the mean						
	T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	**1.339**	1.246	**1.163**	0.819	**1.119**	0.866
tree	1.589	0.851	1.521	0.740	1.472	0.631
re-em	1.487	1.384	1.248	0.912	1.178	0.931
bag	1.473	0.668	1.432	0.666	1.403	0.670
rf	1.430	0.594	1.392	0.602	1.367	0.612
boost	1.657	1.221	1.523	1.182	1.452	1.177
svm	1.422	0.264	1.417	0.211	1.412	0.191
svmk	1.466	0.450	1.447	0.376	1.437	0.345
nn	1.434	0.326	1.422	0.247	1.414	0.212
nntanh	1.433	0.320	1.423	0.243	1.414	0.208
(b) Misspecification of the mean						
	T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	2.384	2.330	1.982	1.636	1.944	1.657
tree	1.619	0.845	1.554	0.710	1.526	0.644
re-em	1.683	1.533	**1.368**	1.007	**1.274**	1.004
bag	1.536	0.724	1.498	0.694	1.473	0.685
rf	**1.491**	0.716	1.422	0.667	1.385	0.657
boost	1.740	1.320	1.550	1.199	1.462	1.181
svm	2.095	1.563	2.083	1.538	2.069	1.524
svmk	2.155	1.645	2.119	1.587	2.101	1.570
nn	1.743	1.036	1.725	0.995	1.719	0.987
nntanh	1.817	1.143	1.794	1.097	1.782	1.082

approximately constant mean response through the following period and a slow rise towards the end.

Piecewise regression is a special type of linear regression that arises when a single line is not sufficient to model a data set. Piecewise regression breaks the domain into potentially many 'segments' and fits a separate line through each one. Breakpoints are the values where the slope of the linear function changes. The value of the breakpoints are unknown and must be estimated. In some cases, the breakpoints can be specified by us. In other words, it is obvious to the naked eyes when one linear trends give way to other. However, this is not fit for all the cases. For some data set, it is not easy to detect the breakpoints just from eyes. In statistics, the popular way is compare the errors with different breakpoints, which means minimize the errors

between each segment's regression and the observed data points.

Diggle et al. [2002] used the following mean response profiles model:

$$\mu_i = \begin{cases} \beta_{0i} + \beta_1 t & \text{if } t \leq 3, \\ \beta_{0i} + 3\beta_1 + \beta_2(t-3) + \beta_3(t-3)^2 & \text{if } t > 3, \end{cases}$$

where $i = 1, 2, 3$ denotes treatment group with an exponential correlation function $\text{Cov}(\epsilon_j, \epsilon_k) = \sigma^2 \exp(-\phi|t_j - t_k|)$.

But the quadratic term was not significant and the breakpoint was not necessarily to be a integer. According to the mean square error, the breakpoint we chose for this milk protein data is 2.6. So we use the piecewise mixed model with the mean response profiles model as follows:

$$\mu_i = \begin{cases} \beta_{0i} + \beta_1 t + b_{i1} + t b_{i2} & \text{if } t \leq 2.6, \\ \beta_{0i} + 2.6\beta_1 + \beta_2(t-2.6) + b_{i1} + t b_{i2} & \text{if } t > 2.6, \end{cases}$$

where $i = 1, 2, 3$ denotes treatment group. The corresponding one-step prediction and two-step prediction results are presented in Table 4.7(a). We can see that the piecewise linear mixed model has the best performance in one-step prediction. RE-EM trees also has advantages. Tree-based methods have smaller RMSE values than support-vector machine and neural network methods.

4.4.2 Case study 2: wages data

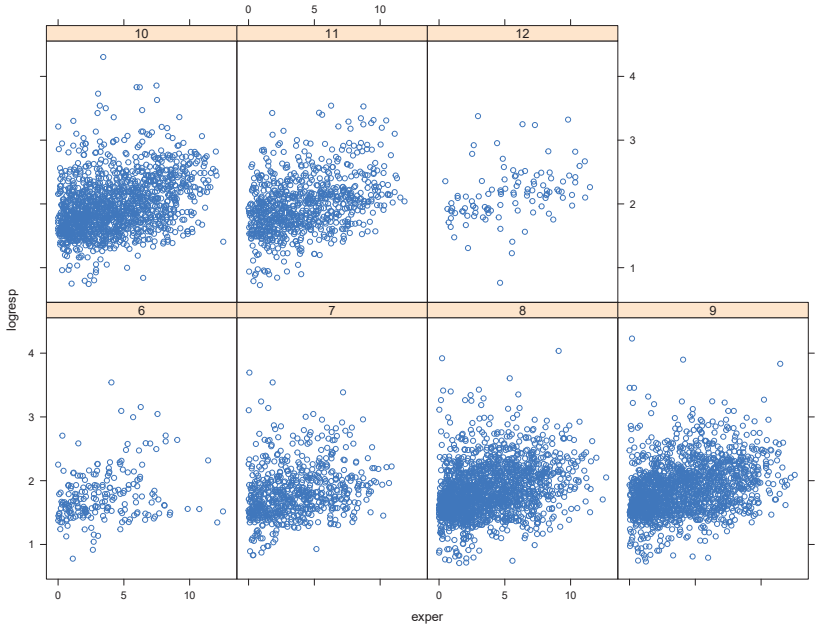
Wages data came from the National Longitudinal Survey of Youth (NLSY), which was previously studied by Singer and Willett [2003], Eo and Cho [2014] and Fu and Simonoff [2015]. The data has the information of 888 individuals' hourly wage, which is the response variable. Each individual has the different observation times, ranged from 1 to 13. Figure 4.1 present the plots of the time variable (exper, which is the duration of the working experience) and the log of wages at different race and hgc. The individual's races are White, Black and Hispanic. The variable hgc means the highest grade completed by the individual. We used the eight cross-validation method to compare the prediction performances between statistical models and

machine learning methods. According to Table 4.7(b), RE-EM methods has the smallest RMSE. Tree-based methods, support-vector machine and neural networks cannot compare.

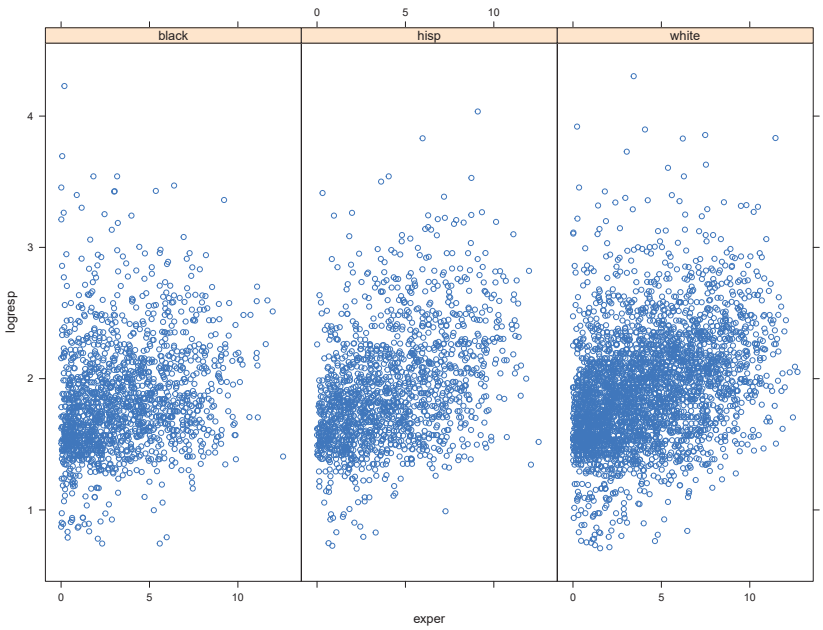
4.5 Conclusions and discussion

We have presented the performances of the statistical models and six machine learning methods for the longitudinal data. The machine learning methods we used in the work were all basic forms without modifications. The results showed that the statistical models were better than the machine leaning methods when the mean function of models were correctly specified. In the simulations, we knew the truth model. The performances under the scenarios of the mean function and the different correlation structures (exchangeable and AR(1)) were compared. Otherwise, even with the milk dataset (a real world dataset), the statistical model (especially, the piecewise linear mixed model) still performed better than the machine learning methods. This means that the piecewise linear mixed model provided an adequate fit to the original data. It can also be concluded that the model diagnostics are very important before making decisions regarding performance. One thing we should mention is that statistical models usually have more assumptions than machine learning methods. However, this is a double-edged sword. Machine learning methods are usually recognised as having a 'black box' aspect, which means there is less attention paid to the processes between their inputs and outputs. Real data sets are usually complex, and it is worthwhile to investigate more about the data before definitive decisions are made.

There are few references about how to measure the predictive power of methods in longitudinal data. The prediction accuracy according to a cross-validation method are not reasonable because longitudinal data are always sequential. In this work, we used one-step and two-step prediction along with future observation, new object and future new observation prediction. The performances of all kinds of methods were demonstrated comprehensively. In addition, we also presented the differences between RMSE and TRMSE values in the predictions. It was not surprising to see that the TRMSE values were smaller than the RMSE values in one prediction because we measured that differences between the predictions and true values without noise.



(a) Different hgc



(b) Different race

Figure 4.1: The wages data

However, this is not always true. According to the formula of mean square error,

$$\begin{aligned}
\text{RMSE} &= \sqrt{\text{ave}_{x\in\text{test}}(\hat{y}-y)^2} \\
&= \sqrt{\text{ave}_{x\in\text{test}}(\hat{y}-(\mu+\epsilon))^2} \\
&= \sqrt{\text{ave}_{x\in\text{test}}[(\hat{y}-\mu)^2-2(\hat{y}-\mu)\epsilon+\epsilon^2]} \\
&= \sqrt{\text{TMSE}+\text{ave}_{x\in\text{test}}[-2(\hat{y}-\mu)\epsilon+\epsilon^2]}
\end{aligned}$$

where $\hat{y}$ is the prediction value, y is the observed value, μ is the true value and ϵ is the error. TMSE is the square of TRMSE. If the error is independent of the observations, in other words, $\text{ave}_{x\in\text{test}}[-2(\hat{y}-\mu)\epsilon]=0$, the RMSE values would be larger than the TRMSE values. Otherwise, in our prediction (especially in the case of future observation prediction), we used the previous observations in one object to predict the following observations.

There are still some limitations in this study. The predictions between the different methods were discussed rather than the parameter estimates and inferences in the longitudinal data. Misspecified models, including the mean function were considered in this work. Wang and Lin [2005] also investigated the effects of variance function and correlation structure misspecification in the analysis of longitudinal data. In this work, we only investigated the popular exchangeable and AR(1) correlation structures that are appropriate for equally spaced (in time) longitudinal data. However, unequally spaced observations and time-dependent correlated errors deserves more attention by researchers [Nunez-Anton and Woodworth, 1994]. It would be of great interest to evaluate machine learning performance in these settings. There are also other modified methods that combine mixed-effects models and tree methods [Eo and Cho, 2014, Fu and Simonoff, 2015, Loh et al., 2013] that deserve further examination. An extended comparison with more recently developed machine learning methods, such as deep learning, would be of interest. As for prediction in longitudinal data, one-step and two-step prediction are proposed but the distribution of the data must be consistent even as it increases.

Table 4.5: The one-step prediction for different methods in simulated data which correlated with the exchangeable and AR(1) structure

(a) Correct specification of the mean

	(i) EXC							
	T2		T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	**0.947**	0.877	**0.823**	0.582	**0.792**	0.612	**0.775**	0.632
tree	1.095	0.463	1.052	0.343	1.025	0.260	1.015	0.219
re-em	0.962	0.897	0.836	0.599	0.805	0.629	0.790	0.649
bag	1.143	0.573	1.150	0.583	1.148	0.584	1.153	0.588
rf	1.019	0.242	1.011	0.194	1.004	0.172	1.004	0.160
boost	1.262	0.785	1.269	0.799	1.239	0.748	1.214	0.696
svm	1.010	0.188	1.003	0.146	0.999	0.132	0.999	0.122
svmk	1.023	0.247	1.010	0.186	1.004	0.163	1.003	0.150
nn	1.008	0.187	1.003	0.148	0.999	0.132	0.999	0.124
nntanh	1.010	0.193	1.004	0.152	1.000	0.135	0.999	0.125
	(ii) AR(1)							
	T2		T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	**0.952**	0.883	**0.918**	0.579	**0.943**	0.533	**0.962**	0.493
tree	1.096	0.470	1.053	0.345	1.028	0.261	1.018	0.211
re-em	0.968	0.902	0.930	0.598	0.954	0.552	0.972	0.514
bag	1.144	0.577	1.151	0.585	1.152	0.586	1.154	0.587
rf	1.019	0.245	1.011	0.199	1.007	0.173	1.007	0.157
boost	1.265	0.787	1.273	0.800	1.242	0.749	1.216	0.699
svm	1.009	0.191	1.004	0.153	1.001	0.132	1.003	0.120
svmk	1.023	0.251	1.011	0.192	1.006	0.165	1.007	0.148
nn	1.010	0.194	1.004	0.154	1.001	0.132	1.002	0.119
nntanh	1.011	0.198	1.006	0.159	1.002	0.134	1.002	0.120

(b) Misspecification of the mean

	(i) EXC							
	T2		T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	2.208	2.180	1.776	1.558	1.748	1.559	1.739	1.567
tree	1.205	0.687	1.155	0.602	1.142	0.571	1.138	0.558
re-em	1.307	1.201	**1.026**	0.784	**0.937**	0.760	**0.889**	0.752
bag	1.157	0.599	1.146	0.584	1.145	0.574	1.149	0.576
rf	1.255	0.771	1.232	0.738	1.226	0.728	1.226	0.724
boost	1.326	0.881	1.282	0.818	1.242	0.748	1.218	0.702
svm	1.866	1.583	1.832	1.543	1.815	1.526	1.817	1.526
svmk	**1.148**	0.553	1.102	0.463	1.090	0.428	1.080	0.408
nn	1.585	1.237	1.548	1.191	1.533	1.174	1.535	1.172
nntanh	1.661	1.329	1.618	1.277	1.603	1.263	1.599	1.250
	(ii) AR(1)							
	T2		T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	2.203	2.173	1.801	1.564	1.790	1.546	1.791	1.543
tree	1.203	0.683	1.157	0.601	1.146	0.573	1.138	0.554
re-em	1.308	1.201	**1.089**	0.778	**1.072**	0.715	**1.066**	0.669
bag	1.158	0.596	1.148	0.580	1.147	0.575	1.149	0.575
rf	1.254	0.764	1.236	0.736	1.230	0.727	1.228	0.725
boost	1.322	0.876	1.285	0.818	1.245	0.752	1.217	0.701
svm	1.862	1.575	1.835	1.541	1.822	1.530	1.819	1.527
svmk	**1.151**	0.560	1.109	0.470	1.091	0.430	1.085	0.407
nn	1.584	1.229	1.551	1.186	1.540	1.175	1.543	1.179
nntanh	1.654	1.316	1.623	1.277	1.611	1.262	1.601	1.255

Table 4.6: The two-step prediction for different methods in simulated data which correlated with the exchangeable and AR(1) structure

(a) Correct specification of the mean

	(i) EXC					
	T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	**0.950**	0.880	**0.821**	0.577	**0.789**	0.612
tree	1.098	0.468	1.052	0.349	1.022	0.259
re-em	0.966	0.899	0.834	0.594	0.803	0.628
bag	1.146	0.576	1.149	0.588	1.148	0.585
rf	1.021	0.244	1.008	0.196	1.002	0.173
boost	1.268	0.788	1.271	0.802	1.239	0.747
svm	1.011	0.195	1.001	0.149	0.998	0.134
svmk	1.024	0.251	1.007	0.191	1.003	0.167
nn	1.010	0.190	1.000	0.149	0.997	0.136
nntanh	1.013	0.197	1.002	0.154	0.998	0.138
	(ii) AR(1)					
	T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	1.154	0.879	1.044	0.580	1.044	0.536
tree	1.106	0.473	1.058	0.344	1.031	0.258
re-em	1.170	0.899	1.055	0.598	1.054	0.555
bag	1.153	0.577	1.157	0.585	1.154	0.585
rf	1.027	0.244	1.017	0.194	1.011	0.168
boost	1.273	0.787	1.279	0.801	1.246	0.747
svm	1.016	0.192	1.010	0.150	1.007	0.129
svmk	1.029	0.251	1.016	0.191	1.011	0.162
nn	**1.015**	0.190	**1.009**	0.149	**1.006**	0.128
nntanh	1.017	0.196	1.010	0.153	1.007	0.130

(b) Misspecification of the mean

	(i) EXC					
	T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	2.195	2.173	1.775	1.560	1.762	1.561
tree	1.207	0.684	1.154	0.597	1.141	0.571
re-em	1.312	1.204	**1.029**	0.779	**0.938**	0.763
bag	1.159	0.597	1.147	0.576	1.145	0.574
rf	1.250	0.764	1.231	0.736	1.233	0.731
boost	1.325	0.879	1.278	0.810	1.246	0.753
svm	1.853	1.571	1.826	1.539	1.826	1.529
svmk	**1.154**	0.558	1.105	0.467	1.091	0.425
nn	1.581	1.229	1.551	1.192	1.544	1.179
nntanh	1.650	1.317	1.621	1.281	1.612	1.266
	(ii) AR(1)					
	T3		T4		T5	
	RMSE	TRMSE	RMSE	TRMSE	RMSE	TRMSE
lme	2.304	2.178	1.822	1.560	1.818	1.544
tree	1.204	0.680	1.160	0.599	1.146	0.572
re-em	1.441	1.197	1.179	0.774	1.155	0.717
bag	1.158	0.595	1.149	0.577	1.150	0.577
rf	1.252	0.763	1.234	0.735	1.233	0.727
boost	1.321	0.872	1.284	0.812	1.250	0.755
svm	1.861	1.575	1.829	1.539	1.828	1.529
svmk	**1.151**	0.560	**1.106**	0.464	**1.087**	0.424
nn	1.580	1.223	1.550	1.189	1.541	1.175
nntanh	1.651	1.315	1.619	1.274	1.611	1.261

Table 4.7: The RMSE values for different methods in Milk data and Wages data

(a) Milk data

	One step prediction								
	plme	diggle	re-em	tree	bag	rf	boost	svm	nn
Week 8	**0.211**	0.303	0.227	0.306	0.311	0.303	0.314	0.368	0.324
Week 9	**0.226**	0.292	0.237	0.301	0.291	0.293	0.292	0.360	0.310
Week 10	**0.247**	0.291	0.264	0.282	0.275	0.275	0.276	0.300	0.294
Week 11	**0.291**	0.322	0.300	0.308	0.307	0.307	0.306	0.304	0.317
Week 12	**0.273**	0.281	0.284	0.289	0.280	0.283	0.280	0.281	0.295
Week 13	**0.223**	0.293	0.248	0.286	0.296	0.287	0.296	0.306	0.297
Week 14	**0.230**	0.320	0.296	0.315	0.318	0.316	0.320	0.320	0.315
Week 15	**0.282**	0.317	0.304	0.316	0.315	0.315	0.315	0.330	0.315
Week 16	0.342	0.363	**0.339**	0.353	0.340	0.347	0.341	0.372	0.387
Week 17	**0.264**	0.309	0.272	0.307	0.284	0.291	0.284	0.319	0.294
Week 18	**0.234**	0.302	0.260	0.301	0.293	0.290	0.294	0.307	0.292
Week 19	**0.263**	0.315	**0.263**	0.314	0.308	0.309	0.305	0.314	0.310

	Two step prediction								
	plme	diggle	re-em	tree	bag	rf	boost	svm	nn
Week 9	0.259	0.294	**0.247**	0.303	0.311	0.299	0.309	0.419	0.341
Week 10	0.276	0.312	**0.270**	0.284	0.273	0.275	0.275	0.359	0.272
Week 11	0.333	0.383	0.312	**0.307**	0.314	0.310	0.315	0.315	0.315
Week 12	0.329	0.321	0.298	0.290	0.288	0.288	0.285	**0.284**	0.295
Week 13	**0.266**	0.289	0.261	0.287	0.292	0.285	0.291	0.309	0.284
Week 14	**0.265**	0.319	0.311	0.316	0.318	0.315	0.318	0.328	0.331
Week 15	**0.311**	0.317	0.314	0.316	0.316	0.316	0.316	0.333	0.313
Week 16	0.377	0.355	0.351	0.354	0.355	0.353	**0.348**	0.384	0.381
Week 17	0.314	0.324	**0.285**	0.309	0.293	0.302	0.296	0.334	0.296
Week 18	**0.268**	0.304	0.270	0.303	0.299	0.295	0.294	0.315	0.332
Week 19	0.292	0.315	**0.272**	0.315	0.311	0.311	0.310	0.315	0.319

(b) Wages data

	lme	re-em	tree	bag	rf	boost	svm	svmk	nn
RMSE value	0.399	**0.314**	0.407	0.439	0.402	0.432	0.401	0.401	0.398

plme: Piecewise Linear Mixed-Effects Model; diggle: Diggle's Model;
re-em: RE-EM trees with random intercept;
tree: Decision Tree Method; bag: Bagging Method;
rf: Random Forest Method; boost: Boosting Method;
svm: Support-Vector Regression Method; nn: Neural Network Method

Chapter 5

Inclusion of Features Derived from a Mixture of Time Window Sizes Improved Classification Accuracy of Machine Learning Algorithms for Sheep Grazing Behaviours

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Abstract

Inertial motion sensors located on the animal have been used to study the behaviour of ruminant livestock, where patterns in the sensor signal can be correlated to the observed behaviours. The time window size of segmented signal data can significantly affect the classification accuracy of animal behaviours. To date, there have been no studies evaluating the impact of a mixture of time window size features on the accuracy of behaviour classification. In this study, sensor data were collected from accelerometers attached to the neck of 17 Merino sheep over two days. A ground truth dataset of matched behaviour recordings (grazing, ruminating, walking, and standing) were collected over the same period. We applied three machine learning algorithms, random forest (RF), support-vector machine (SVM) and linear discriminant analysis (LDA), to investigate the ability of these different approaches to classify sheep behaviour accurately. The results clearly show that simultaneous inclusion of features derived from mixed time window sizes significantly improved the behaviour classification accuracy, in comparison to those determined from a single unique time window size. Importantly, using features derived from time windows of different length may provide the context needed to accurately identify certain behaviours. Overall, the RF algorithm performed the best among the three ML methods

tested. Both RF and SVM had improved F1 score for the walking behaviour, which is a rare behaviour in this dataset accounting for only 1% ground truth data. In addition, using the features combining mixed window sizes and cumulative effects improved the classification accuracy of sheep behaviours.

Keywords: Mixture of time window sizes; Features ranking; Classification algorithm; Machine learning; Sheep behaviour; Accelerometer.

5.1 Introduction

Change in animal behaviour can be used for early detection and quantitative assessment of the animal's health status [Barwick et al., 2018, Martiskainen et al., 2009]. Grazing and ruminating are two important behaviours for ruminants and continuous monitoring of animal eating behaviour provides vital information about ruminant health, productivity, and welfare [Mansbridge et al., 2018]. The traditional methods of animal monitoring are based on direct observation by human operators or assessment of video recording. Both approaches are labour-intensive, time-consuming, and prone to human error [Alvarenga et al., 2016, Watanabe et al., 2008]. The rapid development of sensor technologies provides great opportunities for remotely monitoring animals in a range of applications [Brown et al., 2013]. Sensor devices, especially accelerometer that generates the information in both gravitational and inertial acceleration that are associated with movement (usually on three different axes), can give a good insight into individual animal behaviour patterns [Fogarty et al., 2020].

To date, several studies have used sensors to study the behaviour of ruminant animals, especially in cattle. Greenwood et al. [2018] investigated the possibility of predicting pasture intake based on behaviour classification. Using the sensor data, they developed a simple algorithm to predict pasture intake of individual cattle. Rahman et al. [2018] compared the cattle behaviour classification using the information from sensors located on different parts of the body (collar, halter and ear) and found that different sensor placement can still achieve good classification accuracy providing that the feature variation between the training and testing animals is very small. For sheep behaviour classification, Guo et al. [2018] compared grazing behaviour of sheep on pasture with different sward surface heights using an inertial measurement unit sensor. They found that a high accuracy ($\geq 95\%$) of identifying grazing behaviour from non-grazing behaviour could be achieved for all epochs (5s, 10s and 15s) with 10s being the best, regardless of sward surface heights. There are a few commercially available monitoring systems that may be used to capture feeding behaviours for dairy cattle, such as Lely [Bar and Solomon, 2010] and MooMonitors [Verdon et al., 2018]. However, for sheep, these automatic systems cannot be directly applied as there are likely differences in accelerometer signal patterns between species and in many cases, the sensor is impractical or unsafe for deployment on sheep due to limitations involving sensor size, shape, weight or method of attachment.

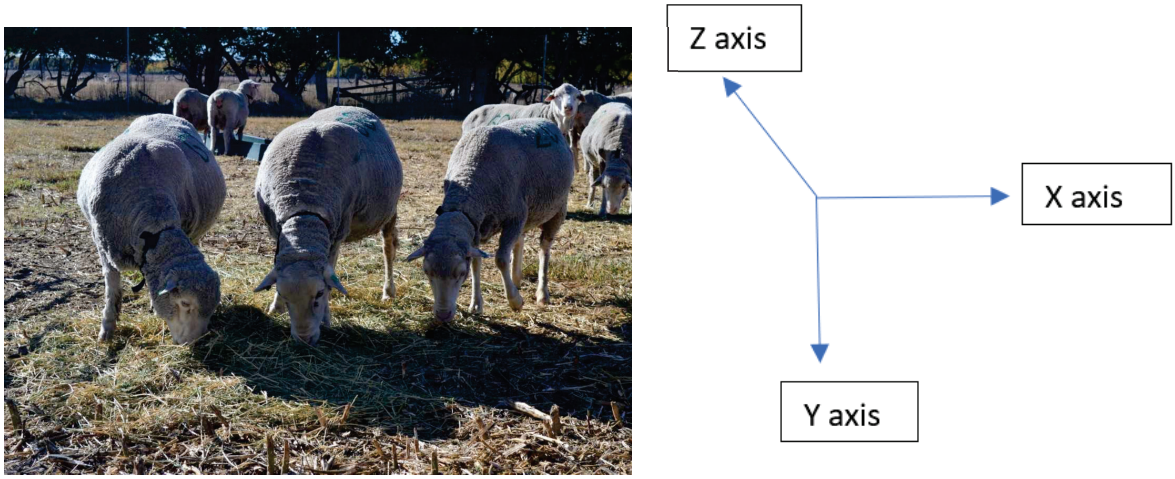
Recently machine learning algorithms have become very popular and offer great potential in

animal behaviour classification, largely because of their abilities in dealing with 'large p, small n' problems associated with high dimension datasets (such as sensor data) and provide high prediction accuracy for complex phenotypes. For example, Dutta et al. [2015] applied six supervised machine learning methods, binary tree, LDA, naive Bayes, k nearest neighbour (kNN) and adaptive neuro fuzzy inference system (ANFIS), to classify five major cattle behaviours. These include Grazing, Ruminating, Resting, Walking and other behaviour. They achieved a high accuracy (96%) of classification by using the bagging ensemble classification with tree learner. For sheep behaviour classification, Mansbridge et al. [2018] found that RF performed the best when compared with SVM, kNN and adaptive boosting (AdaBoost) for sheep data collected by accelerometer/gyroscope sensor attached to the ear and collar. Guo et al. [2018] also reported that the LDA classifier was the best performer compared to binary tree, naive Bayes, kNN and ANFIS on classifying grazing activities. Window size for signal segmentation is one of the crucial factors influencing on activity recognition. Banos et al. [2014] evaluated the impact of different window sizes (0.25–7 s) on human activity classification with accelerometer data and found that the interval 1-2 s was the best trade-off between speed and accuracy of recognition. In sheep, Walton et al. [2018] investigated the effects of sensor position (ear and collar), sampling frequency (8, 16 and 32 Hz) of triaxial accelerometer and gyroscope sensor, and window size (3, 5 and 7 s) on behaviour classification, and concluded that the combination of 16Hz with 7 s window would produce the benefits of energy efficient and reasonable classification accuracy (91-93%) in a real-time sheep monitoring system. Smith et al. [2016] built a separate classifier for each of five cattle behaviours (grazing, walking, ruminating, resting and 'other') using a 'one vs all' ensemble on 24 Holstein-Friesian dairy cows. Of nine window sizes evaluated (1.5, 2.5, 5, 7.5, 10, 15, 20, 25, 30 s), they found that 'the grazing, resting and rumination behaviours produced their highest mean F-score for the longest window of the study (30 s)'. A question is whether the mixture of different time window sizes can be used to improve the classification accuracy of livestock grazing behaviour because there is no consensus about window sizes for signal segmentation. In this study, we aimed to: (1) Evaluate the impact of mixed time window size features in classification versus that of single unique time window size features in the imbalanced dataset; (2) Compare the performance of three machine learning methods, RF, SVM and LDA in behaviour classification; (3) Investigate if new features including the mixed window sizes and cumulative effects can improve the classification performance.

5.2 Materials and methods

5.2.1 Experiment design and data collection

Data collection was conducted on animals enrolled in a grazing trial, according to the Australian Code for the Use and Care of Animals in Research and Teaching, and approved protocols were approved by the CSIRO Armidale Animal Ethics Committee (Animal Research Authority 18-13). A total of 20 Merino ewes, habituated to human presence, were kept in a square mixed sward pasture paddock of 70 m x 70 m. A subgroup of 17 animals was randomly chosen for device deployment and behavioural annotation. Devices were attached around the neck of the animals with an elasticated strap (Figure 5.1) for a period of 48 hours. The sensor datasets were collected from Actigraph wGT3X-BT sensor devices fitted with collars at the neck of Merino ewes, each containing a triaxial microelectromechanical systems (MEMS) accelerometer. The accelerometer sampled at a frequency of 30 Hz. The X-axis aligned approximately with the vertical or dorsoventral direction, the Y-axis with the craniocaudal direction, and the Z axis with the transverse or mediolateral direction (see Figure 5.1 for illustration).



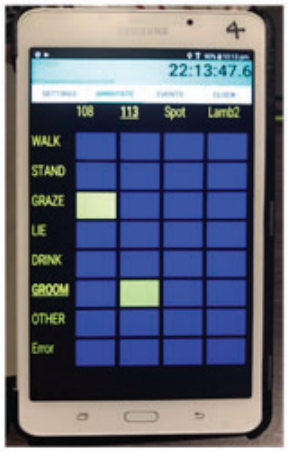
(a) (b)

Figure 5.1: Location of sensor and its orientation on sheep

Annotation of behaviours was performed by direct observation of animals within the paddock environment. Trained operators were equipped with tablet devices with a custom-designed annotation application (CSIRO AnnoLOG v 1.0.23) installed on a Samsung Galaxy Tab A 7.0 (Samsung, Seoul, Korea). The app displayed a grid of observed animals and behaviours to be annotated (Figure 5.2 a). Observers tapped fields corresponding to a behaviour for a specific

animal whenever a change in behaviour occurred. Behaviours were recorded retrospectively, that is, a tap to a field recorded the selected behaviour as having been displayed by the selected individual animal in the preceding time interval. This approach effectively eliminated the difficulty for the observer to predict which behaviour would be displayed next. The output of the behaviour log was in tabular form (see Figure 5.2 b). It contained the number of animals falling into individual categories of behaviour observations during two representative days, 21 and 22 February 2019. Four main behaviours, Grazing, Ruminating, Walking and Standing, were recorded. Recording time differed between animals. Overall, the final dataset contained approximately 30 minutes of records for each sheep with annotated behaviour information.

a)



b)

Clock date	Clock start time	Animal ID	Behaviour	Clock end time
21/02/2019	8:43:54 AM	1	Graze	8:49:10 AM
21/02/2019	8:49:10 AM	1	Walk	8:49:16 AM
21/02/2019	8:49:16 AM	1	Stand	8:49:47 AM
21/02/2019	8:49:47 AM	1	Walk	8:50:08 AM
21/02/2019	8:50:08 AM	1	Stand	8:51:01 AM
21/02/2019	8:51:01 AM	1	Walk	8:51:05 AM
21/02/2019	8:51:05 AM	1	Graze	8:56:15 AM
21/02/2019	8:56:15 AM	1	Walk	8:56:26 AM
21/02/2019	8:56:26 AM	1	Stand	8:58:16 AM
21/02/2019	8:58:16 AM	1	Walk	8:58:28 AM
21/02/2019	8:58:28 AM	1	Stand	8:59:08 AM
21/02/2019	8:59:08 AM	1	Stand	9:00:26 AM

Figure 5.2: a) Operator interface of the annotation tool CSIRO AnnoLOG v. 1.0.23. b) Tabular output of annotated behaviours for animal ID1

5.2.2 Consolidation of sensor data and ground truth dataset

We aligned the raw sensor data and the behaviour observations together via the time stamps (i.e. windows, every 1/30 second). A total of 1,052,475 data points was obtained. Figure 5.3 demonstrates the distribution of four different behaviour classes from 17 Merino sheep. Note that Walking has a small representation.

5.2.3 Feature extraction from sensor data

A feature extraction involves the process of transforming raw signals that measure the magnitude of acceleration to different domains of representation, such as time/frequency or statistical

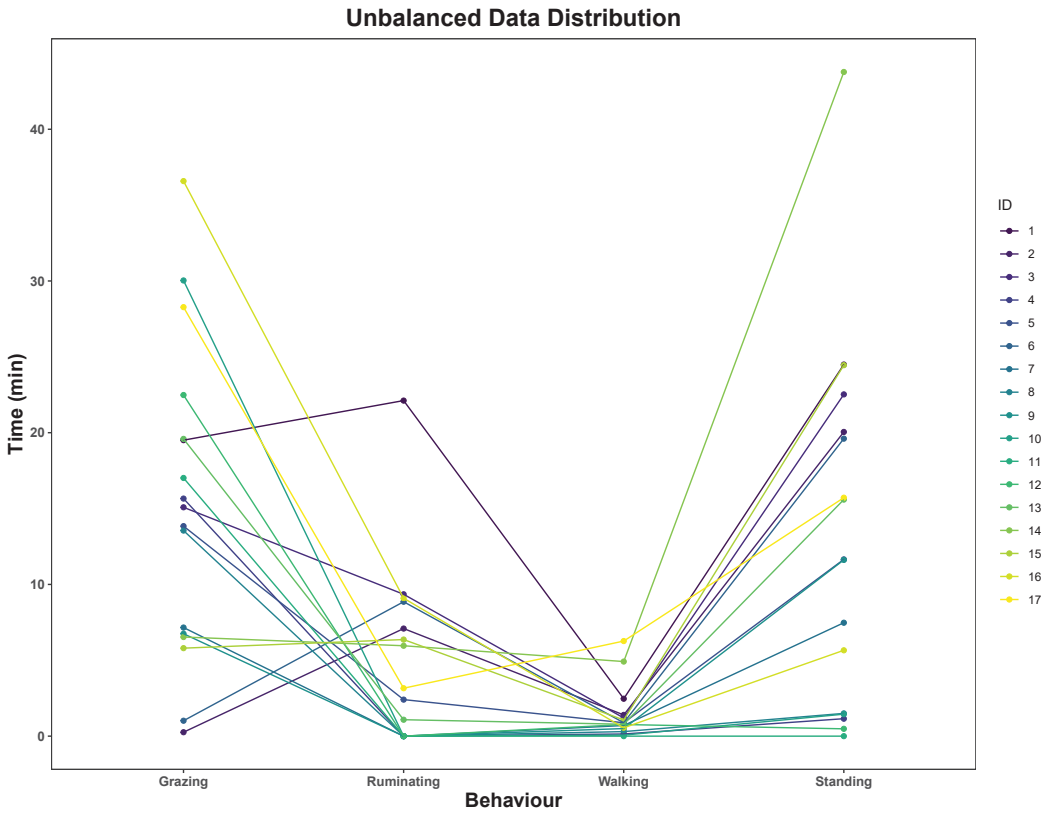


Figure 5.3: Unbalanced behaviour dataset

or mathematical functions, that can be later used as the input to a classifier or reasoner for activity recognition [Banos et al., 2014], such as classification of animal behaviour. In this study, five different time window sizes, 1s, 2s, 5s, 10s and 15s, were used for signal segmentation. For each time window, six basic statistical features, minimum, maximum, mean, standard deviation, skewness, and kurtosis, were computed for each of three axes (x, y and z) acceleration data (see Table 5.1).

Instead of treating the features from five different time window sizes in isolation, we developed a new method that enabled us to conduct the classification analysis with all features from different time windows together in the same dataset (see Table 5.2). In brief, a time window of 1s was used as the basis. A total number of 28,425 intervals were generated with the 1s window. Using the average of acceleration magnitude values for X-axis as an example of a statistical feature, the corresponding average values for all intervals were denoted as $a_1, a_2, \ldots, a_{28425}$. When using 2s time window, a total number of intervals will be 14,213 with the average values being $b_1, b_2, \ldots, b_{14213}$. To combine the features of average values from 1s and 2s windows together in the same dataset, individual average values of 2s window were used twice to meet 1s

Table 5.1: Illustration of deriving average features from mixed time window sizes

Number of bins	Time window				
	1s	2s	5s	10s	15s
1	a_1	b_1	c_1	d_1	e_1
2	a_2	b_1	c_1	d_1	e_1
3	a_3	b_2	c_1	d_1	e_1
4	a_4	b_2	c_1	d_1	e_1
5	a_5	b_3	c_1	d_1	e_1
6	a_6	b_3	c_2	d_1	e_1
...	...	...	...	...	...
...	...	...	...	...	...
28423	a_{28423}	b_{14212}	c_{5685}	d_{2843}	e_{1895}
28424	a_{28424}	b_{14212}	c_{5685}	d_{2843}	e_{1895}
28425	a_{28425}	b_{14213}	c_{5685}	d_{2843}	e_{1895}

window requirement (see the second column in Table 5.2). By doing so, the average value from a 2s window remained unchanged. The same concept applies to the average values derived for the time window sizes of 5s, 10s and 15s (See Table 5.2) as well as other statistical features (i.e. minimum, maximum, standard deviation, skewness and kurtosis values for X-axis acceleration magnitude measurement).

Table 5.2: Illustration of calculation of additional features of cumulative effects of X-axis Acceleration magnitude measurement. T: the total number of intervals for a given time window; t: a particular interval

Time	X	Xsum	Xvelocity	Xsummean	Xsum2	Xdis
1	x_1	x_1	$x_1/30$	$x_1/1$	x_1	$x_1/30$
2	x_2	$x_1 + x_2$	$(x_1 + x_2)/30$	$(x_1 + x_2)/2$	$2x_1 + x_2$	$(2x_1 + x_2)/30$
...	...	...	...	...	...	...
...	...	...	...	...	...	...
T	x_T	$\sum_{t=1}^{T} x_t$	$(\sum_{t=1}^{T} x_t)/30$	$(\sum_{t=1}^{T} x_t)/T$	$(\sum_{t=1}^{T}(T-t+1)x_t)$	$(\sum_{t=1}^{T}(T-t+1)x_t)/30$

Time	X	Xsum2mean	acc	dyz	interyz
1	x_1	x_1/1	$\sqrt{x_1^2 + y_1^2 + z_1^2}$	$\sqrt{y_1^2 + z_1^2}$	$y_1 z_1$
2	x_2	$(2x_1 + x_2)$/2	$\sqrt{x_2^2 + y_2^2 + z_2^2}$	$\sqrt{y_2^2 + z_2^2}$	$y_2 z_2$
...	...	...	...	...	...
...	...	...	...	...	...
T	x_T	$(\sum_{t=1}^{T}(T-t+1)x_t)/T$	$\sqrt{x_T^2 + y_T^2 + z_T^2}$	$\sqrt{y_T^2 + z_T^2}$	$y_T z_T$

Apart from six basic statistics features, we also explored new features of cumulative effects of raw data X_t, Y_t and Z_t. Table 5.2 illustrates how the cumulative effects are derived for the corresponding features named Xsum, Xvelocity, Xsummean, Xsum2, Xdis and Xsum2mean, using the time series data from X-axis acceleration magnitude measurements. The same methods were also applied for the computation of the features for the Y and Z axes acceleration magnitude measurements. In addition, the squared acceleration magnitude (acc) that considers the joint effects of X, Y and Z measurements were also included in this study [Rahman et al., 2018]. As the Y-axis detected the motion from front to back and the Z-axis detected the motion from the side to side, the interaction between Y and Z axis measurements were further examined using the features named dyz and interyz. For each window size and each statistics feature, there were 24 different metrics (including X, Y and Z axis data). A total of 720 features were generated.

5.2.4 Machine learning (ML) algorithms for classification

RF is a tree-based ensemble method that builds a large collection of decision trees using training datasets, and validate predictions using testing datasets [Breiman, 2001]. The advantage of RF is it is far less prone to overfitting than other algorithms and provides robust classification performance [Hastie et al., 2009]. The algorithm starts building decision trees by randomly sampling m features each time (with replacement) as split candidates from the full dataset of M predictors (features). This forces each split to consider only a subset of the predictors. After the forest is grown, the predictor is obtained by the majority vote. A variable importance measure (VIM) is computed using the mean decrease in Gini index. The importance of the variable is measured by the mean decrease in the Gini index. A larger positive value of the Gini index means that a variable is more important in training the classifier [James et al., 2013].

SVM constructs a linear partition of the high-dimensional space into two subspaces for classification or regression [James et al., 2013]. Intuitively, as the larger margins tend to provide lower classification errors, a good separation is obtained by a hyperplane that has the largest distance to the nearest training data. It often happens that the dataset is not linearly separable. Then a kernel function is used to transform the non-linearity system in the input space to the linearity system in the feature space. In this study, both the linear and radial kernel functions were applied.

LDA is a discriminant analysis that can separate a dataset into two or more classes. LDA assumes that the data within each class are drawn from a multivariate Gaussian distribution with a class-specific mean vector and a covariance matrix that is common to all classes [James et al., 2013]. Guo et al. [2018] found that LDA was an efficient classifier on detection of grazing activity in sheep sensor and behaviour recording datasets. In this study, we used the LDA classifier as a benchmark to compare the classification performance of RF and SVM. For all three methods, a five-fold stratified cross-validation scheme was applied. That is, the dataset from 17 sheep were randomly partitioned into 5 subsets. Each subset was in turn used as a test dataset while the other 4 subsets were used as the training dataset. The process was repeated five times. For each training dataset, hyperparameters were examined with a range of values to determine the minimum parameter values required for the final analysis. The library ranger in R [Wright and Ziegler, 2015] was applied for determining hyperparameters in RF. For the SVM and LDA method, the caret function in R [Kuhn et al., 2008] was applied.

5.2.5 Performance of the classification

Similar to the approach previously used for cattle behaviour classification [Rahman et al., 2018], we chose four metrics, namely overall accuracy, precision, recall (also called sensitivity) and F1-score, to assess the classification performance of individual ML algorithms. The four metrics were calculated as follows:

The overall accuracy can be considered the most intuitive metric to measure the classification performance as it represents the total number of animals correctly classified to the behaviour categories with truly observed phenotypes. High precision indicates a low false positive rate. Recall reflects sensitivity of identifying true positives. Since F1-score is the weighted average of Precision and Recall, it is a very useful metric when there is an uneven class distribution, especially when the number of false positives and false negatives estimates are remarkably different from each other. In this case, F1-score would be the preferred choice to measure the classification performance.

5.3 Results

5.3.1 Classification of behaviours using individual unique time window sizes

Table 5.3 presents the general effects of different window sizes on the classification performance of three ML methods, RF, SVM (with linear kernel), SVM (with radial kernel) and LDA classifiers, when ignoring cumulative effects, squared acceleration magnitude and interaction effects between Y and Z axes. When the window size increased from 2s, 5s, 10s to 15s, the classification performance of all ML classifiers for grazing behaviour showed a continued improvement except in SVM method with radial kernel (Table 5.3). The window size 15s gave the best classification performance for the grazing behaviour, although the difference between 10s and 15s was small. Among all methods, RF showed the highest F1-score (0.876 – 0.889). However, when considering ruminating, increasing the window size resulted in a reduction in performance in both RF and SVM with radial kernel at 2s being the best window size. SVM with linear kernel and LDA had no ability or a weak power to identify the ruminating behaviour (Table 5.3). In all cases, none of the ML methods had the ability to recognize walking behaviour regardless of the time window applied, except a very weak power identified by RF (0.173) or LDA (0.129) at 2s window.

When comparing the overall accuracies of three ML methods in correct identification of all behaviour classes (Figure 5.4), the RF performed best of all classifiers regardless of time window size. However, the window size did impact on the performance of SVM and LDA classifiers. For both SVM (linear kernel) and LDA, increasing the window size improved the accuracy value, with 15s giving the highest value (Figure 5.4).

5.3.2 Classification of behaviours using mixed time window sizes

Figure 5.5 illustrates the composition of different behaviours in the newly formed mixed time window dataset. Grazing behaviour accounted for the largest percentage (47%) of the data and the walking behaviour had the least representation at 1% of data.

Table 5.3: Effects of individual time window sizes on the behaviour recognition performance F1-score, when using RF, SVM and LDA. NA - not available

		F1 score			
Method	Behaviour	2s	5s	10s	15s
RF	Grazing	0.876	0.879	0.886	0.889
	Ruminating	0.655	0.582	0.545	0.550
	Walking	0.173	NA	NA	NA
	Standing	0.794	0.782	0.783	0.781
SVM(linear)	Grazing	0.832	0.850	0.864	0.875
	Ruminating	NA	NA	NA	NA
	Walking	NA	NA	NA	NA
	Standing	0.706	0.721	0.731	0.741
SVM(radial)	Grazing	0.839	0.844	0.842	0.846
	Ruminating	0.286	0.289	0.254	0.224
	Walking	NA	NA	NA	NA
	Standing	0.720	0.719	0.705	0.692
LDA	Grazing	0.823	0.838	0.851	0.863
	Ruminating	0.129	0.057	NA	NA
	Walking	NA	NA	NA	NA
	Standing	0.697	0.709	0.719	0.724

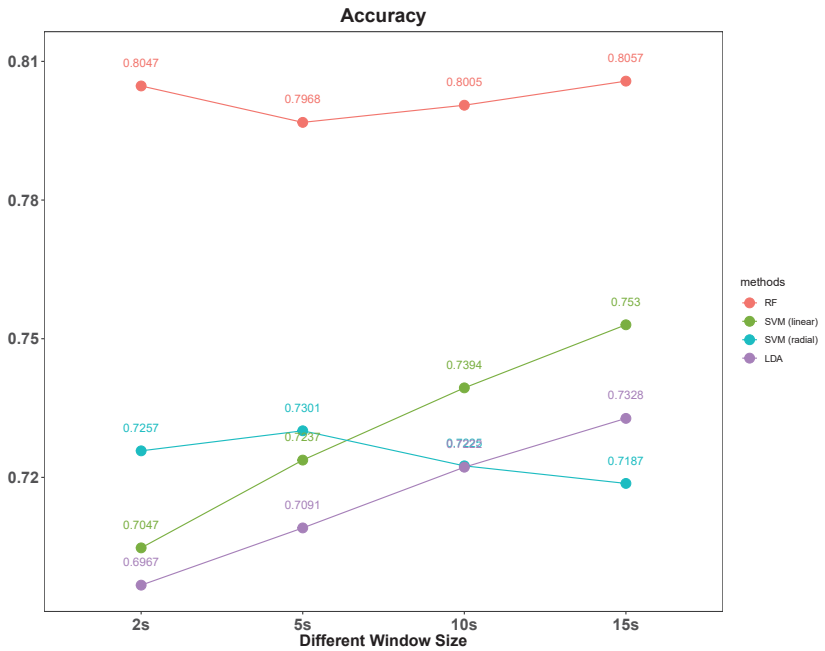


Figure 5.4: The change of overall accuracy for individual ML methods with different time window sizes

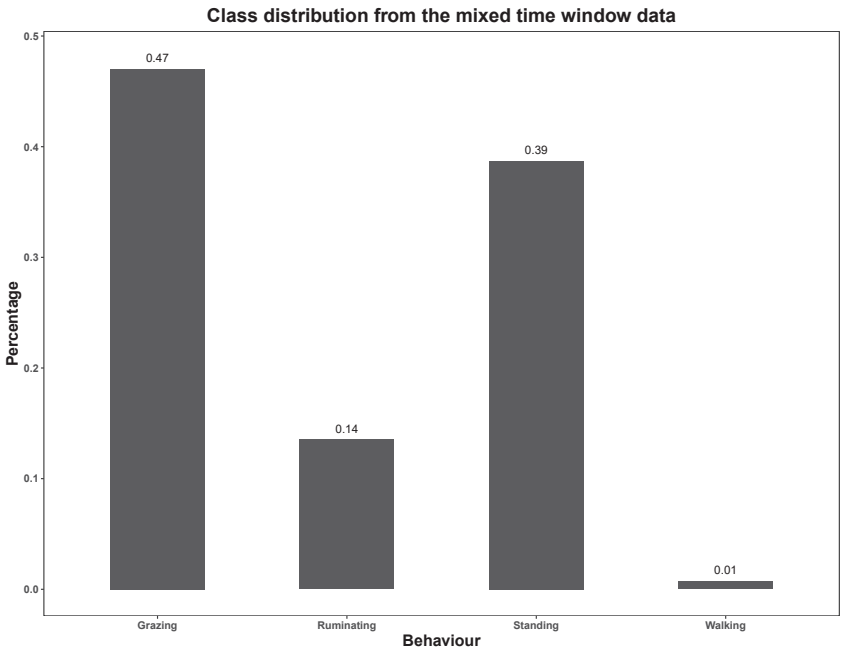


Figure 5.5: The percentage distribution of four behaviours in the mixed time window data

Random forest (RF)

Of the 720 features examined with the mixed window size dataset, the top 36 features (with the highest Gini index values) identified by the RF are shown in Figure 5.6. Among the top 36 features, the features derived from different window sizes (1s, 2s, 5s, 10s and 15s) all contributed to the classification accuracy of different behaviours. One other noteworthy observation was that the features derived from using the cumulative effects (i.e. with "sum" in the labels) were also among the top contributing features, e.g. Ysummean with different window sizes.

Next, we compared the behaviour classification performance of a different number of top ranked features from RF with that of 720 features. The top 3 ranked features from the RF produced an overall accuracy of 0.946 (Figure 5.7 a). The accuracy value increased to 0.986 with the top 9 features and to 0.999 with the top 27 features. Similar trends were observed irrespective of the performance metric used (i.e. the precision, recall or F1 score). For all four behaviour classes, the RF identified the correct individual behaviour classes with high precision (> 0.970), sensitivity (Recall > 0.950) and F1-score (> 0.970) when either the top 9 or 27 features were applied (Figure 5.7). Even with the top three features, the results show that for the most difficult behaviour to classify – walking, there was 0.910 (precision), 0.780 (sensitivity)

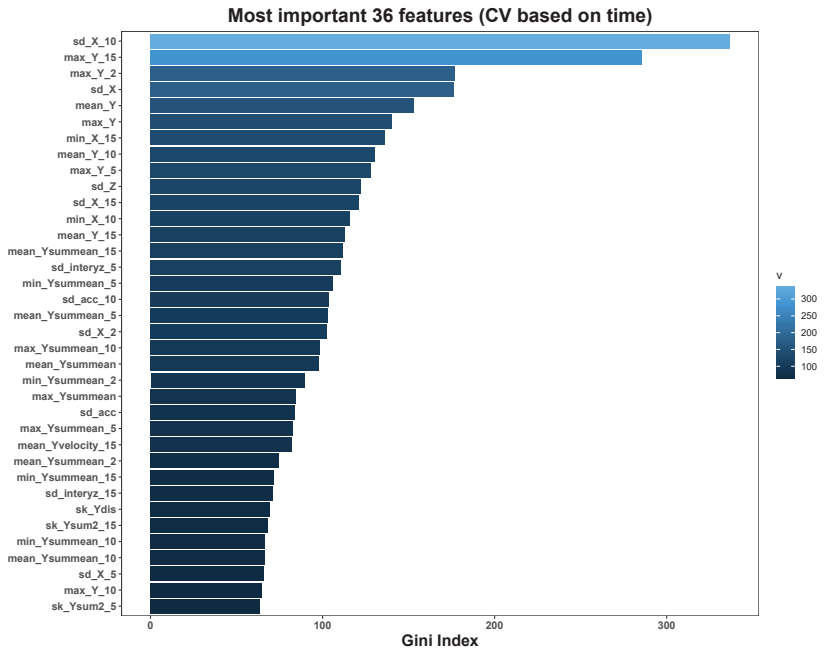


Figure 5.6: The list of top 36 features selected by RF

or 0.850 (F1-score) achieved.

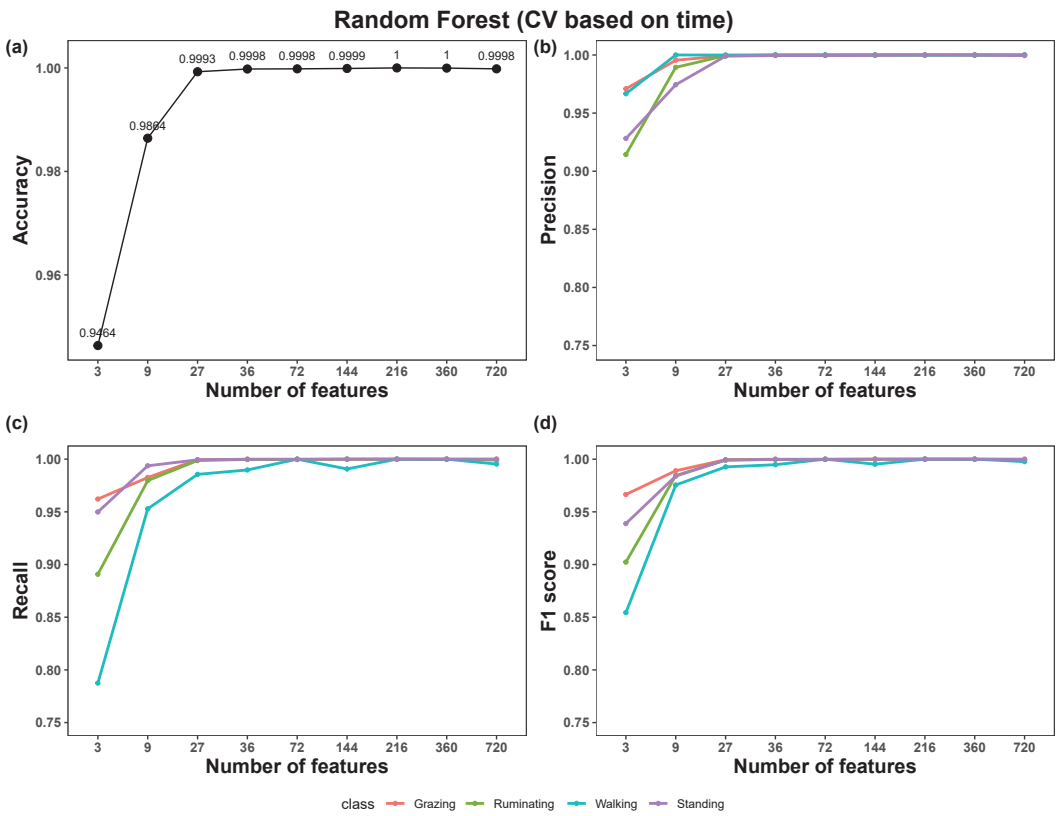


Figure 5.7: The overall accuracy, precision, recall and F1 score values from using the different number of top ranked features chosen from RF for behaviour classification

Support-vector machine (SVM)

The SVM classifier was evaluated with both linear and radial kernels. The overall accuracy of SVM (linear kernel) was largely dependent on the number of features applied for classification (Figure 5.8 a). While the lowest accuracy was 0.741 with three top features, the highest accuracy (0.933) was achieved when all the 720 features were applied. When investigating the classification performance of SVM (linear kernel) for individual behaviour classes, increasing the number of features slightly improved the classification performance for grazing and standing behaviours (Figure 5.8 b,c,d). However, for walking and ruminating, the change in the number of features had a significant impact on the classification performance.

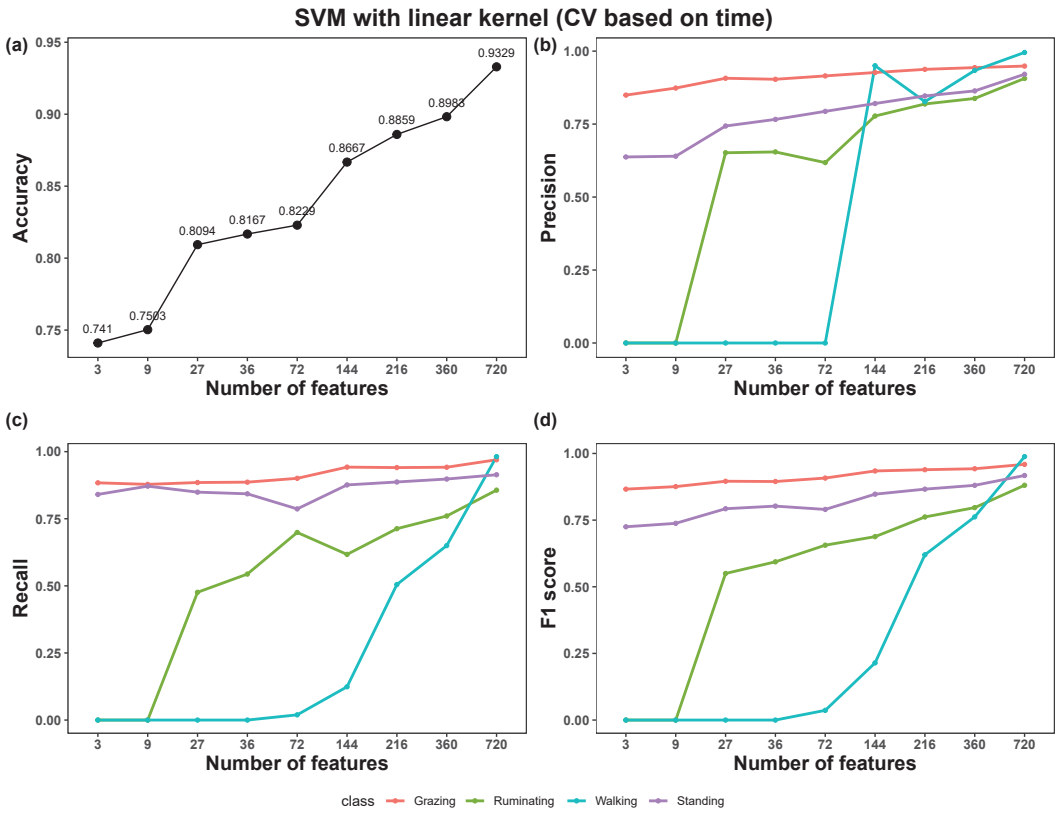


Figure 5.8: The overall accuracy, precision, recall and F1 score values from the different number of top ranked features chosen from SVM with a linear kernel function and the mixed time window sizes

When comparing SVM (radial kernel) with SVM (linear kernel), the overall accuracy was significantly improved by 8.40% with top three features, and by 17.70% with 27 features (Figure 5.9 a vs Figure 5.8 a). SVM (radial kernel) performed well in classification of all individual behaviours when the number of features was more than 27. This can be demonstrated by high

precision (> 0.900), reasonable sensitivity (> 0.750) and medium to high F1 score (> 0.850) in Figure 5.9. The most noticeable results are for the walking behaviour.

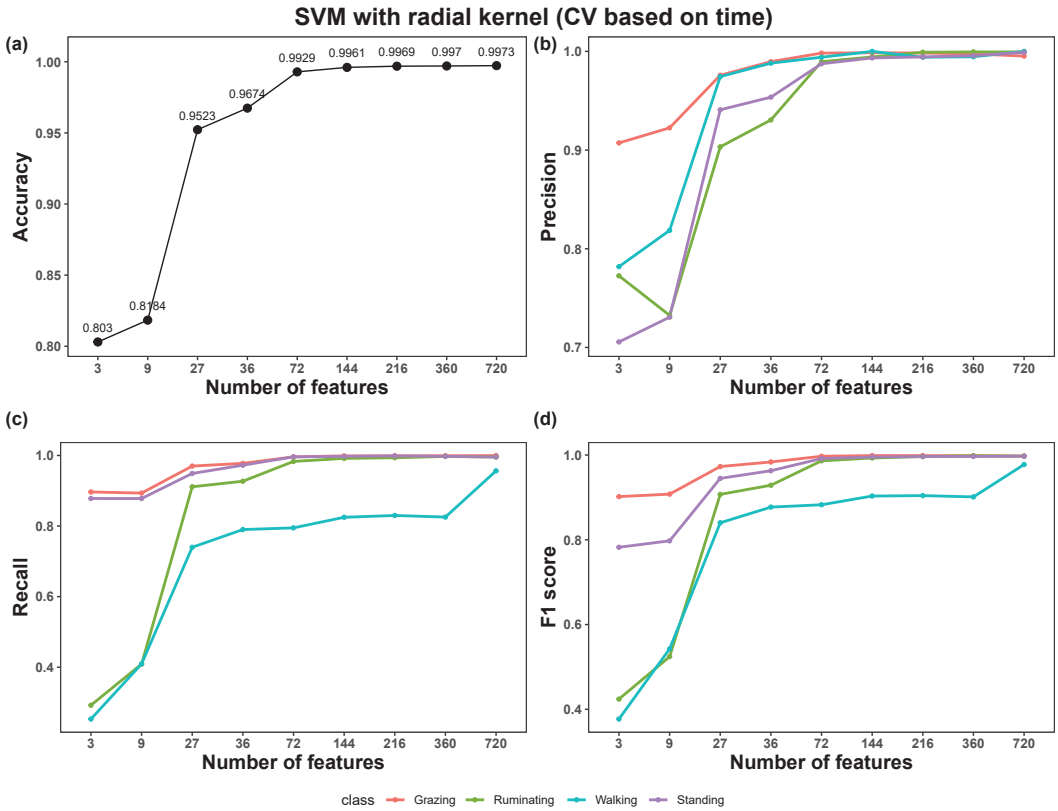


Figure 5.9: The overall accuracy, precision, recall and F1 score values from the different number of top ranked features chosen from SVM with a radial kernel function and the mixed time window sizes

Linear discriminant analysis (LDA)

Figure 5.10 present the classification results from the LDA classifier. The overall accuracy from the LDA classifier followed a similar trend as the SVM classifier with linear kernel (Figure 5.10 a vs Figure 5.8 a). However, the SVM (linear kernel) still gave 2.750% (with 3 features) - 7.010% (with 720 features) better accuracy values than the LDA classifier. When comparing the classification performance for individual behaviour, again, LDA had a very similar performance to the SVM with linear kernel for the grazing, ruminating, and standing behaviours. For the walking behaviour, although the LDA classifier had an overall low precision (< 0.30), low recall (< 0.50) and low F1 score (< 0.375), surprisingly, it did show some ability to recognise the walking behaviour when the number of features were less than 72. This was a stark contrast to the SVM classifier with linear kernel for the walking behaviour.

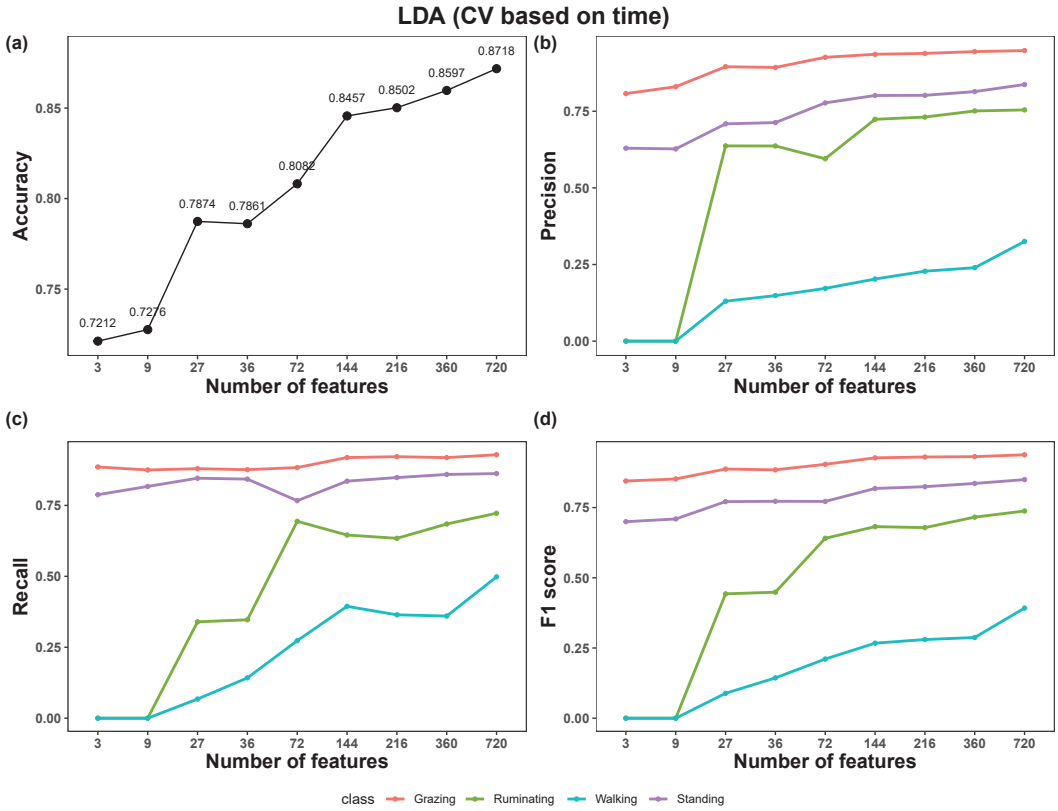


Figure 5.10: The overall accuracy, precision, recall and F1 score values from the different number of top ranked features chosen from LDA with the mixed time window sizes

5.4 Discussion

The application of accelerometer sensors is an appealing approach to monitor animal behaviour because it has the potential to enable monitoring of large numbers of animals remotely, continuously and in real time [Brown et al., 2013]. For this reason, there has been a rapid development in the analytical methods used for analysing and interpreting these datasets. Accurate classification of animal behaviour from sensor derived data is influenced by a number of factors, including experimental design, sensor placement position, data sample rate, signal segmentation window size, feature selection and different ML methods applied [Banos et al., 2014, Barwick et al., 2018, 2020, Fogarty et al., 2020, Mansbridge et al., 2018, Rahman et al., 2018, Walton et al., 2018]. Among them, window size, feature selection and analytical methods play critical roles.

5.4.1 Window size

It is a common practice to spend extensive amount of time to evaluate a range of time window sizes in order to identify an optimal size that produces reasonable classification accuracy from sensor datasets. However, the classifiers produced from this approach only considers features derived from a single time window. Such an approach may be influenced by individual analyst's biases and it also ignores the possibility that more valuable information can be gained by using features derived from time windows of different duration. Smith et al. [2016] evaluated the effect of nine different window sizes on average F1 scores of individual classifier for each cattle behaviour when using a 'one vs all' approach and their method resulted in a 5% improvement in F1 score over standard time series classifiers. Barwick et al. [2020] examined the effect of applying a moving window size classification algorithm (3, 5 and 10 s) on categorising data collected from collar, leg and ear mounted accelerometers. Although there was improvement in classifying different animal behaviours, there was no significant difference found in model accuracy between the three moving window lengths. In this study, developing a new combined window size approach enabled us not only to minimise subjective selection of window sizes, but also to systematically and simultaneously consider the features from multiple window sizes together to capture the irregular duration of animal behaviours, that occur both within and across behaviours. The analysis performed in this study was retrospective. This might impact application of the approach where immediate or real time data is required. However, the delay is unlikely to be greater than the length of the longest window (15 seconds here) plus calculation time since the feature selection procedure can be easily automated in practice.

Most importantly, we have demonstrated that using mixed window sizes in combination with the ML method RF, significantly improved the classification accuracies for all behaviour classes, especially for walking and ruminating. The overall accuracy, precision, recall and F1 score for individual behaviour, when mixed window sizes were applied, were in strong contrast to those when individual time window size was applied. This is largely due to the features derived from different window sizes being inter-related. The correlations between these features of different window sizes provide additional information for ML methods to correctly identify individual animal behaviours. The biological basis for this can be explained by the need to classify specific short duration movements as a component of a longer movement. For instance, grazing might involve the lowering of the head and several biting events. Only classifying the

lowering of the head would likely lead to inaccurate classification as would the classification of biting-like behaviours alone. The combination of the two is however likely to be much more informative. In contrast, when analysing the datasets with the features from one window size only, many feature correlations were not accounted for, especially in the cases where mixed behaviours occurred in a given time window (i.e. unequal length of animal behaviour), errors were expected to arise.

Limitations of the study that may have influenced the results, include the small number of animals and limited paddock space (70 m x 70 m) used during the experiment. However, the study aimed to serve as a proof of concept that incorporation of features calculated across time windows of different lengths has the potential to improve classification accuracy. We believe this principle has been demonstrated and the broader applicability of the approach can be tested in future trials involving larger numbers of animals.

5.4.2 Feature selection

To date an optimum number of features that can be used for classifying animal behaviours has varied greatly between studies, depending on the nature of sensor and behaviour data and analytical methods applied. For example, by analyzing the data collected by collar mounted motion sensors, Guo et al. [2018] found that the top 5 features, mean of accelerometer Z-axis, entropy of accelerometer Y-axis, entropy of accelerometer Z-axis, mean of gyroscope X-axis and mean of gyroscope Y-axis, can be used in classifying the grazing versus non-grazing activities in sheep. Mansbridge et al. [2018] identified 39 being the optimum number of features that can be used successfully in the classification of eating behaviors in sheep with a high accuracy (91% for ear and 92% for collar data). These features ranged from dominant frequency, zero crossings, signal area, spectral entropy, to basic statistics such as mean, min, max, standard deviation, and kurtosis.

In this study for each axis acceleration magnitude measurements, apart from the common features derived from 6 basic statistics and the squared acceleration magnitude (acc), we also evaluated the effects of new features of cumulative effects of measurements for a given time window size on classification performance. The reasons for using these features include: (1) to properly evaluate the efficiency of the new approach - a mixed time window sizes, it is crucial to compare the new method with conventional methods using commonly used features; (2) using

new features from the accumulative effects was to examine if they could better capture actual change of motion movements for mixed behaviours. When applying the RF with the mixed window sizes, of 720 features, we found that the optimum number of features was 27. The top 9 features that contributed the most in the classification accuracy were all related to basic statistics (e.g. , see Figure 5.6) of X and Y-axis measurements. The X and Y-axis in this study aligned with upward and downward, and front to back movements of the neck, respectively hence kinetically related to grazing and ruminating behaviors. Among the 27 top ranking features, there were also 9 features derived from the average of the accumulative effects of Y axis (Ysummean, Figure 5.6) for different window sizes. This indicates that the average cumulative values of Y axis from mixed window sizes may have better-reflected changes of acceleration, therefore contribute to additional improvement in the accuracy of behaviour classification.

5.4.3 Machine learning methods

Different ML methods could yield different outcomes when dealing with unbalanced behaviour classes. Among three ML approaches (RF, SVM and LDA) evaluated, the RF performed the best with over 99% accuracy, followed by the SVM (radial kernel), the SVM (linear kernel) and LDA classifiers. RF has been known to produce good classification accuracies in sheep behaviour [Alvarenga et al., 2016], especially classification of grazing and rumination behaviour [Mansbridge et al., 2018, Walton et al., 2018]. This is mainly due to its great ability in handling non-linearly correlated data and robustness to noise [Mansbridge et al., 2018]. SVM (radial kernel) performed better than the SVM (linear kernel) and LDA classifiers, also because its radial kernel function can non-linearly separate the sensor signals associated with irregular length of individual behaviours. There are other ML methods that can also be applied to provide good classification accuracy of sheep behaviour, depending on the sources of signal information of sensor placement (e.g. ear, collar, or leg, [Barwick et al., 2018]), and trade-off between energy consumption and classification accuracy [Le Roux et al., 2018]. Future work needs to be carried out with ensemble classifiers in which several different classifiers are trained simultaneously, and their classification decisions can be combined at the end [D'Este et al., 2014, Dutta et al., 2015].

All the results presented in this study were obtained using a five-fold stratified cross-validation scheme, rather than the cross-validation based on individual sheep. The initial analysis using a

five-fold cross-validation approach based on subsets of sheep produced much worse results (see supplementary Figure 1s) than that of a stratified cross-validation approach. For example, when applying the cross-validation strategy of leaving one animal out each time, the best overall accuracy achieved with the RF was 0.75 when the number of features was 36 (Figure 1s). The primary reason was due to the small number of animals used in this study and the big variation between individual sheep behaviours in feature space. Therefore, it is difficult to obtain the consistent results in testing datasets when different subsets of sheep were used as training datasets. To minimize the impact of individuality on classification accuracy of animal behaviours in future sensor application, it is crucial to: (1) apply more cost-effective sensor devices so that a large population can be used for the study; (2) explore and validate results using a number of repeated k-fold CV to improve the prediction on both population and individual results.

5.5 Conclusions

This study demonstrated that the sheep behaviours of grazing, ruminating, walking, and standing can be differentiated with a high accuracy using machine learning algorithm RF and a mixed window size approach. This is especially the case in which the walking behaviour only accounted for 1% of the ground truth data and a conventional approach of applying a machine learning method to sensor data with single unique time window size failed to recognise the behaviour. This can be explained with the fact that using features derived from time windows of different length may provide the context needed to accurately identify certain behaviours.

Chapter 6

Conclusions and Discussion

This chapter contains a summary of the results discussed in the thesis, along with a discussion of the limitations and topics for further research.

6.1 Summary of the research

Longitudinal data have become increasingly popular in many research areas and applications of statistical modelling and statistical learning methods for longitudinal data will continue to increase. In this thesis, the statistical models GEE and LMM have been investigated for longitudinal data. By way of comparison, statistical learning methods–the tree-based method, the support-vector machine (SVM), the neural network (NN) and linear discriminant analysis–have been applied to the longitudinal data.

The first question proposed in section 1.2 aimed to investigate the performance of variance estimators in the analysis of longitudinal data with a large cluster size. This question was addressed in chapter 3. The sandwich estimators relied on the residual products as a predictor of the true covariance of the responses, which became singular and sparse as the cluster size became larger. Extensive simulation studies and two example of real data analysis were conducted to address this question. In our simulation studies, the sample size was fixed to 30, and the cluster size increased from five to 1,000. The sandwich estimator exhibited considerable bias and underestimated the sampling variance when the cluster size was relatively small. However, in the longitudinal data with large cluster sizes, the sandwich estimators were still valid based on our simulations. The improved sandwich estimators ($\hat{V}_{WL}$) and the regularised

sandwich estimators ($\hat{V}_W$) performed more robustly. The smooth bootstrap method ($\hat{V}_B$) was more efficient in a longitudinal data analysis with large cluster sizes but had a larger variation. The core in the smooth bootstrap method involves the choice of $w_i, i = 1, ..., K$, which is generated from the Pois(1) distribution. Choosing the appropriate w_i for the specific data is particularly important.

Chapter 4 addressed the second research question. In this chapter, the performances of the statistical models and the six machine learning methods for the longitudinal data were presented. The results showed that the statistical models were better than the machine learning methods when the models were correctly specified, including in terms of both the mean function and the correlation structure. The performances under the scenarios of the mean function and correlation structure–correctly specified and misspecified–were compared. Otherwise, even with the milk dataset (a real-world dataset), the statistical model (especially, the piecewise linear mixed model) still performed better than the machine learning methods. This means that the piecewise LMM provided an adequate fit for the original data. It can also be concluded that the model diagnostics are crucial for making decisions regarding performance. One thing we should mention is that statistical models usually entail more assumptions than machine learning methods. However, this is a double-edged sword. Machine learning methods are usually recognised as having a 'black box' aspect, which means that there is less attention paid to the processes between their inputs and outputs.

Another contribution from chapter 4 was the proposal of a different way to measure the predictive power of methods in the longitudinal data. The stratified cross-validation method is not reasonable for longitudinal data because of the sequential correlation. Therefore, we used one-step and two-step prediction along with future observations and new-object and new-observation predictions. A new metric, the true root mean square error, was defined to measure the prediction performance because the closeness of the predictions to the true values without noise is of interest.

Chapter 5 addressed a real-world problem concerning how statistical learning methods can be used for the classification of sheep behaviour data. The collection of sheep sensor data and ground truth data were described in detail in this chapter. The data pre-processing including

the consolidation of the sheep sensor data and the ground truth data as well as feature engineering. The experimental arrangement, which included the cross-validation strategy and hyper-parameter tuning for each statistical learning methods, was also explained. Four metrics–overall accuracy, precision, recall and the F1 score–were used to access the classification performance. The results showed that the simultaneous inclusion of features derived from mixed time-window sizes of sensor signal data significantly improved the sheep behaviour classification accuracy, in contrast to those from a single, unique time window. Using features derived from time windows of different lengths provided the information needed to accurately identify different behaviours involving multiple movements of unequal duration. The random forest (RF) with a mixed window-size approach significantly improved the ability to identify the walking behaviour, which only accounted for 1% of the ground truth data.

6.2 Topics for further research

There are limitations to any research work. Some of the limitations that motivate future research in this case are summarised below.

First, there are many other modified sandwich estimators that are not covered in this research even though they are proposed for the bias reduction in longitudinal data with a small cluster size [Kauermann and Carroll, 2001, Wang et al., 2016]. Li and Wang [2008] discussed how their resampling method only captures the first and second-order moments. The random weights for the smooth bootstrap method may depend on other factors. How to choose the appropriate random weights for a specific data set requires further investigation. The regularised sandwich estimator and smooth bootstrapping estimator have been compared through extensive simulation studies in the longitudinal study. However, it is still necessary to investigate which method would be appropriate for various aspects of real applications. A novel theory on the methods for the selection criteria is necessary for the longitudinal data set with a large cluster size.

Second, in our comparison of the performances of the LMM and statistical learning methods in longitudinal data, the predictions of the different methods were investigated rather than the parameter estimates and inferences. Misspecified models, including the mean function and correlation structure–misspecified either simultaneously or separately–were considered in our research. However, Wang and Lin [2005] also investigated the effects of variance function

misspecification in the analysis of longitudinal data. They claimed that the correct specification of the variance function can improve estimation efficiency, even if the correlation structure is misspecified. The popular exchangeable and AR(1) correlation structures are considered because they are appropriate for equally spaced (in time) longitudinal data. Unequally spaced observations and time-dependent correlated errors also deserve more attention by researchers [Nunez-Anton and Woodworth, 1994].

Third, GEE and LMM were discussed in this thesis for the modelling of longitudinal data. As an alternative to the mean regression model, the application of the quantile regression model to longitudinal data has received extensive attention as a result of its flexibility and ability to describe the entire conditional distribution of a response variable [Fu and Wang, 2016, Geraci and Bottai, 2007, Koenker, 2004, Wang and Fu, 2011]. Another characteristics of quantile regression is that it does not require the assumption of error distribution. It would be of great interest to incorporate quantile regression with statistical learning methods, such as the tree-based method.

Finally, deep or reinforcement learning has become increasingly popular recently. Research has already been conducted on the mixed-effects trees, the mixed-effects RF and the mixed-effects SVM for longitudinal data. It would be interesting to combine the GEE method with the statistical learning methods or even the deep learning methods for longitudinal data.

To conclude this thesis, real data sets are usually complex, and it is worthwhile investigating more about the data before any definitive decision is made.

Appendix A

Example: Analysis for Machine Learning Methods

This is an example analysis of how to use the machine learning method for classification. McTavish et al. [2013] investigated the ancestry of cattle throughout the world. We took a subset of cattle (691 of 15 different breeds) from 1.8k dataset. The 691 heads of cattle were genotyped for 1,814 single nucleotide polymorphisms (SNPs). The breed names and their respective sample sizes are shown below:

Table A.1: Breeds included in the analysis

S.No	Name	Region of origin	Sample size
1	Shorthorn	Great Britain	100
2	Angus	Great Britain	90
3	Hereford	Great Britain	100
4	Simmental	Northern Europe	80
5	Red Angus	Great Britain	15
6	Limousin	Southern Europe	100
7	Finnish Ayrshire	Northern Europe	10
8	Guernsey	Channel Islands	10
9	Norwegian Red	Northern Europe	21
10	Scottish Highland	Great Britain	9
11	Holstien	Northern Europe	84
12	Charolais	Southern Europe	55
13	Hanwoo Korean	Asia	7
14	Japanese Black	Asia	10

The tree-based method

For a classification tree, we can predict that each observation belongs to the most commonly occurring class of training observations in its region. We split the entire data set: 400 heads of cattle were randomly selected as a training data set and the remaining 291 heads were regard as the test data set.

The classification error rate was used as a criterion for evaluating prediction accuracy. Using the *tree* function in R, the results are as follows:

```
> summary(tree.m)

Classification tree:
tree(formula = factor(BRCol) ~ ., data = data.frame(x1, BRCol),
    subset = train)
Variables actually used in tree construction:
 [1] "X22805" "X6879"  "X44330" "X34696" "X13190" "X18870" "X6639"  "X2790"  "X11261" "X15537" "X45587"
[12] "X39818" "X42736" "X16058" "X4455"  "X9754"  "X28081" "X54743" "X4373"  "X34918" "X12904" "X10255"
[23] "X9828"  "X53920"
Number of terminal nodes:  25
Residual mean deviance:  1.116 = 418.6 / 375
Misclassification error rate: 0.21 = 84 / 400
```

Figure A.1: The summary result from the tree

The results listed the variables (SNPs) that were used as internal nodes in the tree, the number of terminal nodes and the training error. The number of terminal nodes was 27 and the training error rate was 21%, which is small. To evaluate the performance of the classification tree for these data, we had to estimate the test error rather than simply computing the training error. Therefore, we used the test data to make predictions. The results showed that around 51.5% of the cattle had been classified correctly in the test data set, which was not a satisfying result.

Next, we considered pruning the tree to improve the results. According to Figure A.2 , the tree with seven terminal nodes resulted in the lowest cross-validation error rate. Then, the classification accuracy improved slightly from 51.5% to 54.3%.

The RF is an ensemble learning method. When building the decision trees, each time a split in a tree was considered, a random sample of predictors was chosen as split candidates from the full set of predictors. In other words, at each split in the tree, the algorithm forced consideration of a mere subset of the predictors. This efficiently prevented choosing the same strong predictor in the top split every time. This process made the resulting trees more reliable. The results of the cattle data analysis also demonstrated this. The classification accuracy grew to 89.7%.

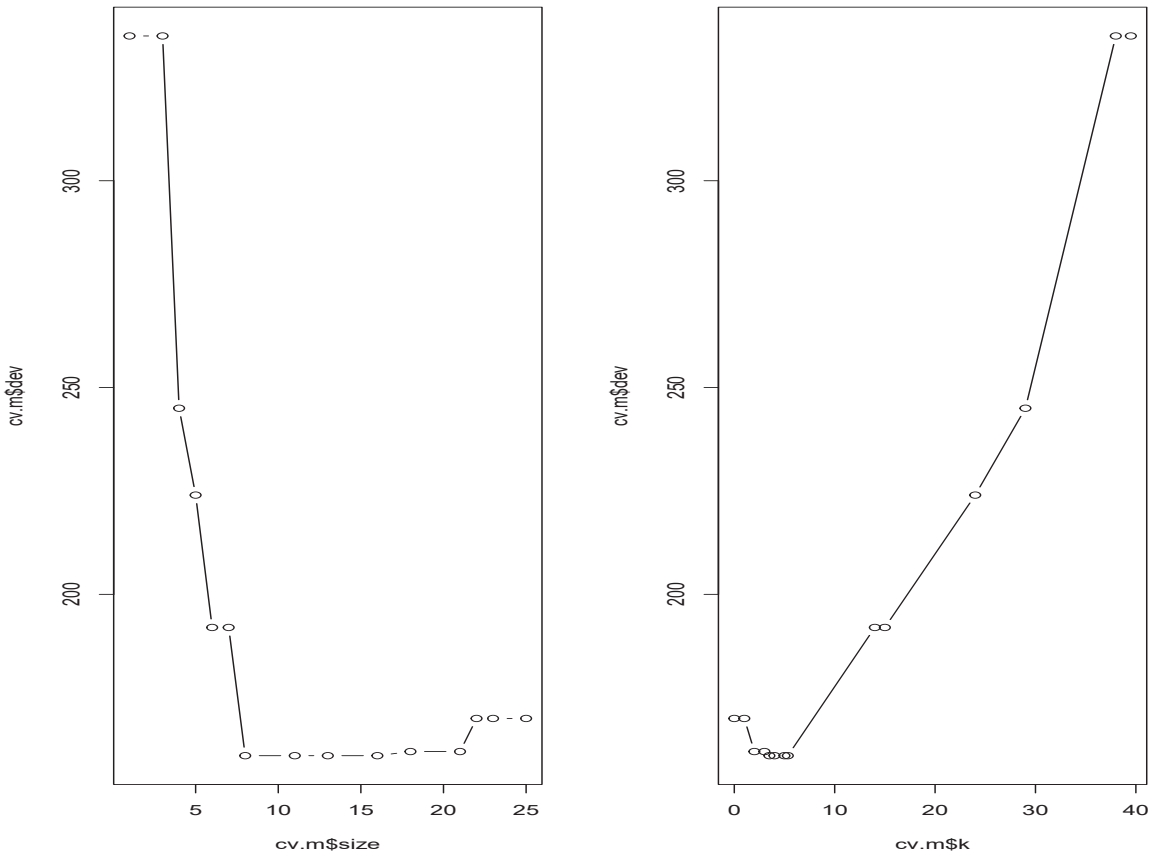


Figure A.2: The error rate of pruning the tree

SVC method

We used the *svm* function in R to conduct the SVC. A 'cost' argument allowed us to specify the cost of a violation to the margin, which was the constant C of the regularisation term in the Lagrange formulation. When the C was small, the margins would be wide, and many support vectors would either be on the margin or would violate it. When the C was large, the margins would be narrow and there would be few support vectors on the margin or violating the margin. We used the default cost ($C = 1$).The number of support vectors was 381 and the classification accuracy was 97%. When we increased the C to 2, the number of support vectors and the classification accuracy remained. When $C = 0.1$, the number of support vectors increased to 383 and the classification accuracy was 98%. If we decreased the value of C to 0.01, the number of support vectors would increase to 380 but the classification accuracy would still be 98%. This means that we need to balance of the value of C. The cross-validation method is usually used to obtain the optimal choice of C.

Appendix B

Supplementary Material for Chapter 3

Various simulation results following Warton [2011] are presented. We describe the simulation arrangement as follows.

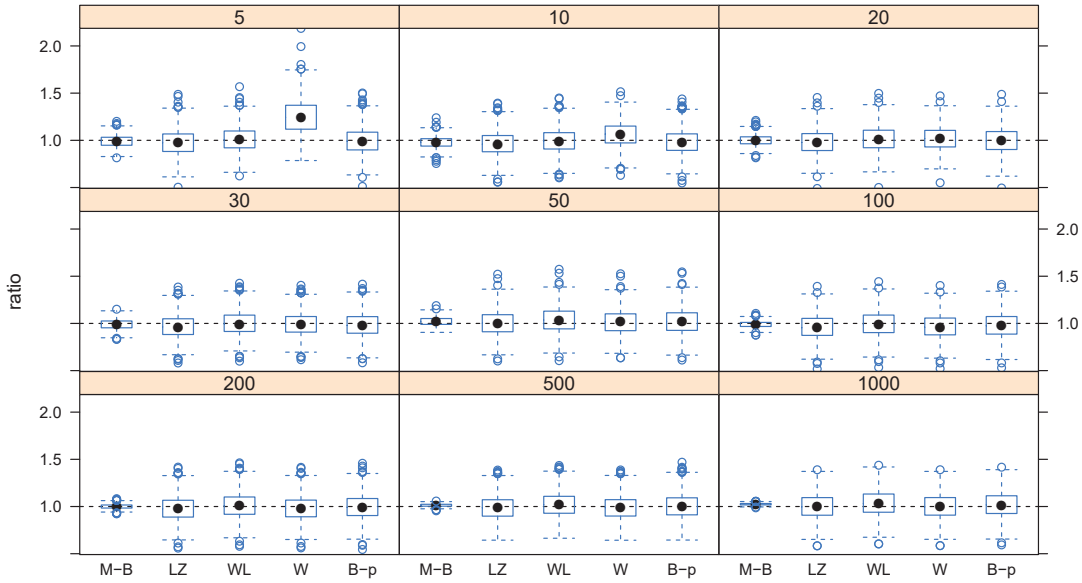
The Poisson data with the multivariate-lognormal mean and exchangeable correlation was generated:

$$y_{ij}|\mu^*_{ij} \sim Poisson(\mu^*_{ij}),$$

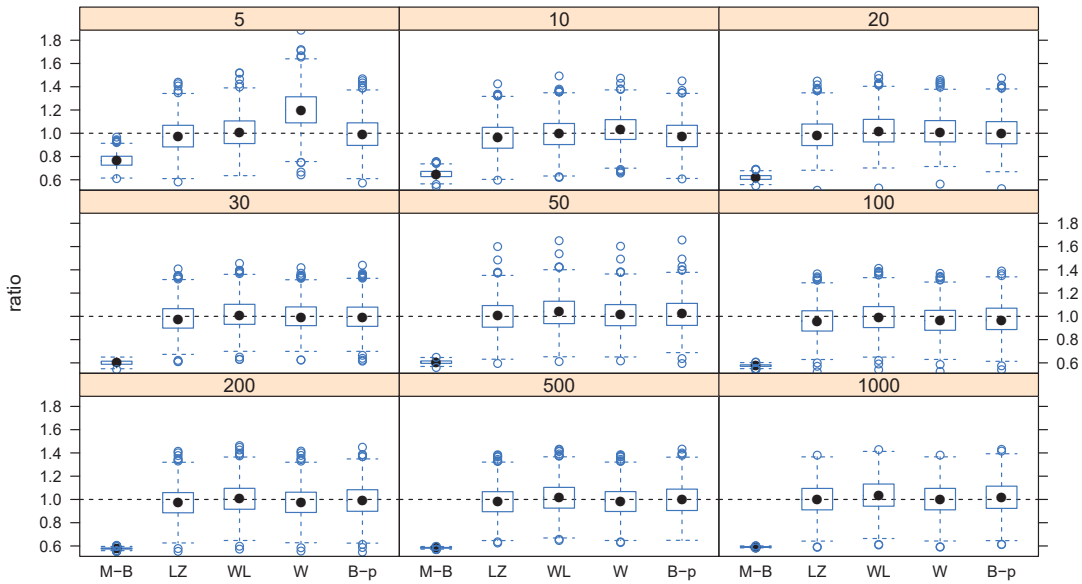
$$\log(\boldsymbol{\mu}^*_i) \sim MVN(\log(\boldsymbol{\mu}_i - \log(2)/2, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\mu}^*_i = (\mu^*_{ij}), j = 1, ..., n$. If the diagonal elements of the covariance matrix $\boldsymbol{\Sigma}$ are equal to $\log(2)$, then $E(y_{ij}) = \mu_{ij}$ and Var$(y_{ij}) = V(\mu_{ij}) = \mu_{ij} + \mu^2_{ij}$. Figure 2(a), from Warton [2011], presents the results when the cluster size was fixed ($n = 8$) and the number of clusters (K) increased from 12 to 96. The two independent samples revealed that the $K/2$ clusters were generated to obtain a mean vector $\boldsymbol{\mu}_1$ and the remaining samples provide a constant vector $\boldsymbol{\mu} = 2\boldsymbol{I}$. The definition of $\log(\boldsymbol{\mu}_1)$ is $\log(\boldsymbol{\mu}_1) = \log(2)\boldsymbol{I} + \frac{\sqrt{n}}{2}\,\Delta_{jj}\boldsymbol{\nu}_j$, where Δ_{jj} is the jth eigenvalue of the true correlation matrix $\boldsymbol{R}_0$ and $\boldsymbol{\nu}_j$ is the corresponding jth eigenvector. The correlation coefficient was 0.25. Then, the number of clusters was fixed ($K = 20$) and the cluster sizes varied from 2 to 30. This dominant shift means that the first eigenvector had the maximum eigenvalue of the true correlation matrix. The R codes for generating this kind of correlated data are listed for reference. Other R codes for the simulation studies and data sets are available upon request to the author.

```
mvrpois_warton1<-function(K,n,rho)
{
```

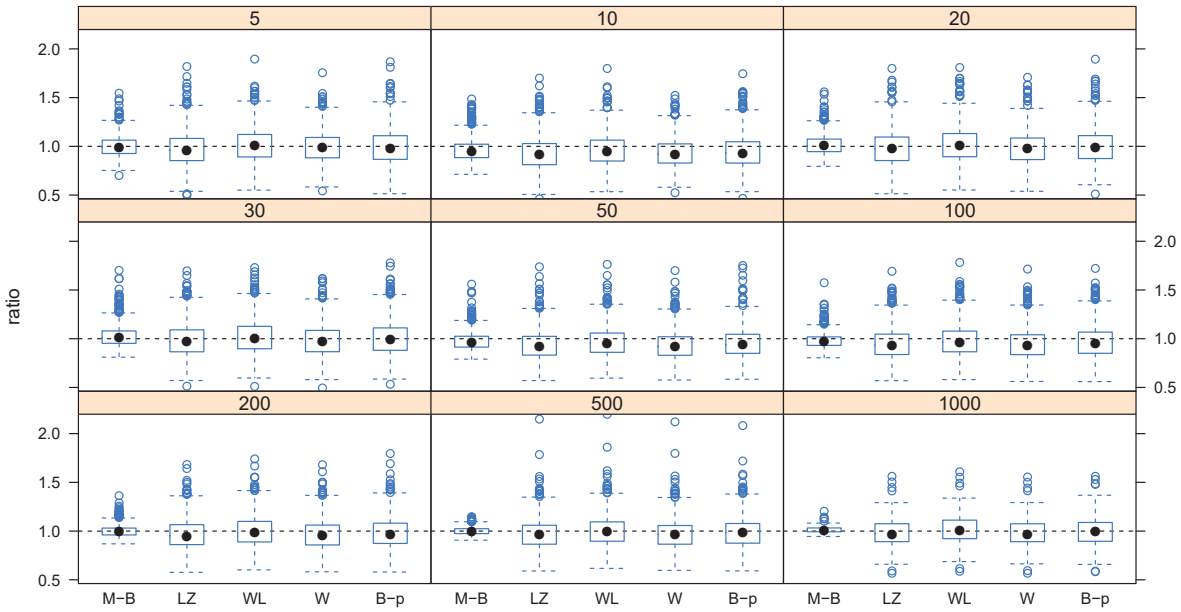


(a) AR(1) working correlation matrix

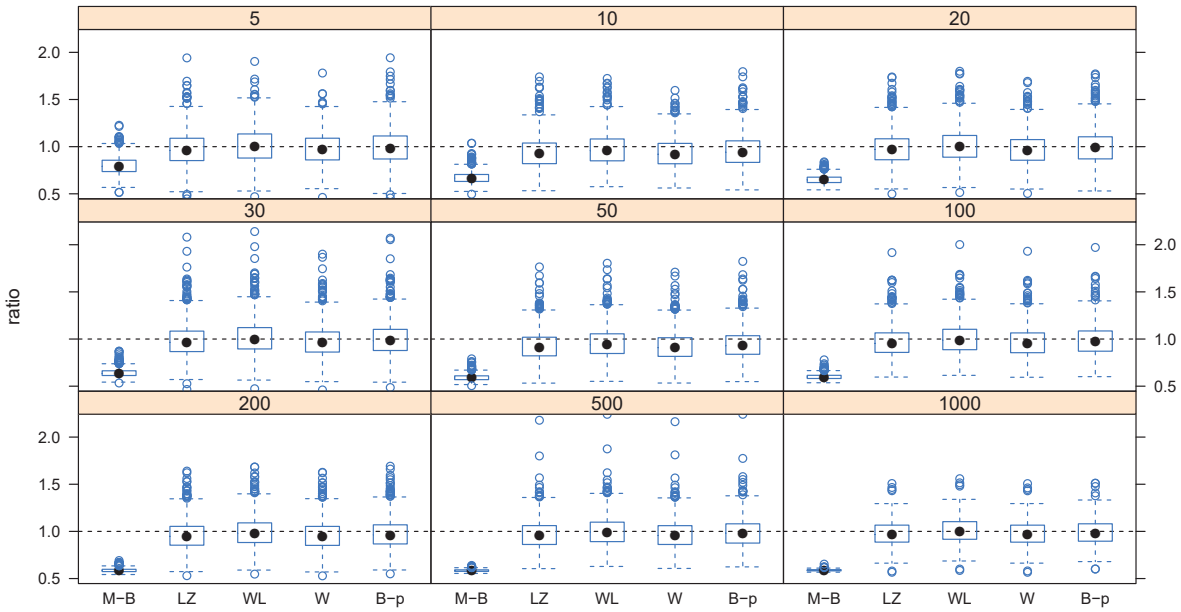


(b) Exchangeable working correlation matrix

Figure B.1: Boxplots for ratios of standard error by different methods versus the true standard error for normal data with sample size 30. The cluster sizes are 5, 10, 20, 30, 50, 100, 200, 500 and 1000 separately. M-B: model-based method, LZ: sandwich method, WL: method from Wang and Long [2011], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution.

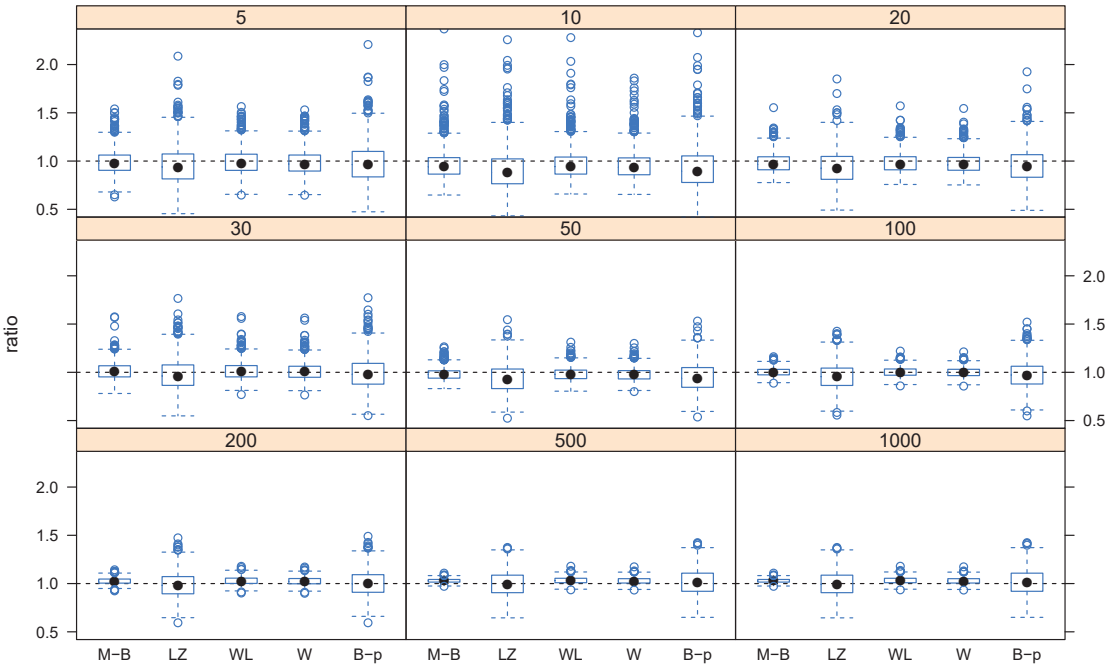


(a) AR(1) working correlation matrix

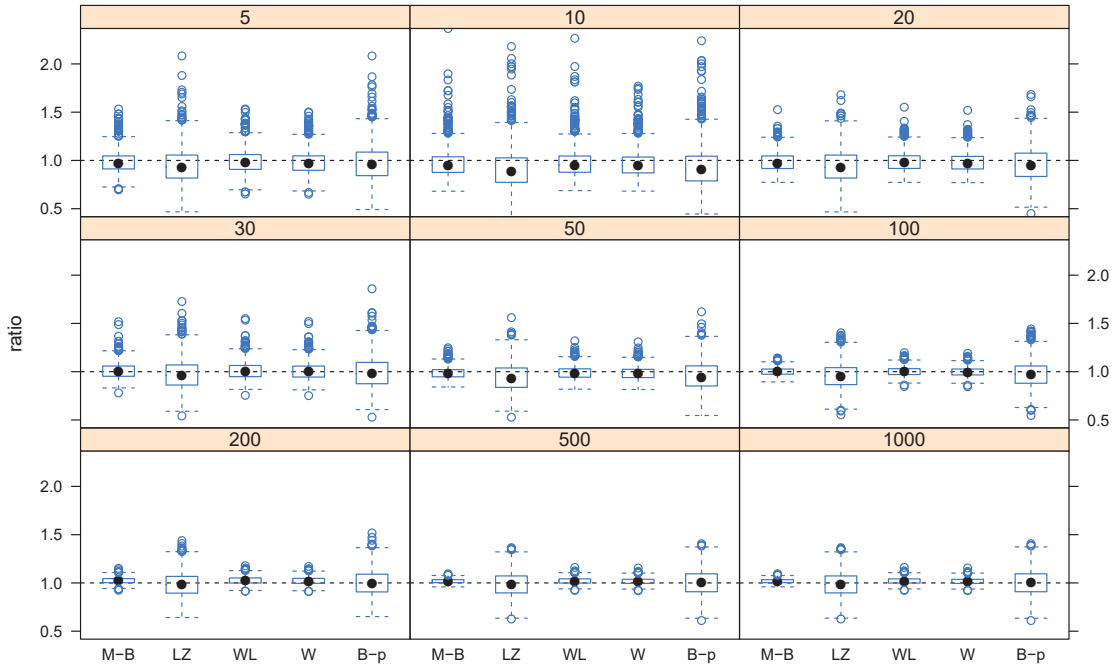


(b) Exchangeable working correlation matrix

Figure B.2: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data with sample size 30. The cluster sizes are 5, 10, 20, 30, 50, 100, 200, 500 and 1000 separately. M-B: model-based method, LZ: sandwich method, WL: method from Wang and Long [2011], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution.



(a) AR(1) working correlation matrix



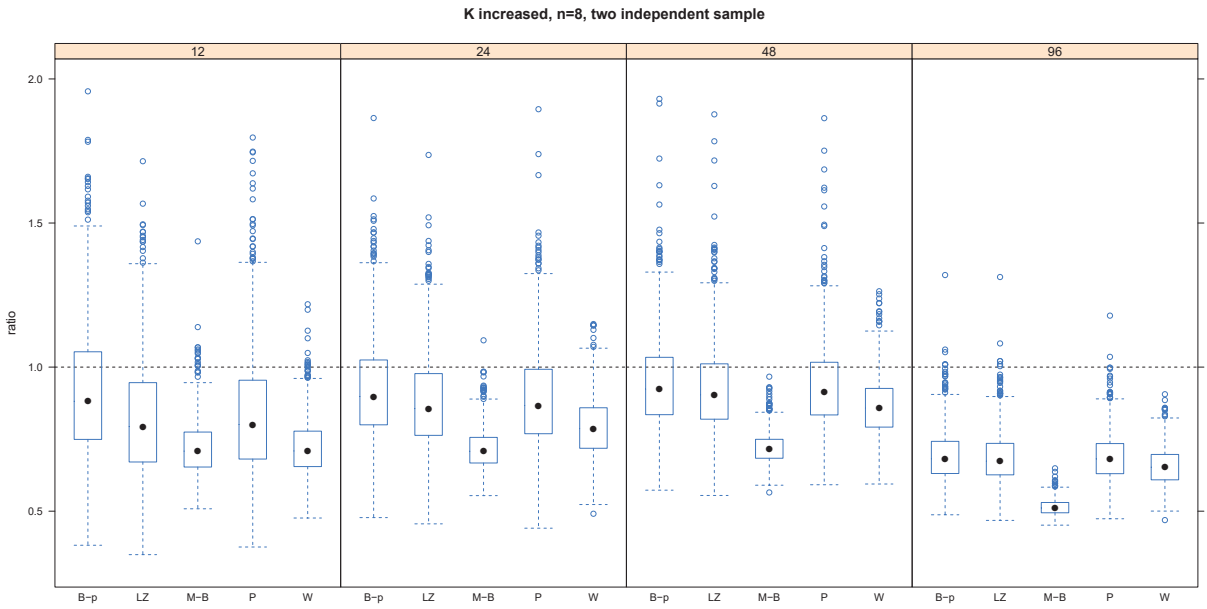
(b) Exchangeable working correlation matrix

Figure B.3: Boxplots for ratios of standard error by different methods versus the true standard error for binomial data with sample size 30. The cluster sizes are 5, 10, 20, 30, 50, 100, 200, 500 and 1000 separately. M-B: model-based method, LZ: sandwich method, WL: method from Wang and Long [2011], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution.

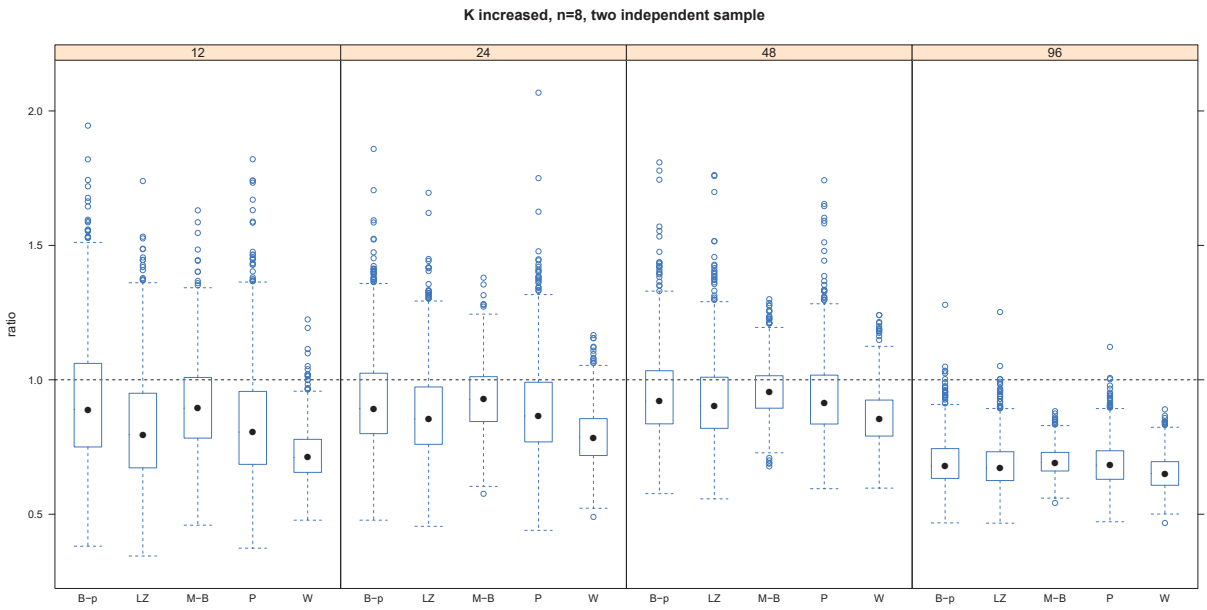
```
  y<-matrix(NA,K*n,5)
  dimnames(y)<-list(NULL,c("id","yy","xx","time","mu"))
  Omega=eigen(exc(rho,n))
  ##the first eigenvector
  xx=Omega$vectors[,1]
  btrue=c(log(2),sqrt(n)/2*Omega$values[1])
  ##using the first eigenvalue
  mui<-exp(btrue[1]+btrue[2]*xx)
  Sigma=diag(log(2),n)  ##define the diagnoal
  for(j in 1:n){
    Sigma[j,-(1:j)]=log(1+rho*sqrt(mui[j]+mui[j]^2)*sqrt(mui
        [-(1:j)]+mui[-(1:j)]^2)/(mui[j]*mui[-(1:j)]))
  }
  ##define the lower triangle
  Sigma[lower.tri(Sigma)]=t(Sigma)[lower.tri(Sigma)]
  ##Sigma is the same for the different cluster
    for (i in 1:K){
    id<-rep(i,n)
    time<-1:n
    mi=exp(mvrnorm(1,mu=log(mui)-log(2)/2,Sigma))
    yy=rpois(n,mi)
    #yy=sapply(mi,rpois,n=1)
    y[((i-1)*n+1):(i*n),] <-cbind(id,yy,xx,time,mui)
  }
  y
}

mvrpois_warton2<-function(K,n,rho)
{
  y<-matrix(NA,K*n,5)
  dimnames(y)<-list(NULL,c("id","yy","xx","time","mu"))
  Omega=eigen(exc(rho,n))
  xx=rep(0,n)
```

```
  btrue=c(log(2),sqrt(n)/2*Omega$values[1])
  mui<-exp(btrue[1]+btrue[2]*xx)
  Sigma=diag(log(2),n)
  ##define the diagnoal
  for(j in 1:n){
    Sigma[j,-(1:j)]=log(1+rho*sqrt(mui[j]+mui[j]^2)*sqrt(mui
        [-(1:j)]+mui[-(1:j)]^2)/(mui[j]*mui[-(1:j)]))
  }
  ##define the lower triangle
  Sigma[lower.tri(Sigma)]=t(Sigma)[lower.tri(Sigma)]
  ##Sigma is the same for each cluster
    for (i in 1:K){
    id<-rep(i,n)
    time<-1:n
    mi=exp(mvrnorm(1,mu=log(mui)-log(2)/2,Sigma))
    yy=rpois(n,mi)
    y[((i-1)*n+1):(i*n),] <-cbind(id,yy,xx,time,mui)
  }
  y
}

####### EXC correlation structure ##########
exc<- function(r,n)
{
 out <- diag(n)*.5
 out[row(out)!=col(out)]<-r*.5
 out+t(out)
}

####### Generate correlated data ##########
K=20
n=30
rho=.25
```

```
y1<-as.data.frame(mvrpois_warton1(K/2,n,rho))
y2<-as.data.frame(mvrpois_warton2(K/2,n,rho))
y=rbind(y1,y2)
X=y$xx;  Y=y$yy
```

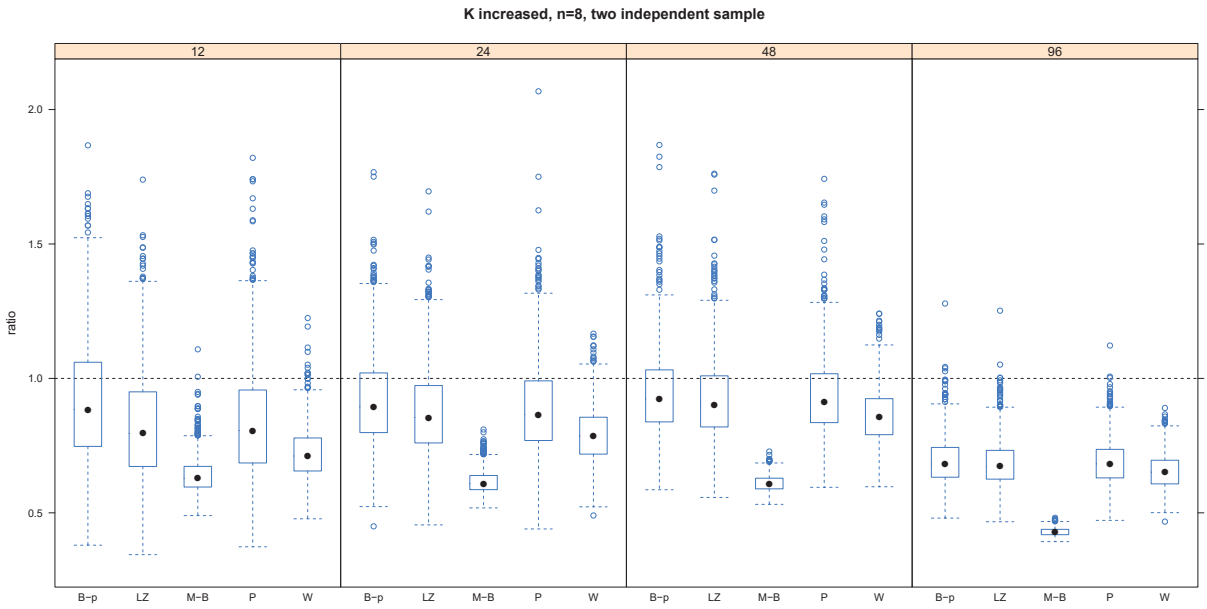


(a) AR(1) working correlation matrix

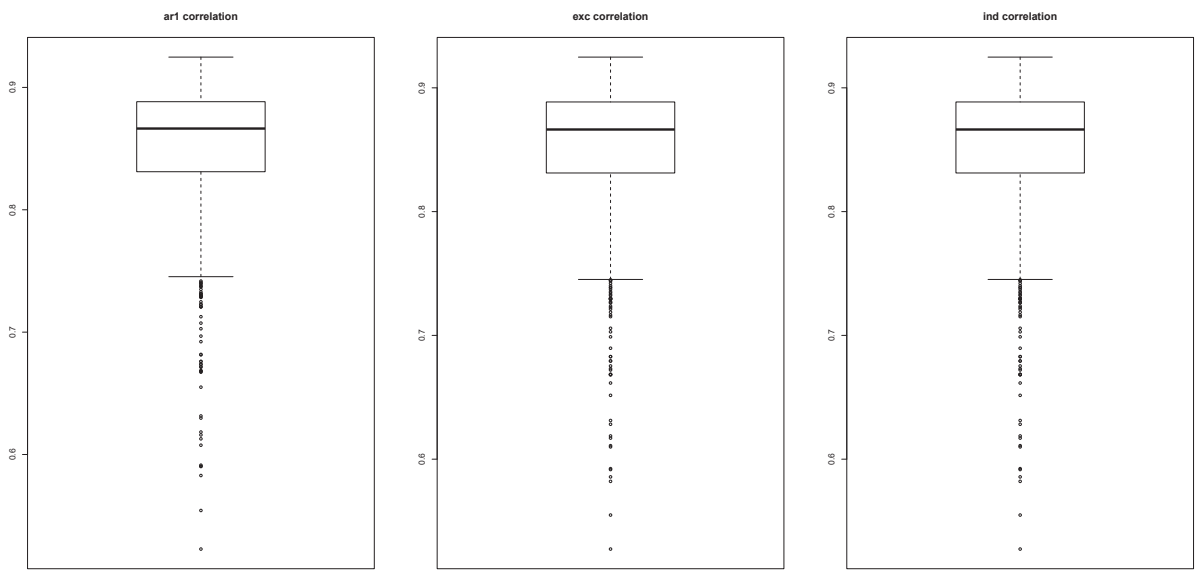


(b) Exchangeable working correlation matrix

Figure B.4: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data when number of clusters increased. The cluster sizes is 8. There are two independent samples. M-B: model-based method, LZ: sandwich method, P: method from Pan [2001], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution

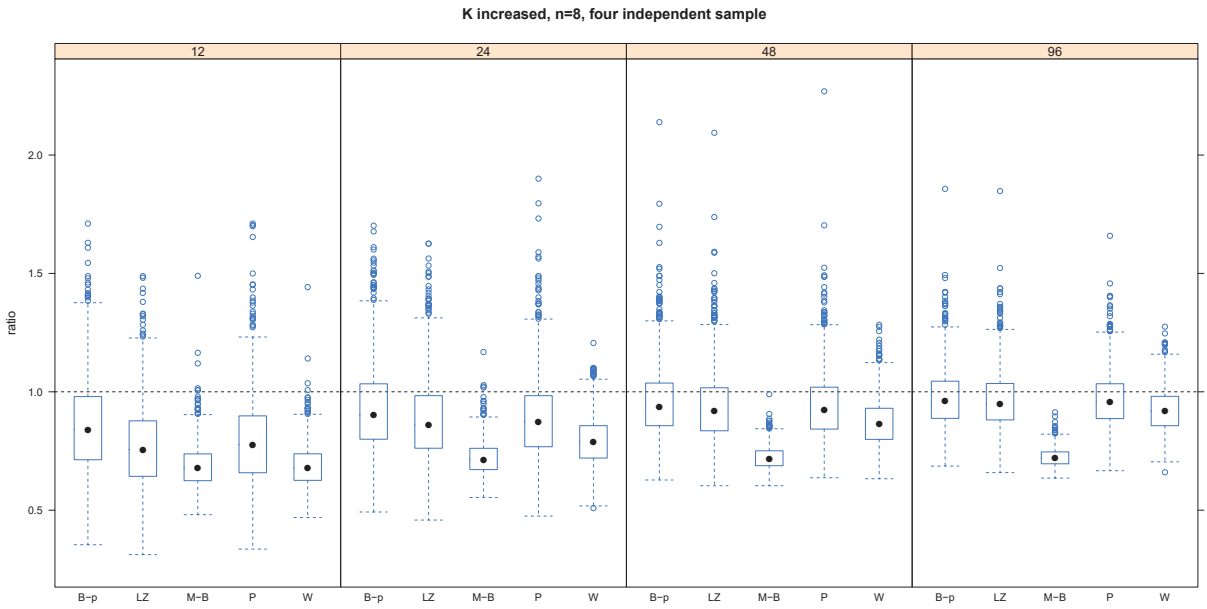


(a) Independent working correlation matrix

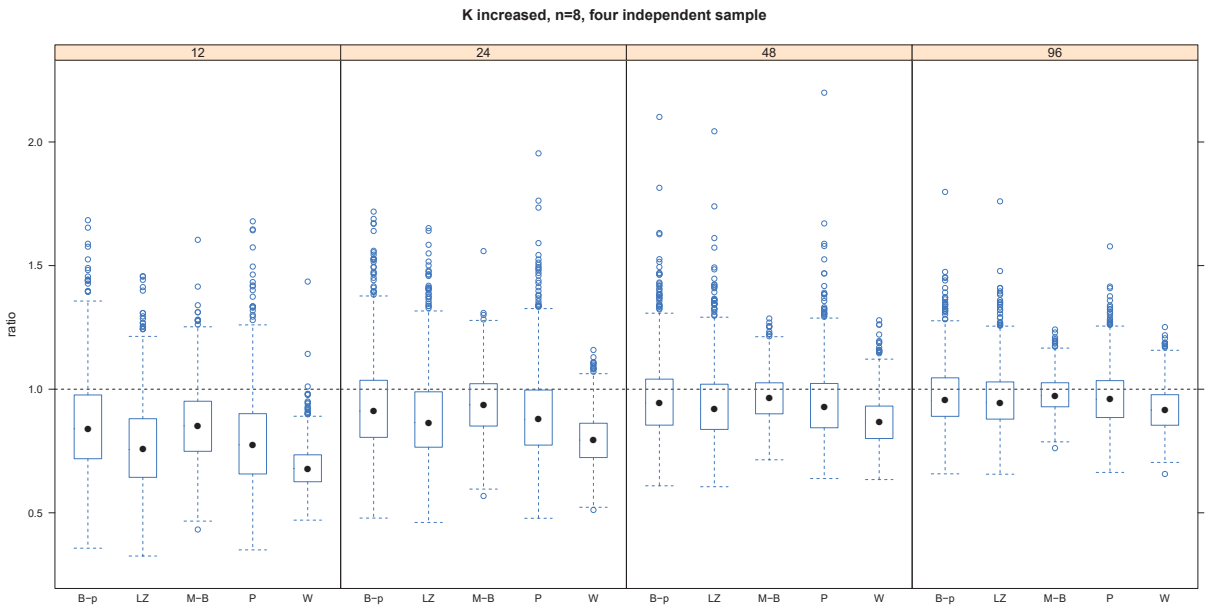


(b) Boxplot for regularized parameter

Figure B.5: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data when number of clusters increased. The cluster sizes is 8. There are two independent samples. M-B: model-based method, LZ: sandwich method, P: method from Pan [2001], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution

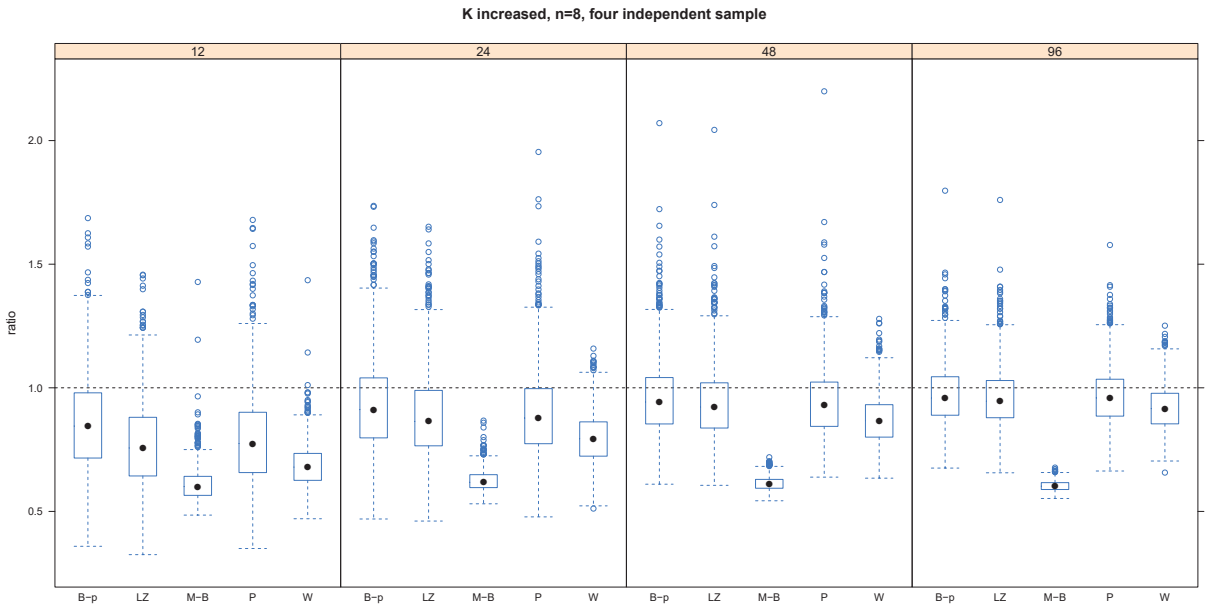


(a) AR(1) working correlation matrix

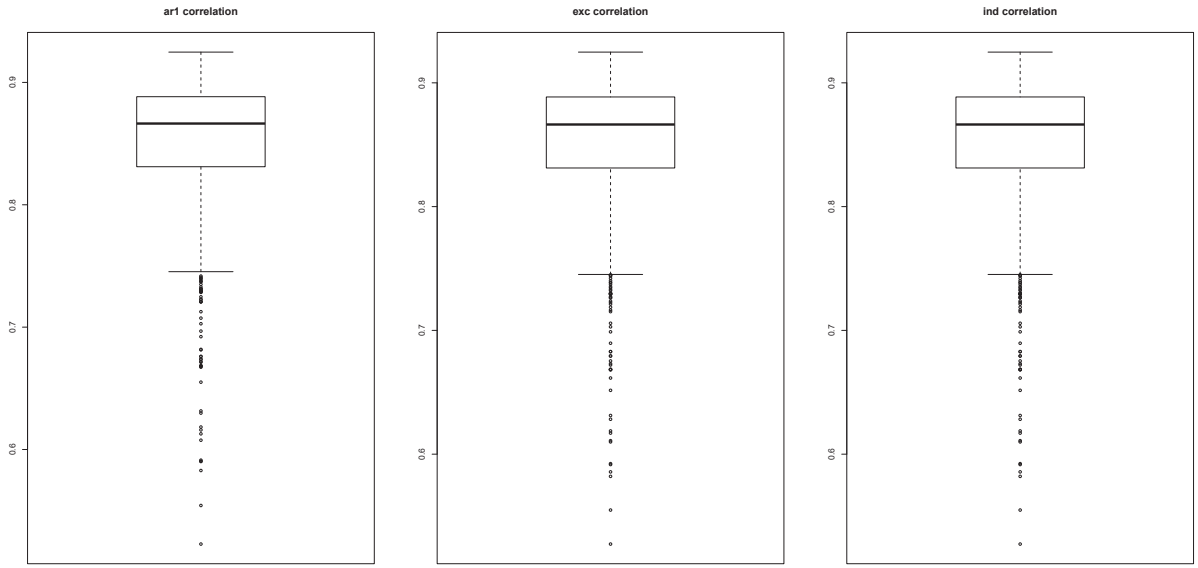


(b) Exchangeable working correlation matrix

Figure B.6: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data when number of clusters increased. The cluster sizes is 8. There are four independent samples. M-B: model-based method, LZ: sandwich method, P: method from Pan [2001], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution

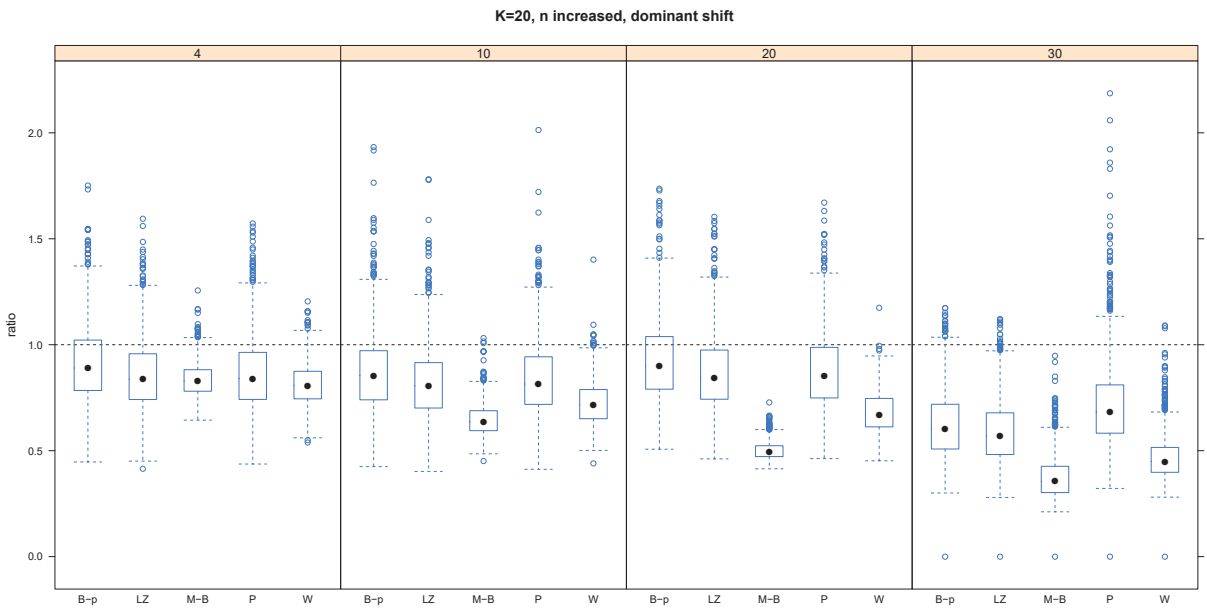


(a) Independent working correlation matrix

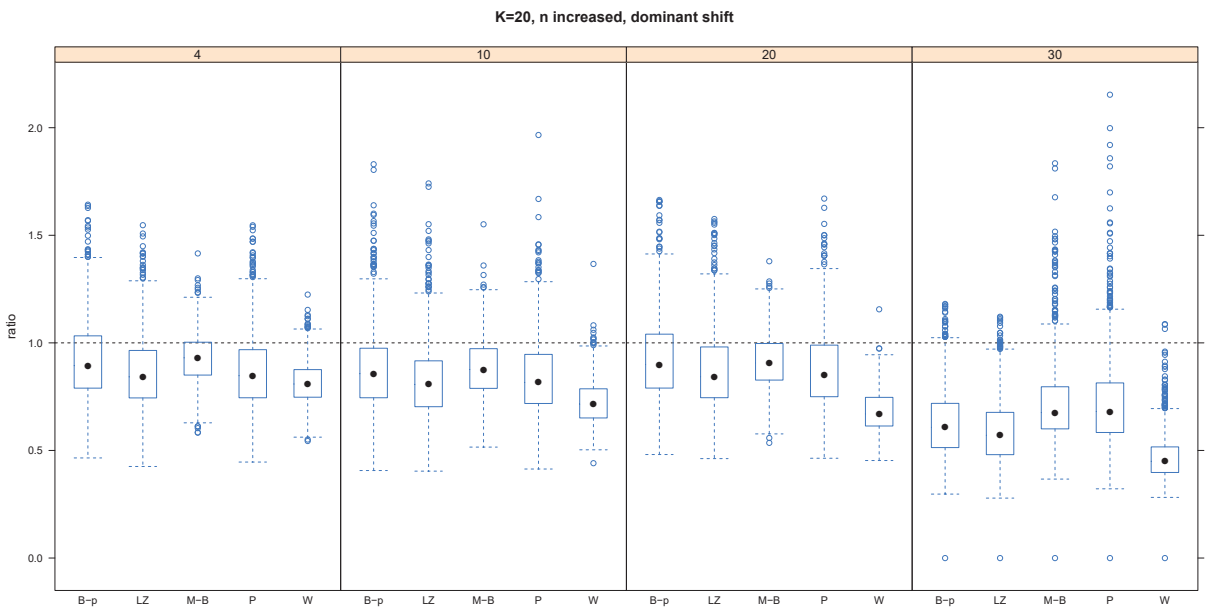


(b) Boxplot for regularized parameter

Figure B.7: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data when number of clusters increased. The cluster sizes is 8. There are four independent samples. M-B: model-based method, LZ: sandwich method, P: method from Pan [2001], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution

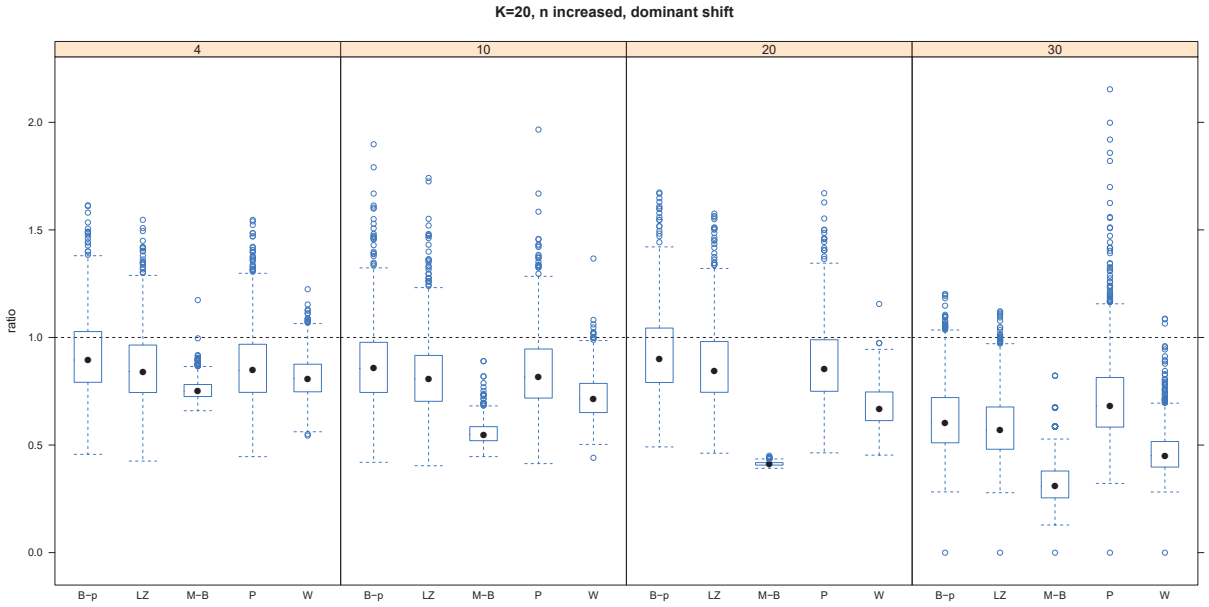


(a) AR(1) working correlation matrix

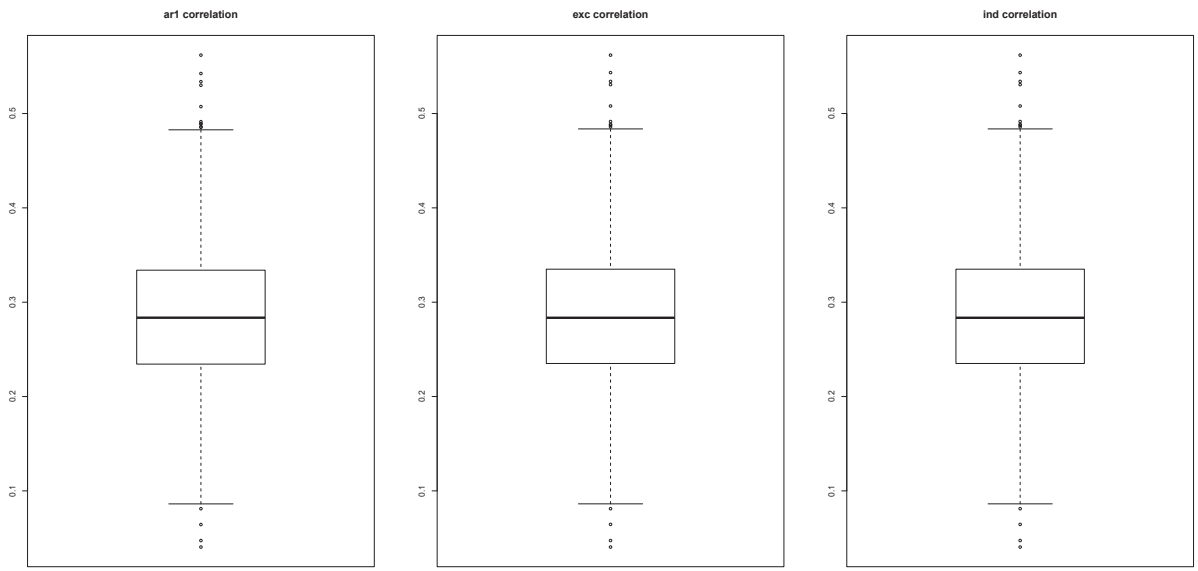


(b) Exchangeable working correlation matrix

Figure B.8: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data with the cluster size increased from 4 to 30. The number of cluster size is 20. M-B: model-based method, LZ: sandwich method, P: method from Pan [2001], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution

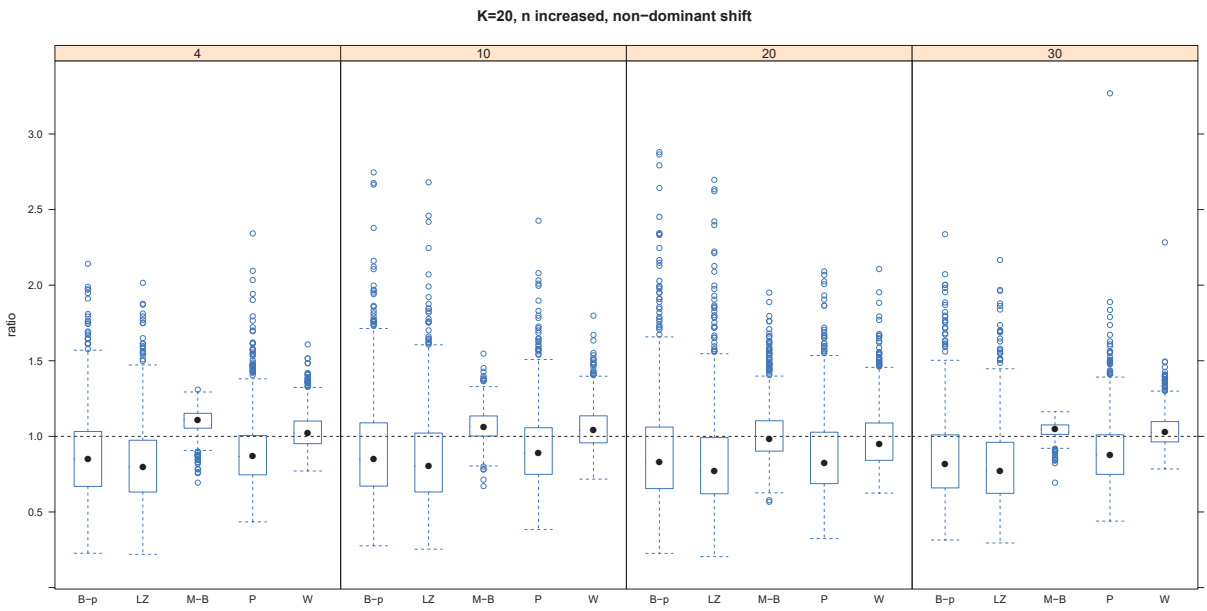


(a) Independent working correlation matrix

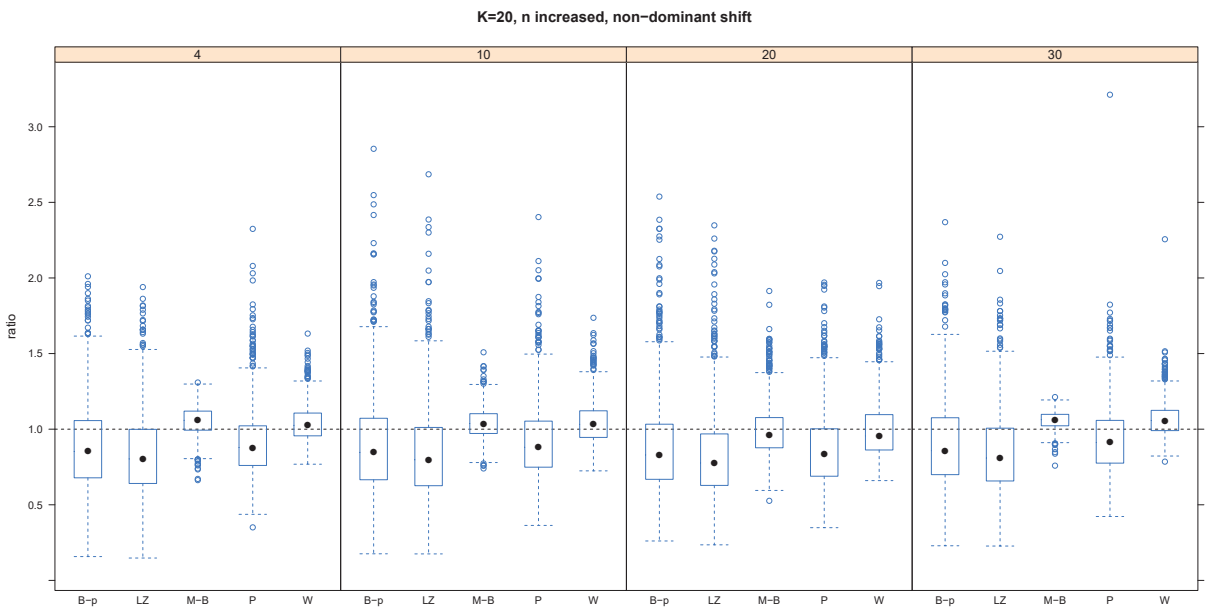


(b) Boxplot for regularized parameter

Figure B.9: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data with the cluster size increased from 4 to 30. The number of cluster size is 20. M-B: model-based method, LZ: sandwich method, P: method from Pan [2001], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution

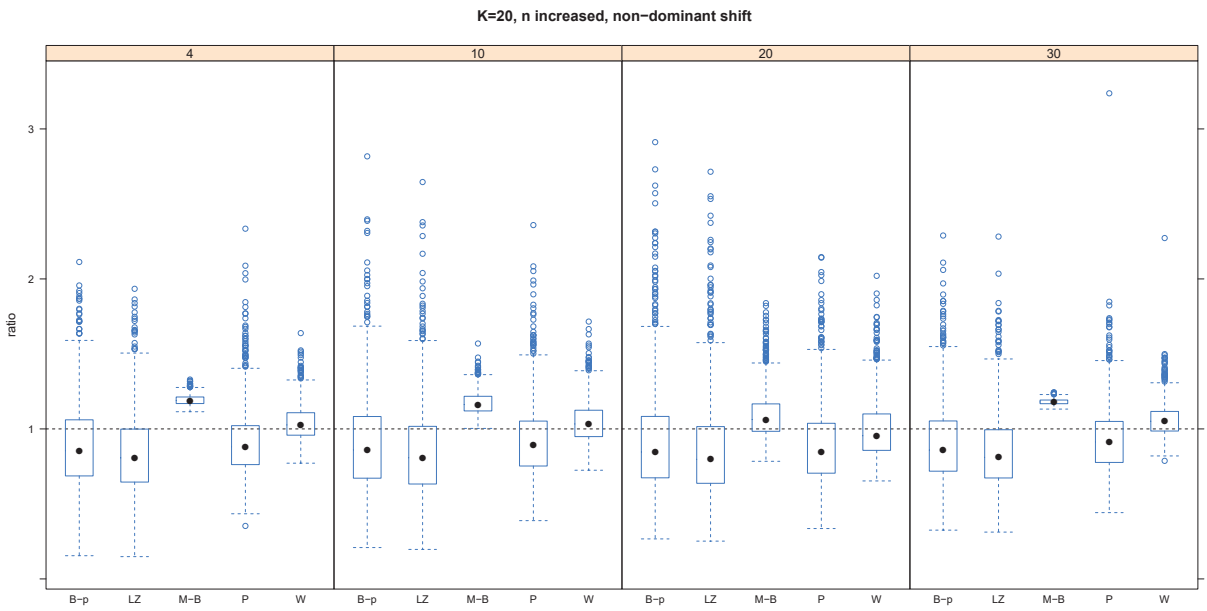


(a) AR(1) working correlation matrix

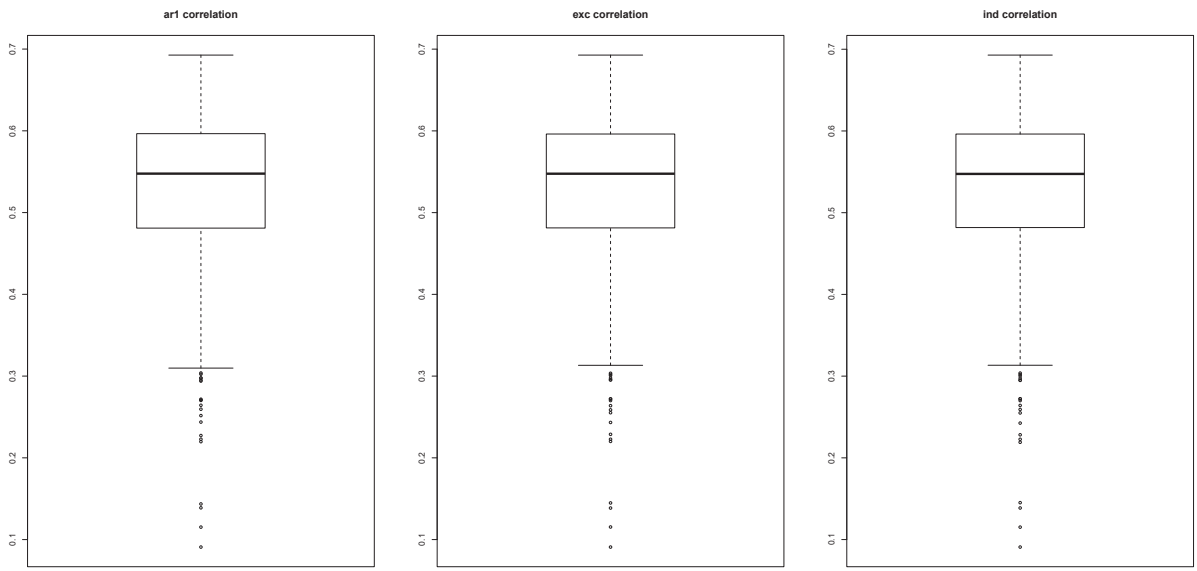


(b) Exchangeable working correlation matrix

Figure B.10: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data with the cluster size increased from 4 to 30. The number of cluster size is 20. M-B: model-based method, LZ: sandwich method, P: method from Pan [2001], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution



(a) Independent working correlation matrix



(b) Boxplot for regularized parameter

Figure B.11: Boxplots for ratios of standard error by different methods versus the true standard error for Poisson data with the cluster size increased from 4 to 30. The number of cluster size is 20. M-B: model-based method, LZ: sandwich method, P: method from Pan [2001], W: regularized method from Warton [2011], B-p: bootstrap method with Poisson distribution

Appendix C

Supplementary Material for Chapter 5

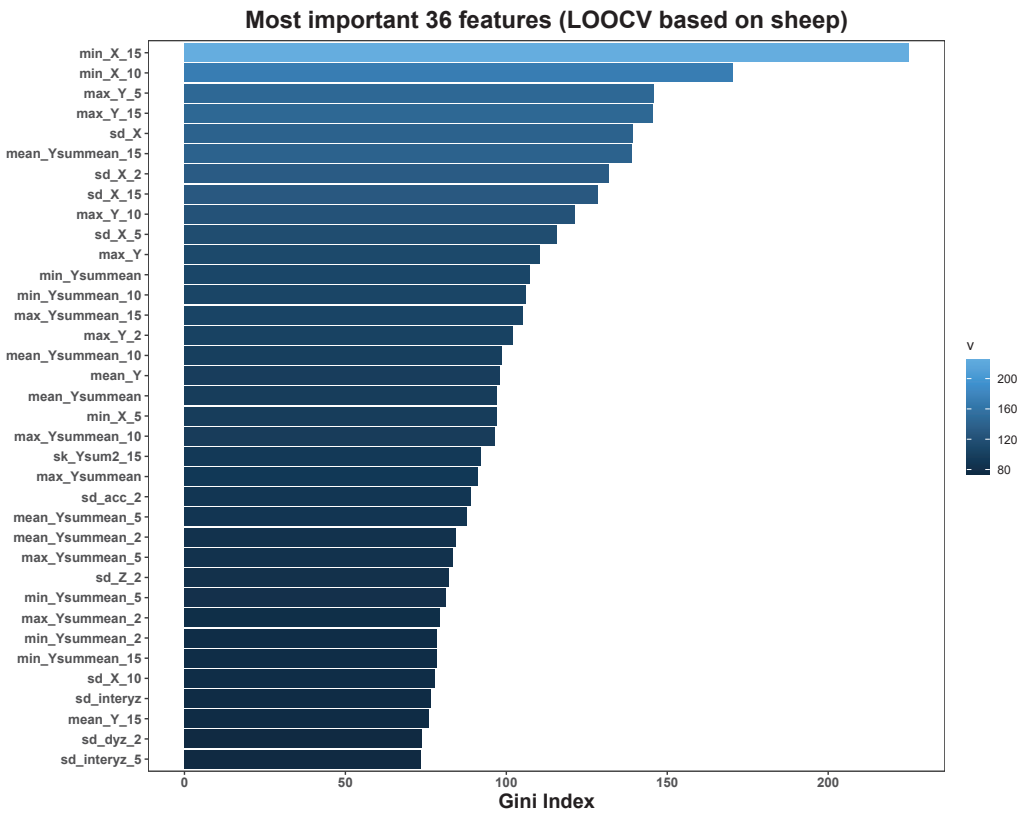


Figure C.1: The list of 36 of the most important features selected by RF based on the leaving one animal out cross-validation scheme

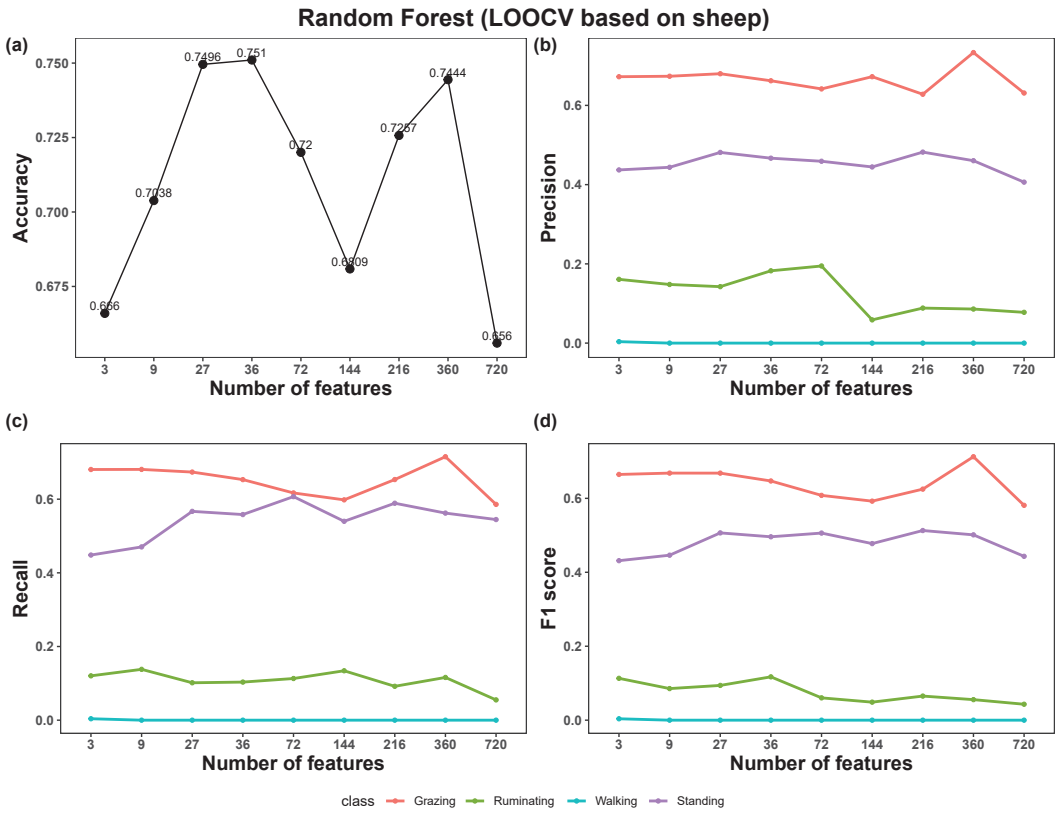


Figure C.2: The over accuracy, precision, recall and F1 score values from the different number of top ranked features chosen from Random Forest (RF) when using the mixed time window approach and leaving one animal out cross-validation scheme

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